Note for MATH 550 - Elements of Topology

Hsu-Hsiang Tsao

Fall 2026

Introduction

This text serves as my notes for the course MATH 550 - Elements of Topology instructing by Dr. Ryan Blair in Fall 2026 at CSULB, and has as a reference *Topology*, second edition, by Munkres.

The development of the notes primarily follows Munkres's text, and additional topics and lecture materials are incorporated in sections where I deem fit.

Unless otherwise specified, we

- denote $\mathbb{N} := \{1, 2, \dots\}$ and $\mathbb{W} := \{0, 1, 2, \dots\}$ the sets of natural numbers and whole numbers, respectively;
- assume functions have non-empty domains and codomains;
- denote, respectively, $\mathcal{P}(X)$ and $|X|$ to be the power set and cardinality of a set X;
- denote n and m to be natural numbers;
- denote $A/B := \{a \in A : a \notin B\}$ for sets A and B;

Due to the inability of the writer to observe the obvious, countably many mundane or lengthy verifications are included in the text. In attempts to avoid drowning in arguments, such verifications are (often) enclosed in the boxed environments or stated as lemmas, which may be disregarded.

Contents

1 Topological Spaces and Continuous Functions **4**

1.1 Topological Spaces 4

1.1.1 Elementary Examples 4

1.2 Basis for a Topology 6

1.2.1 Basis 6

1.2.2 Subbasis 8

1.2.3 Topologies Generated by Subsets 10

1.3 Order Topology 11

1.3.1 Ordered Sets 11

1.3.2 Order Topology 12

1.4 Product Topology 17

1.5 Subspace Topology 20

1.6 Closed Sets and Limit Points 23

1.6.1 Closed Sets 23

1.6.2 Closure and interior 24

1.6.3 Closure and Limit Point 26

1.6.4 Hausdorff Space 28

1.6.5 Convergence and Limit 30

1.6.6 Exercises 31

1.7 Continuous Functions 35

1.7.1 Elementary Definitions and Examples 35

1.7.2 Elementary Properties 36

1.7.3 Homeomorphism 38

1.7.4 Constructing Continuous Functions 38

1.7.5 Exercises 40

1.7.6 *Continuity of Vector Operations 43

1.7.7 *Topological Group 45

1.8 *Initial Topology 50

1.9 Product Topology (Revisited) 54

2 Metric Spaces **60**

1 Topological Spaces and Continuous Functions

1.1 Topological Spaces

Definition 1.1.1 (Topology, topological space)**.** *(a) Suppose X is a set. A* ***topology*** *of X is a collection τ of subsets of X such that*

(a.1) $X, \emptyset \in \tau$,

(a.2) τ is closed under union, and

(a.3) τ is closed under finite intersection.

(b) A ***topological space*** *is a pair (X, τ) of set X and a topology τ on X.*

Definition 1.1.2 (Open set, closed set)**.** *Suppose (X, τ) is a topological space.*

(a) An ***open set of (X, τ) or τ-open set*** *is subset of X contained τ.*

(a.1) A ***neighborhood in (X, τ) of a point (subset)*** *of X is an open set of (X, τ) containing the point (subset).*

(b) A ***closed set of (X, τ) or τ-closed set*** *is a subset of X whose complement (relative to X) is an open set of (X, τ).*

Definition 1.1.3 (Comparing topologies)**.** *Suppose X is a set with topologies τ_0 and τ_1. Then,*

(a) τ_0 is ***finer (stronger/larger) than*** *τ_1 if $\tau_0 \supset \tau_1$.*

(a.1) τ_0 is ***strictly finer (stronger/larger) than*** *τ_1 if $\tau_0 \supsetneq \tau_1$.*

(b) τ_0 is ***coarser (weaker/smaller) than*** *τ_1 if $\tau_0 \subset \tau_1$.*

(b.1) τ_0 is ***strictly coarser (weaker/smaller) than*** *τ_1 if $\tau_0 \subsetneq \tau_1$.*

(c) τ_0 and τ_1 are ***comparable*** *if $\tau_0 \subset \tau_1$ or $\tau_1 \subset \tau_0$.*

1.1.1 Elementary Examples

Remark 1.1.1. *Shrinking sequences of subsets are nice in constructing topologies, since, as we now prove, closures under finite intersection and union are automatic.*

Lemma 1.1.1. *Suppose X is a set. If $(A_j)_{j\in\mathbb{N}}$ is a shrinking sequence of subsets of X (that is, $A_1 \supset A_2 \supset \dots$), then*

$$\tau(\{A_j : j \in \mathbb{N}\}) := \{A_j : j \in \mathbb{N}\} \cup \{X, \emptyset\}$$

is a topology on X.

Proof. Let $(A_j)_{j\in\mathbb{N}}$ be a shrinking sequence of subsets of X and $A_0 := X$, so $(A_j)_{j\in\mathbb{W}}$ remains a shrinking sequence. For brevity, denote $\tau(\{A_j : j \in \mathbb{N}\})$ by τ. By construction, $X, \emptyset \in \tau$, so it remains to show the closures of τ under unions and finite intersections.

Fix $\mathcal{A} \subset \tau$. Without loss of generality, we may assume that $\mathcal{A}$ does not contain the empty set. It follows that, for any $U \in \mathcal{A}$, $U = A_{j(U)}$ for some $j(U) \in \mathbb{W}$. Since $(A_j)_{j\in\mathbb{W}}$ is shrinking,

$$\cup_{U\in\mathcal{A}} U = \cup_{U\in\mathcal{A}} A_{j(U)} = A_{\min\{j(U):U\in\mathcal{A}\}} \in \tau.$$

Now, fix $n \in \mathbb{N}$ and $U_1, \ldots, U_n \in \tau$. If U_J is empty for some $J \in \{1, \ldots, n\}$, then $\cap_{j=1}^n U_j = \emptyset \in \tau$. Thus, we suppose $U_1, \ldots, U_n$ are non-empty. Again, for $j \in \{1, \ldots, n\}$, $U_j = A_{k(j)}$ for some $k(j) \in \mathbb{W}$ by the construction of τ. Since $(A_j)_{j \in \mathbb{W}}$ is shrinking,

$$\cap_{j=1}^n U_j = \cap_{j=1}^n [A_{k(j)}] = A_{\max\{k(j): j \in \{1,\ldots,n\}\}} \in \tau.$$

Indeed, τ is closed under unions and finite intersections. By definition, τ is a topology of X. $\square$

Example 1.1.2 (Finite Topology)**.** *Suppose $X = \{0, 1, 2, 3\}$. By a preceding lemma, X has the topologies*

$$\{\emptyset, X, \{0\}, \{0, 1\}\}, \ \{\emptyset, X, \{1\}, \{0, 1\}, \{0, 1, 3\}\}, \text{ and } \{\emptyset, X, \{2\}, \{0, 1, 2\}\}.$$

Lemma 1.1.3 (Discrete and Indiscrete Topologies)**.** *Suppose X is a set. Then, $\mathcal{P}(X)$ and $\{X, \emptyset\}$ are topologies on X.*

Remark 1.1.2. *Since the power set of a set gives rise to the discrete topology, by considering the cardinality 2^n of the power set of a set X with $n \in \mathbb{N}$ elements, one may show via the closure of a topology under union and finite intersections that there is no topology consisting of exactly $2^n - 1$ elements.*

Lemma 1.1.4 (Finite Complement and Countable Complement Topology)**.** *Suppose X is a set. Then,*

$$\mathcal{T}_f := \{U \subset X : X/U \text{ is finite or } U = \emptyset\}$$

and

$$\mathcal{T}_C := \{U \subset X : X/U \text{ is countable or } U = \emptyset\}$$

are topologies of X.

1.2 Basis for a Topology

1.2.1 Basis

Definition 1.2.1 (Cover)**.** *Suppose X is a set.*

*(a) A **cover** of X is a collection $\mathcal{A}$ of subsets of X such that $X = \cup_{U\in\mathcal{A}} U$.*

*(a.1) A collection $\mathcal{A}$ of subsets of X **covers** X if it is a cover of X.*

Definition 1.2.2 (Basis for a set)**.** *Suppose X is a set.*

*(a) A **basis** of X is a collection $\mathcal{B}$ of subsets of X such that*

(a.1) for any $x \in X$, there exists $B_x \in B$ containing x (equivalently, $\mathcal{B}$ covers X), and

(a.2) for any $x \in X$, if $x \in B_1 \cap B_2$ for some $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in B$ such that $x \in B_3 \subset B_1 \cap B_2$.

*(b) Suppose $\mathcal{B}$ is a basis of X. A **basis element** of $\mathcal{B}$ is an element of $\mathcal{B}$.*

Definition 1.2.3 (Topology generated by basis, basis for a topology)**.** *Suppose X is a set.*

*(a) The **topology generated by a basis** $\mathcal{B}$ of X is the collection*

$$\tau(\mathcal{B}) := \{U \subset X : \forall x \in U, \text{ there is a basis element } B_x \in \mathcal{B} \text{ with } x \in B_x \subset U\}.$$

(b) Suppose τ is a topology on X.

*(b.1) A basis $\mathcal{B}$ of X **generates** τ if $\tau = \tau(\mathcal{B})$.*

*(b.2) A **basis for** τ is a basis of X that generates τ.*

Lemma 1.2.1 (Topology Generated by Basis)**.** *Suppose X is a set with a basis $\mathcal{B}$. Then,*

$$\tau(\mathcal{B}) := \{U \subset X : \forall x \in U, \text{ there is a basis element } B_x \in \mathcal{B} \text{ with } x \in B_x \subset U\}$$

is a topology of X, and $\mathcal{B} \subset \tau(\mathcal{B})$.

Proof. Since $\mathcal{B}$ covers X, $X \in \tau(\mathcal{B})$. We observe that $\emptyset \in \tau(\mathcal{B})$, since $\emptyset$ satisfies the condition

$$\forall x \in U, \text{ there is a basis element } B_x \in \mathcal{B} \text{ with } x \in B_x \subset U$$

vacuously, as it has no elements.

[Closure under union] Fix $\mathcal{A} \subset \tau(\mathcal{B})$. Fix $x \in \cup_{U\in\mathcal{A}} U$, so $x \in U_0$ for some $U_0 \in \mathcal{A}$. Since U_0 lives in $\tau(\mathcal{B})$, there is a basis element $B_x \in \mathcal{B}$ containing x with

$$x \in B_x \subset U \subset \cup_{U\in\mathcal{A}} U.$$

That is, $\cup_{U\in\mathcal{A}} U \in \tau(\mathcal{B})$ by definition. Indeed, $\tau(\mathcal{B})$ is closed under union.

[Closure under finite intersection] Fix elements U, W of $\tau(\mathcal{B})$. Fix $x \in U \cap W$, so there are basis elements B_U and B_W containing x with $B_U \subset U$ and $B_W \subset W$. That is, $x \in B_U \cap B_W$, so there is a bais element B_x containing x with $B_x \subset B_U \cap B_W \subset U \cap W$, by the definition of a basis. By the definition of $\tau(\mathcal{B})$, $U \cap W \in \tau(\mathcal{B})$. Indeed, $\tau(\mathcal{B})$ is closed under finite intersection.

By definition, $\tau(\mathcal{B})$ is a topology of X.

[$\mathcal{B} \subset \tau(\mathcal{B})$] For any $B \in \mathcal{B}$, $\forall x \in X$, $x \in B \subset B$, so $B \in \tau(\mathcal{B})$ by the construction of $\tau(\mathcal{B})$. That is, $\mathcal{B} \subset \tau(\mathcal{B})$. □

Lemma 1.2.2 (Topologies Sharing Basis Coincide)**.** *Suppose X is a set with topologies τ_1, τ_2. If $\mathcal{B}$ is a basis for τ_1 and τ_2, then $\tau_1 = \tau_2$.*

Proof. Suppose $\mathcal{B}$ is a basis for τ_1 and τ_2, then $\tau_1 = \tau_2$. By definition, τ_1 and τ_2 coincide with $\tau(\mathcal{B})$. Particularly, $\tau_1 = \tau(\mathcal{B}) = \tau_2$. □

Lemma 1.2.3 (Characterization of Topology Generated by Basis)**.** *Suppose X is a set with a basis $\mathcal{B}$. Then, $\tau(\mathcal{B}) = \{\cup_{U\in\mathcal{A}} U : \mathcal{A} \subset \mathcal{B}\}$.*

Proof. **[$\subset$]** Fix an open set U of $\tau(\mathcal{B})$. By the construction of $\tau(\mathcal{B})$, for any $x \in U$, there is a basis element B_x containing x with $B_x \subset U$. That is, $U = \cup_{x\in U} B_x$. Indeed, $\tau(\mathcal{B}) \subset \{\cup_{U\in\mathcal{A}} U : \mathcal{A} \subset \mathcal{B}\}$.

[$\supset$] We recall that basis elements are open sets of $\tau(\mathcal{B})$, so the closure of $\tau(\mathcal{B})$ under union ensures that all unions of elements of $\mathcal{B}$ remain open sets. That is, $\tau(\mathcal{B}) \supset \{\cup_{U\in\mathcal{A}} U : \mathcal{A} \subset \mathcal{B}\}$.

Indeed, the two collections coincide. □

Lemma 1.2.4 (Sufficient Condition for Basis of Topology)**.** *Suppose (X, τ) is a topological space and $\mathcal{C}$ is a collection of subsets of X. If*

(a) $\mathcal{C} \subset \tau$ and

(b) for every open set U of τ, $\forall x \in U$, there exists $C_x \in \mathcal{C}$ such that $x \in C_x \subset U$ (equivalently, U is the union of some elements of $\mathcal{C}$),

then $\mathcal{C}$ is a basis for τ.

Proof. Suppose assumption (a) and (b) hold. We show that [1] $\mathcal{C}$ is a basis for X and [2] τ coincides with $\tau(\mathcal{C})$.

[1] Since X is an open set, assumption (b) ensures that X is the union of some elements of $\mathcal{C}$. That is, $\mathcal{C}$ covers X.

Fix $x \in X$. Suppose x is contained in $C_1 \cap C_2$ for some $C_1, C_2 \in \mathcal{C}$. Since $\mathcal{C} \subset \tau$ by assumption, $C_1 \cap C_2$ remains open in (X, τ). It follows from assumption (b) that there is $C_x \in \mathcal{C}$ with $x \in C_x \subset C_1 \cap C_2$. By definition, $\mathcal{C}$ is a basis for X as desired.

[2] Since $\mathcal{C}$ is a basis for X, $\tau(\mathcal{C})$ is defined. We observe that assumption (b) ensures that $\tau \subset \tau(\mathcal{C})$, since every τ-open set is a union of some $\tau(\mathcal{C})$-open sets.

Conversely, every open set of $\tau(\mathcal{C})$ is the union of some elements of $\mathcal{C}$ by 1.2.3 and, therefore, of τ by assumption (a). That is, $\tau(\mathcal{C}) \subset \tau$ by the closure of τ under union.

Indeed, the two topologies on X coincide. By definition, $\mathcal{C}$ is a basis for τ. □

Lemma 1.2.5 (Characterization of Finer Topology via Basis)**.** *Suppose X is a set with topologies τ_1 and τ_2 having basis $\mathcal{B}_1$ and $\mathcal{B}_2$, respectively. Then, τ_1 is finer than $\tau_2 \iff$ for every $x \in X$, for every neighborhood $B_2 \in \mathcal{B}_2$ of x, there is a neighborhood $B_1 \in \mathcal{B}_1$ of x contained in B_2.*

Proof. **[$\implies$]** Suppose τ_1 is finer than τ_2. That is, $\tau_1 \supset \tau_2$. Fix $x \in X$ and a basis element $B_2 \in \mathcal{B}_2$ containing x.

Since B_2 is open in τ_2 and, therefore, in τ_1, we may decompose B_2 into the union of some basis elements of $\mathcal{B}_1$. Particularly, x lives in a basis element of $\mathcal{B}_1$ that is contained in B_2.

[$\impliedby$] Suppose that for every $x \in X$, for every neighborhood $B_2 \in \mathcal{B}_2$ of x, there is a neighborhood $B_1 \in \mathcal{B}_1$ of x contained in B_2. Fix a non-empty open set U of τ_2 and fix $x \in U$.

Since $\mathcal{B}_2$ is a basis of τ_2, we may decompose $U = \cup_{V\in\mathcal{A}} V$ for some $\mathcal{A} \subset \mathcal{B}_2$. Particularly, x lives in a basis element $V_x \in \mathcal{B}_2$ residing in U, so there exists a neighborhood $B_x \in \mathcal{B}_1$ of x contained in V_x by assumption.

That is, $x \in B_x \subset V_x \subset U$, which ensures the decomposition $U = \cup_{x\in U} B_x$. It follows that U is the union of open sets of τ_1, so $U \in \tau_1$. Indeed, $\tau_2 \subset \tau_1$. □

Definition 1.2.4 (Topologies on $\mathbb{R}$)**.** *(a) The* ***standard topology*** $\tau_{\mathbb{R}}$ ***of*** $\mathbb{R}$ *is the topology generated by the basis*

$$\mathcal{B}_{\mathbb{R}} := \{(a,b) : a,b \in \mathbb{R} \text{ and } a < b\}.$$

(b) The ***lower limit topology*** τ_ℓ ***of*** $\mathbb{R}$ *is the topology generated by the basis*

$$\mathcal{B}_\ell := \{[a,b) : a,b \in \mathbb{R} \text{ and } a < b\}.$$

(c) The K***-topology*** τ_K ***of*** $\mathbb{R}$ *is the topology generated by the basis*

$$\mathcal{B}_K := \{(a,b) : a,b \in \mathbb{R} \text{ and } a < b\} \cup \{(a,b)/K : a,b \in \mathbb{R} \text{ and } a < b\},$$

where $K := \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$.

We denote $\mathbb{R}_\ell := (\mathbb{R}, \tau_\ell)$ *and* $\mathbb{R}_K := (\mathbb{R}, \tau_K)$. *If no ambiguity may arise, we denote by* $\mathbb{R}$ *the topological space* $\mathbb{R}$ *topologised by the standard topology of* $\mathbb{R}$.

Lemma 1.2.6 (Topologies on $\mathbb{R}$)**.** *The topologies* $\tau_{\mathbb{R}}, \tau_\ell, \tau_K$ *are topologies of* $\mathbb{R}$.

Proof. It suffices to verify that the corresponding collections $\mathcal{B}_{\mathbb{R}}, \mathcal{B}_\ell$, and $\mathcal{B}_K$ of the topologies are actually bases for $\mathbb{R}$. □

Lemma 1.2.7 (Fineness of $\mathbb{R}_\ell$ and $\mathbb{R}_k$)**.** *(a) The topologies of* $\mathbb{R}_\ell$ *and* $\mathbb{R}_K$ *are strictly finer than that of the standard topology of* $\mathbb{R}$.

(b) The topologies of $\mathbb{R}_\ell$ *and* $\mathbb{R}_K$ *are not comparable.*

1.2.2 Subbasis

Definition 1.2.5 (Subbasis)**.** *Suppose* X *is a set. A* ***subbasis*** *for* X *is a cover of* X.

Definition 1.2.6 (Topology generated by subbasis)**.** *Suppose* X *is a set.*

(a) The ***topology of*** X ***generated by a subbasis*** $\mathcal{S}$ *of* X *is the collection*

$$\tau(\mathcal{S}) := \left\{\cup_{U\in\mathcal{A}} U : \mathcal{A} \subset \left\{\cap_{j=1}^n S_j : n \in \mathbb{N}, S_1, \ldots, S_n \in \mathcal{S}\right\}\right\}.$$

(b) Suppose τ *is a topology on* X.

(b.1) A subbasis $\mathcal{S}$ *of* X ***generates*** τ *if* $\tau = \tau(\mathcal{S})$.

(b.2) A ***subbasis for*** τ *is a subbasis of* X *that generates* τ.

Lemma 1.2.8 (Topology Generated by Subbasis)**.** *Suppose* X *is a set and* $\mathcal{S}$ *is a subbasis for* X. *Then,*

$$\tau(\mathcal{S}) := \left\{\cup_{U\in\mathcal{A}} U : \mathcal{A} \subset \left\{\cap_{j=1}^n S_j : n \in \mathbb{N}, S_1, \ldots, S_n \in \mathcal{S}\right\}\right\}$$

is a topology on X.

Proof via basis. For brevity, denote $\mathcal{I} := \left\{\cap_{j=1}^{n} S_j : n \in \mathbb{N}, S_1, \ldots, S_n \in \mathcal{S}\right\}$.

It suffices to show that $\mathcal{I}$ is a basis for X, since a preceding characterization lemma (1.2.3) shows that $\tau(\mathcal{S})$ is precisely the topology generated by $\mathcal{I}$ as a basis of X.

Since $\mathcal{S}$ covers X by definition, $\mathcal{I}$ also covers X for $\mathcal{S} \subset \mathcal{I}$. Fix $x \in X$ and suppose that $x \in I_1 \cap I_2$ for some $I_1, I_2 \in \mathcal{I}$.

Since the intersection of two finite intersections of sets in $\mathcal{S}$ remains a finite intersection of sets in $\mathcal{S}$, $I_1 \cap I_2 \in \mathcal{I}$. By considering $I_3 := I_1 \cap I_2$, we obtain that $x \in I_3 \subset I_1 \cap I_2$. By definition, $\mathcal{I}$ is a basis for X.

As a result, $\tau(\mathcal{I})$ is a topology of X and coincides with $\tau(\mathcal{S})$ by 1.2.3. Indeed, $\tau(\mathcal{S})$ is a topology on X, where $\tau(\mathcal{S})$ has the basis $\mathcal{I}$. □

Direct Proof. For brevity, denote $\mathcal{I} := \left\{\cap_{j=1}^{n} S_j : n \in \mathbb{N}, S_1, \ldots, S_n \in \mathcal{S}\right\}$.

By considering $\mathcal{A} := \emptyset$, we obtain that $\emptyset = \cup_{U \in \mathcal{A}} U \in \tau(\mathcal{S})$. Since $\mathcal{S} = \{S \cap S : S \in \mathcal{S}\}$, $\mathcal{S} \subset \mathcal{I}$. By considering $\mathcal{A} := \mathcal{S}$, we obtain that $X = \cup_{U \in \mathcal{A}} U \in \tau(\mathcal{S})$, by the definition of a subbasis.

[Closure under union] Fix a non-empty subcollection $\mathcal{F}$ of $\tau(\mathcal{S})$. Since every element F of $\mathcal{F}$ has the form $F = \cup_{U \in \mathcal{A}_F} U$ for some $\mathcal{A}_F \subset \mathcal{I}$, we may write

$$\bigcup_{F \in \mathcal{F}} F = \bigcup_{F \in \mathcal{F}} \left[\bigcup_{U \in \mathcal{A}_F} U \right] = \bigcup_{U \in \bigcup_{F \in \mathcal{F}} \mathcal{A}_F} U.$$

Since $\cup_{F \in \mathcal{F}} \mathcal{A}_F$ remains a subset of $\mathcal{I}$, we decomposed $\cup_{F \in \mathcal{F}} F$ into the union of a subcollection of $\mathcal{I}$. That is, $\cup_{F \in \mathcal{F}} F \in \tau(\mathcal{S})$ by construction. Indeed, $\tau(\mathcal{S})$ is closed under union.

[Closure under finite intersection] Fix $A = \cup_{U \in \mathcal{A}} U$ and $B = \cup_{W \in \mathcal{B}} W \in \tau(\mathcal{S})$ for some $\mathcal{A}, \mathcal{B} \subset \mathcal{I}$. We compute

$$\begin{aligned} A \cap B &= \cup_{U \in \mathcal{A}} [U \cap B] = \cup_{U \in \mathcal{A}} [\cup_{W \in \mathcal{B}} (U \cap W)] \\ &= \cup_{H \in \{U \cap W : U \in \mathcal{A}, W \in \mathcal{B}\}} H. \end{aligned}$$

Since the intersection of two finite intersections of sets in $\mathcal{S}$ remains a finite intersection of sets in $\mathcal{S}$, we obtain that $\{U \cap W : U \in \mathcal{A}, W \in \mathcal{B}\} \subset \mathcal{I}$. That is, $A \cap B$ is the union of a subcollection of $\mathcal{I}$ and, therefore, an element of $\tau(\mathcal{S})$ by construction. Indeed, $\tau(\mathcal{S})$ is closed under finite intersection.

By definition, $\tau(\mathcal{S})$ is a topology on X. □

Lemma 1.2.9 (Subbasis Induces Basis Generating the Same Topology)**.** *Suppose X is a set, $\mathcal{S}$ is a subbasis of X, and*

$$\mathcal{I}_{\mathcal{S}} := \left\{\cap_{j=1}^{n} S_j : n \in \mathbb{N}, S_1, \ldots, S_n \in \mathcal{S}\right\}.$$

Then, $\mathcal{I}_{\mathcal{S}}$ is a basis of $\tau(\mathcal{S})$. Particularly $\mathcal{S} \subset \mathcal{I}_{\mathcal{S}} \subset \tau(\mathcal{I}_{\mathcal{S}}) = \tau(\mathcal{S})$.

Proof. The proofs of the preceding lemma contained this result. □

Exercise 1.2.1 (Minimality of Topologies Generated by Basis and Subbasis)**.** *Suppose X is a set and $\mathcal{C}$ is a collection of subsets of X. If $\mathcal{C}$ is a basis or subbasis of X, then $\tau(\mathcal{C})$ is the intersection of all topologies of X that contains $\mathcal{C}$ and is the minimal topology on X containing $\mathcal{C}$.* ***[Munkres 13.5]***

Proof. Denote by $\mathcal{T}$ intersection of all topologies of X that contains $\mathcal{C}$ and suppose $\mathcal{C}$ is a basis or subbasis of X.

Since $\tau(\mathcal{C})$ is a topology of X that contains $\mathcal{C}$, $\mathcal{T} \subset \tau(\mathcal{C})$. It remains to show that $\tau(\mathcal{C}) \subset \mathcal{T}$.

[Case #1] Suppose $\mathcal{C}$ is a basis of X. Fix $U \in \tau(\mathcal{C})$. By 1.2.3, U may be decomposed into $U = \cup_{A \in \mathcal{A}} A$ for some $\mathcal{A} \subset \mathcal{C}$. For any topology τ of X containing $\mathcal{C}$, $\mathcal{A} \subset \tau$ implies that $U \in \tau$. That is, $U \in \mathcal{T}$ by the construction of $\mathcal{T}$. Indeed, $\tau(\mathcal{C}) \subset \mathcal{T}$.

[Case #2] Suppose $\mathcal{C}$ is a subbasis of X. We reduce this case to the first.

Let $\mathcal{I}_{\mathcal{C}}$ denote the collection consisting of all finite intersections of elements of $\mathcal{C}$. By 1.2.9, $\mathcal{I}_{\mathcal{C}}$ is a basis for X, and its generated topology $\tau(\mathcal{I}_{\mathcal{C}})$ on X coincides with $\tau(\mathcal{C})$.

By case #1, $\tau(\mathcal{C}) = \tau(\mathcal{I}_{\mathcal{C}}) \subset \mathcal{T}$, which proves this case.

Indeed, $\tau(\mathcal{C}) = \mathcal{T}$ and the minimality of $\tau(\mathcal{C})$ follows immediately. □

1.2.3 Topologies Generated by Subsets

Remark 1.2.1. *We generated topologies on a set via its bases and subbases. We now generate topologies on a set via arbitrary collections of its subsets.*

For basis and subbasis, the new notion of generated topology coincides with the old, which a previous exercise showed.

The exercise justifies exactly our languages and notations concerning topologies generated by bases and subbases of sets.

Definition 1.2.7 (Topology generated by sets)**.** *Suppose X is a set and $\mathcal{C}$ is collection of subsets of X. The* ***topology*** *$\tau(\mathcal{C})$* ***of*** *X* ***generated by*** *$\mathcal{C}$ is the intersection of all topologies of X that contain $\mathcal{C}$.*

Lemma 1.2.10 (Topology Generated by Sets)**.** *Suppose X is a set and $\mathcal{C}$ is collection of subsets of X. Then, $\tau(\mathcal{C})$ is a topology of X and is the minimal topology of X containing $\mathcal{C}$.*

Proof. Since all topologies of X contains X and $\emptyset$, $\tau(\mathcal{C})$ contains X and $\emptyset$.

Fix a non-empty subcollection $\mathcal{A} \subset \tau(\mathcal{C})$. For any topology τ of X containing $\mathcal{C}$, every $U \in \mathcal{A}$ is contained τ, so $\cup_{U \in \mathcal{A}} U \in \tau$ by the closure of τ under union. That is, $\cup_{U \in \mathcal{A}} U \in \tau(\mathcal{C})$, so $\tau(\mathcal{C})$ is closed under union.

Now, fix $U, W \in \tau(\mathcal{C})$. It follows that, for any topology τ of X containing $\mathcal{C}$, U and W live in τ, which ensures that $U \cap W \in \tau$ by the closure of τ under finite intersection. That is, $U \cap W \in \tau(\mathcal{C})$ by construction. Indeed, $\tau(\mathcal{C})$ is closed under finite intersection and, thus, is a topology of X.

Lastly, $\mathcal{C} \subset \tau(\mathcal{C})$ and the minimality of $\tau(\mathcal{C})$ follow from construction. □

1.3 Order Topology

1.3.1 Ordered Sets

Definition 1.3.1 (Relation)**.** *Suppose X is a set.*

(a) A relation on X is a subset $<$ of $X \times X$.

(b) Suppose $<$ is a relation on X. For any $x, y \in X$, we write

(b.1) $x < y$ if $(x, y) \in <$ and

(b.2) $x \not< y$ if $(x, y) \notin <$.

Definition 1.3.2 (Simply ordered set)**.** *(a) Suppose X is a set. A **(simple) order on** X is a relation $<$ on X satisfying*

(a.1) comparability ($\forall x, y \in X$, if $x \neq y$, then $x < y$ or $y < x$),

(a.2) non-reflexivity ($\forall x \in X$, $x \not< x$), and

(a.3) transitivity ($\forall x, y, z \in X$, if $x < y$ and $y < z$, then $x < z$).

*(b) A **(simply) ordered set** is an ordered pair $(X, <)$ consisting of a set X and a simple order $<$ on X;*

(c) Suppose $(X, <)$ is an ordered set. For any $x, y \in X$, we write $x > y$ if $y < x$.

Remark 1.3.1. *Suppose $(X, <)$ is an ordered set and $<$ is non-empty. Consider $(a, b) \in <$. The non-reflexivity of $<$ ensures that $(a, a) \notin <$. Particularly, $a \neq b$, so X contains at least 2 elements.*

*To avoid triviality and stating repeatedly the hypothesis that the underlying set of an ordered set has more than two elements, unless otherwise stated, **we assume that a simple order is non-empty.***

Definition 1.3.3 (Least and greatest element)**.** *Suppose $(X, <)$ is an ordered set.*

*(a) A **(minimal) least element of** $(X, <)$ is an element $m \in X$ satisfying $\forall x \in X / \{m\}$, $m < x$.*

*(b) A **(maximal) greatest element of** $(X, <)$ is an element $M \in X$ satisfying $\forall x \in X / \{M\}$, $x < M$.*

Lemma 1.3.1 (Uniqueness of Least and Greatest Elements)**.** *Suppose $(X, <)$ is an ordered set.*

(a) If $m, m' \in X$ is are least elements of $(X, <)$, then $m = m'$.

(b) If $M, M' \in X$ is are greatest elements of $(X, <)$, then $M = M'$

Proof. (a) Suppose $m, m' \in X$ is are least elements of $(X, <)$. Suppose, to the contrary, that $m \neq m'$. By the definition of least element, $m < m'$ and $m' < m$. It follows from the transitivity of $<$ that $m < m$. However, the non-reflexitivity ensures that $m \not< m$, which is a contradiction. We conclude that $m = m'$.

(b) This implication follows from a similar, if not identical, argument presented in part (a). □

Lemma 1.3.2. *Suppose $(X, <)$ is an ordered set and $x \in X$.*

(a) If x is not a minimal element of $(X, <)$, then there exists $a_x \in X$ with $a_x < x$.

(b) If x is not a maximal element of $(X, <)$, then there exists $b_x \in X$ with $x < b_x$.

Proof. We show (a) only, since (b) follows from a symmetric argument.

(a) Suppose x is not a minimal element of $(X, <)$.

Suppose, to the contrary, that $\forall a \in X$, $a \not< x$. By the comparability of $<$, for any $a \in X / \{x\}$, $x < a$. That is, x is a minimal element of $(X, <)$ by definition, which is a contradiction. □

1.3.2 Order Topology

Definition 1.3.4 (Intervals)**.** *Suppose $(X, <)$ is an ordered set. Then,*

*(a) an **open interval in** $(X, <)$ is a subset*

$$(a, b) := \{x \in X : a < x < b\}$$

for some $a, b \in X$ and $a < b$;

*(b) a **closed interval in** $(X, <)$ is a subset*

$$[a, b] := \{a\} \cup (a, b) \cup \{b\}.$$

for some $a, b \in X$ and $a < b$;

*(c) a **half-open interval in** $(X, <)$ is a subset*

$$(a, b] := (a, b) \cup \{b\} \text{ or } [a, b) := \{a\} \cup (a, b)$$

for some $a, b \in X$ and $a < b$;

*(d) an **interval in** $(X, <)$ is a subset of X that is an open interval, closed interval, or a half-open interval;*

*(d.1) the **start (end) point of an interval** (a, b) **(**$[a, b]$**,** $[a, b)$**,** $(a, b]$**) in** $(X, <)$ is a (b).*

Definition 1.3.5 (Order topology)**.** *Suppose $(X, <)$ is an ordered set. Then, the **order topology** $\tau_<$ **of** $(X, <)$ is the topology on X generated by the basis $\mathcal{B}_<$ consisting of the collections*

(a) $\{(a, b) : a, b \in X$ and $a < b\}$,

(b) $\{[a_0, b) : b \in X$ and $a_0 < b\}$, if a_0 is a least element of $(X, <)$, and

(c) $\{(a, b_0] : a \in X$ and $a < b_0\}$, if b_0 is a greatest element of $(X, <)$.

Lemma 1.3.3 (Order Topology)**.** *Suppose $(X, <)$ is an ordered set. Then, the order topology on (X, τ) is a topology on (X, τ).*

Proof. It suffices to show that the corresponding basis $\mathcal{B}_<$ is actually a basis for X.

Recall that we assumed that all simple orders are non-empty, so X contains at least 2 distinct elements a, b. By the comparability of $<$, $a < b$ or $b < a$. Without loss of generality, we suppose $a < b$.

Fix $x \in X$. If x is neither a minimal element nor a maximal element of $(X, <)$, then a preceding lemma ensures there exist $a_x, b_x \in X$ such that $x \in (a_x, b_x)$. If x is a minimal (maximal) element of $(X, <)$, then x lives in $[x, b)$ $((a, x])$ by the comparability of $<$. Particularly, $\mathcal{B}_<$ covers X.

Suppose now $x \in B_1 \cap B_2$ for some $B_1, B_2 \in \mathcal{B}_<$. Denote by a_1 and a_2 the start points of B_1 and B_2, respectively. Similarly, denote the end points of B_1 and B_2 by b_1 and b_2, respectively.

If x is neither a minimal element nor a maximal element of $(X, <)$, then the interval $B_3 := (\max\{a_1, a_2\}, \min\{b_1, b_2\}) \in \mathcal{B}_<$ satisfies $x \in B_3 \subset B_1 \cap B_2$.

> Suppose x is neither a minimal element nor a maximal element of $(X, <)$. Since $x \in B_1 \cap B_2$, the assumption ensures that $x \notin \{a_1, a_2, b_1, b_2\}$ and, thus, $x \in (a_1, b_1) \cap (a_2, b_2)$.
>
> To see this, we suppose, to the contrary, that $x \in \{a_1, a_2, b_1, b_2\}$.
>
> [Case #1] Suppose $x = a_j$ for some $j \in \{1, 2\}$. That is, x is the start point of B_j. Particularly, $x \in B_j$, so B_j is an interval in $\mathcal{B}_<$ with endpoint b_j that contains its start point.
>
> It follows that $B_j = [A_j, b_j)$ for some minimal element A_j of $(X, <)$, since, by the uniqueness of minimal element, $[A_j, b_j)$ is the only interval in $\mathcal{B}_<$ with end point b_j that contains its start point.
>
> Since x is the start point of B_j and, thus, $[A_j, b_j)$, $x = A_j$ so that x is a minimal element of $(X, <)$, which contradicts the assumption. We conclude $x \neq a_1, a_2$.
>
> [Case #2] Suppose $x \in b_j$ for some $j \in \{1, 2\}$, so that x is the end point of B_j. Via an argument symmetric to that presented in Case #1, we conclude that $x \neq b_1, b_2$.
>
> As a result, $x \in (a_1, b_1) \cap (a_2, b_2) = (\max\{a_1, a_2\}, \min\{b_1, b_2\}) =: B_3$, where
>
> $$(a_1, b_1) \subset B_1 \text{ and } (a_2, b_2) \subset B_2 \implies B_3 = (a_1, b_1) \cap (a_2, b_2) \subset B_1 \cap B_2$$
>
> as desired.

If x is a minimal element of $(X, <)$, then $B_3 := [x, \min\{b_1, b_2\}) \in \mathcal{B}_<$ satisfies

$$x \in B_3 \subset B_1 \cap B_2.$$

> Suppose x is a minimal element of $(X, <)$.
>
> Since B_1 and B_2 are intervals in $\mathcal{B}_<$ containing a minimal element x, and no intervals of the forms (a, b) and $(a, b]$ for $a, b \in X$ with $a < b$ contain a minimal element of $(X, <)$, we must have $B_1 = [a_1, b_1)$, $B_2 = [a_2, b_2)$ with a_1, a_2 being minimal elements.
>
> By the uniqueness of minimal elements, $x = a_1, a_2$. It follows that $x \in B_3 \subset B_1 \cap B_2$.

Symmetrically, if x is a maximal element of $(X, <)$, then $B_3 := (\max\{a_1, a_2\}, x] \in \mathcal{B}_<$ satisfies $x \in B_3 \subset B_1 \cap B_2$.

By definition, $\mathcal{B}_<$ is a basis for X. □

Lemma 1.3.4 (Lexicographical ordering)**.** *Suppose* $(X_1, <_1)$ *and* $(X_2, <_2)$ *are ordered sets.*

Then, the relation $<$ on $X_1 \times X_2$ defined by

$$\forall (a,b),(c,d) \in X_1 \times X_2,\ (a,b) < (c,d) \iff \begin{cases} a < b & \text{if } a \neq c \\ b < d & \text{if } a = c \end{cases}$$

is a simple order on $X_1 \times X_2$.

Definition 1.3.6 (Lexicographical ordering and topology)**.** *Suppose $(X_1, <_1)$ and $(X_2, <_2)$ are ordered sets.*

*(a) The **lexicographical (dictionary) order of** $X_1 \times X_2$ **(induced by** $<_1$ **and** $<_2$**)** is the simple order $<$ on $X_1 \times X_2$ defined by*

$$\forall (a,b),(c,d) \in X_1 \times X_2,\ (a,b) < (c,d) \iff \begin{cases} a < c & \text{if } a \neq c \\ b < d & \text{if } a = c \end{cases}$$

*(b) The **lexicographical (dictionary) order topology of** $X_1 \times X_2$ is the order topology on $X_1 \times X_2$ induced by the lexicographical order on $X_1 \times X_2$.*

Example 1.3.5 (Lexicographical ordering on $\mathbb{R}^2$)**.** *Consider the lexicographical order $<$ on $\mathbb{R}^2$ induced by the usual ordering $<_{\mathbb{R}}$ of $\mathbb{R}$. Since $(\mathbb{R}, <_{\mathbb{R}})$ has no minimal nor maximal elements, $(\mathbb{R}^2, <)$ contains no minimal nor maximal elements.*

Thus, every basis element of the standard basis for the lexicographical order topology on $\mathbb{R}^2$ is an open interval $((a,b),(c,d))$, with $(a,b) < (c,d)$, and has the form

(a) $((a,b),(c,d)) = \{a\} \times (b,d)$ for $a = c$, which is the straight line in $\mathbb{R}^2$ parallel to the y-axis connecting, but excluding, the points (a,b) and (c,d), and

(b) $((a,b),(c,d)) = [\{a\} \times (b,\infty)] \cup \left[\cup_{x \in (a,c)} (\{x\} \times \mathbb{R})\right] \cup [\{b\} \times (-\infty, d)]$ for $a < c$.

Definition 1.3.7 (Rays)**.** *Suppose $(X, <)$ is an ordered set.*

(a) An open ray in $(X, <)$ is a subset $(a, \infty) := \{x \in X : a < x\}$ or $(-\infty, b) := \{x \in X : x < b\}$ for some $a, b \in X$.

(b) A closed ray in $(X, <)$ is a subset $[a, \infty) := \{a\} \cup (a, \infty)$ or $(-\infty, b] := (-\infty, b) \cup \{b\}$ for some $a, b \in X$.

Lemma 1.3.6 (Open Rays are Open)**.** *Suppose $(X, <)$ is an ordered set. Then,*

(a) every open ray in $(X, <)$ is open in the order topology of $(X, <)$, and

(b) every closed ray in $(X, <)$ is closed in the order topology of $(X, <)$.

Proof. (a) Fix $a \in X$ and consider the open ray (a, ∞).

[Case #1] Suppose $(X, <)$ contains a maximal element $M \in X$. If $a = M$, then $(a, \infty) := \{x \in X : x > a\} = \emptyset$, for otherwise $x \in (a, \infty)$ implies $a < x < a$, which is a contradiction. If $a = M$, then (a, ∞) remains open.

If $a \neq M$, then for any $x \in X$,

$$\begin{aligned} x \in (a, \infty) &\iff a < x \\ &\iff a < x, \text{ where } x < M \text{ or } x = M \\ &\iff x \in (a, M], \end{aligned}$$

so that $(a, \infty) = (a, M]$, which is an element of the usual basis for the order topology and, thus, an open set.

[Case #2] Suppose $(X, <)$ does not contain a maximal element. For any $x \in (a, \infty)$, there exists $b_x \in X$ such that $x < b_x$ and, thus, $x \in (a, b_x) \subset (a, \infty)$. Particularly,

$$(a, \infty) = \cup_{x \in (a,\infty)} (a, b_x),$$

which is a union of some elements of the usual basis for $\tau_<$ and, therefore, open.

Indeed, (a, ∞) is open. The openness of open rays of the form $(-\infty, b)$ for $b \in X$ follows from an argument symmetric to that presented above.

(b) Fix $a \in X$ and consider the closed ray $(-\infty, a]$. For any $x \in X$,

$$x \in X/(-\infty, a] \iff x \neq a \text{ and } x \not< a \iff x > a \iff x \in (a, \infty).$$

Particularly, $X/(-\infty, a] = (a, \infty)$, which is open by part (a). By definition, $(-\infty, a]$ is closed.

The closedness of closed rays of the form $[b, \infty)$ for $b \in X$ follows from an argument symmetric to that presented above. □

Lemma 1.3.7 (Recovering Standard Basis Elements from Open Rays)**.** *Suppose $(X, <)$ is an ordered set and $a, b \in X$.*

(a) If $a < b$, then $(a, b) = (a, \infty) \cap (-\infty, b)$.

(b) If $m \in X$ is a minimal element of $(X, <)$, then $[m, b) = (-\infty, b)$.

(c) If $M \in X$ is a maximal element of $(X, <)$, then $(a, M] = (a, \infty)$.

Proof. (a) Suppose $a < b$. By definition, for any $x \in X$,

$$x \in (a, b) \iff x > a \text{ and } x < b \iff x \in (a, \infty) \text{ and } x \in (-\infty, b).$$

Indeed, $(a, b) = (a, \infty) \cap (-\infty, b)$.

(b) Suppose $m \in X$ is a minimal element of $(X, <)$. For any $x \in X$,

$$x \in (-\infty, b)/\{m\} \iff x \neq m \text{ and } x < b \iff m < x \text{ and } x < b \iff x \in (m, b)$$

so that $(-\infty, b)/\{m\} = (m, b)$. By adjoining m to the set, we conclude that $(-\infty, b) = [m, b)$.

(c) Suppose $M \in X$ is a maximal element of $(X, <)$. This implication follows from an argument symmetric to that presented in part (b). □

Lemma 1.3.8 (Subbasis of Open Rays Generates Order Topology)**.** *Suppose $(X, <)$ is an ordered set. Then,*

$$\mathcal{S}_< := \{(-\infty, b) : b \in X\} \cup \{(a, \infty) : a \in X\}$$

is a subbasis for the order topology of $(X, <)$.

Proof. Recall that we assumed all simple orders are non-empty, so X contains at least 2 distinct points a, b, with which we may assume without loss of generality that $a < b$.

Fix $x \in X$. If x is not a maximal (minimal) element of $(X, <)$, then there exists $b_x \in X$ ($a_x \in X$) such that $x < b_x$ ($a_x < x$) and, thus, x lives in $(-\infty, b_x) \in \mathcal{S}_<$ ($(a_x, \infty) \in \mathcal{S}_<$). That is, $\mathcal{S}_<$ covers X and is a subbasis for X by definition.

The above encompasses the cases in which x is maximal or minimal in $(X, <)$, since x cannot be both maximal and minimal, given X has 2 distinct elements.

Suppose, to the contrary, that x is both a maximal and minimal element of $(X, <)$. We observe that x is distinct from a, for otherwise

$$x = a \implies x < b \implies x \neq b \implies x > b, \qquad \text{(Since } x \text{ is maximal)}$$

which implies via the transitivity of $<$ that $x < x$ and contradicts the non-reflexivity of $<$. However, the maximality and minimality of x ensure that $x < a < x$, which again contradicts the non-reflexivity of $<$ by transitivity. Indeed, x cannot be both maximal and minimal.

Recall that $\mathcal{S}_<$ generates a basis $\mathcal{I}_\mathcal{S} := \{\cap_{S\in\mathcal{A}} S : \mathcal{A} \subset \mathcal{S}_< \text{ is finite}\}$ on X, where the topology $\tau(\mathcal{S}_<)$ generated by $\mathcal{S}_<$ coincides with the topology $\tau(\mathcal{I}_\mathcal{S})$ generated by $\mathcal{I}_\mathcal{S}$ as a basis of X.

To show that $\mathcal{S}_<$ is a subbasis of for the order topology $\tau_<$, it remains to show that $\tau(\mathcal{I}_\mathcal{S}) = \tau_<$, which we do by proving that $\mathcal{I}_\mathcal{S}$ is a basis for $\tau_<$ via 1.2.4.

By a preceding lemma, all open rays are $\tau_<$-open, so $\mathcal{S}_< \subset \tau_<$. By the closure of $\tau_<$ under finite intersections, $\mathcal{I}_\mathcal{S} \subset \tau_<$. Lastly, fix a $\tau_<$-open set U and decompose $U = \cup_{B\in\mathcal{A}} B$ via some subcollection $\mathcal{A}$ of the standard basis for the $\tau_<$.

Via a preceding lemma, any standard basis element of $\tau_<$ coincides with a finite intersection of open rays. Particularly, U is the union of basis elements of $\mathcal{I}_\mathcal{S}$. By 1.2.4, $\mathcal{I}_\mathcal{S}$ is a basis for $\tau_<$. That is, $\tau_< = \tau(\mathcal{I}_\mathcal{S})$, so $\mathcal{S}_<$ is a subbasis for the order topology $\tau_<$ as desired. □

1.4 Product Topology

Lemma 1.4.1 (Compatibility of Intersections and Products)**.** *Let A, B, C, D be sets. Then,*

$$[A \times B] \cap [C \times D] = [A \cap C] \times [B \cap D].$$

Proof. For brevity, denote by L and R the sets $[A\times B]\cap[C\times D]$ and $[A\cap C]\times[B\cap D]$, respectively.

[$L \subset R$] For any $(x, y) \in L$, $x \in A \cap C$ and $y \in B \cap D$ ensures that $(x, y) \in R$ by definition. That is, $L \subset R$.

[$R \subset L$] For any $(x, y) \in R$,

$$\begin{aligned} x \in A \text{ and } y \in B &\implies (x, y) \in A \times B, \text{ where} \\ x \in C \text{ and } y \in D &\implies (x, y) \in C \times D, \end{aligned}$$

which ensure that $(x, y) \in L$. That is, $R \subset L$.

Indeed, the two sets coincide. □

Lemma 1.4.2 (Basis for Product Topology)**.** *Suppose (X_1, τ_1) and (X_2, τ_2) are topological spaces. Then,*

$$\mathcal{B}_{X_1\times X_2} := \{U_1 \times U_2 : U_1 \in \tau_1, U_2 \in \tau_2\}$$

is a basis for $X_1 \times X_2$.

Proof. For brevity, denote $\mathcal{B}_{X_1\times X_2}$ by $\mathcal{B}$.

Since $X_1 \in \tau_1$ and $X_2 \in \tau_2$, $X_1 \times X_2 \in \mathcal{B}$. Particularly, $\mathcal{B}$ covers $X_1 \times X_2$.

Fix $(x_1, x_2) \in X_1 \times X_2$ and suppose that $(x_1, x_2) \in B_1 \cap B_2$ for some $B_1, B_2 \in \mathcal{B}$. For any $k \in \{1, 2\}$, $B_k = U_k \times V_k$ for some $U_k \in \tau_1, V_k \in \tau_2$, by the construction of $\mathcal{B}$. Via a preceding lemma, we compute

$$B_1 \cap B_2 = [U_1 \times V_1] \cap [U_2 \times V_2] = \underbrace{[U_1 \cap U_2]}_{\in\tau_1} \times \underbrace{[V_1 \cap V_2]}_{\in\tau_2},$$

which remains an element of $\mathcal{B}$. By considering $B_3 := B_1 \cap B_2$, we obtain that $(x_1, x_2) \in B_3 \subset B_1 \cap B_2$ and $B_3 \in \mathcal{B}$. By definition, $\mathcal{B}$ is a basis for $X_1 \times X_2$. □

Definition 1.4.1 (Product topology, product space)**.** *Suppose (X_1, τ_1) and (X_2, τ_2) are topological spaces. The **product topology** $\mathcal{P}_{X_1\times X_2}$ **on** $X_1 \times X_2$ is the topology on $X_1 \times X_2$ generated by the basis*

$$\mathcal{B}_{X_1\times X_2} := \{U_1 \times U_2 : U_1 \in \tau_1, U_2 \in \tau_2\}.$$

Unless otherwise specified, we topologize every Cartesian product of underlying sets of topological spaces by the product topology the Cartesian product inherits.

Definition 1.4.2 (Standard topology on $\mathbb{R}^2$)**.** *The standard topology on $\mathbb{R}^2$ is the product topology on $\mathbb{R}^2$ generated the basis*

$$\mathcal{B}_{\mathbb{R}^2} := \{U_1 \times U_2 : U_1, U_2 \text{ are open subsets of } \mathbb{R} \text{ in the standard topology of } \mathbb{R}\}.$$

Theorem 1.4.3 (Canonically Inherited Basis for Product Topology)**.** *Suppose (X_1, τ_1) and (X_2, τ_2) are topological spaces, and $\mathcal{B}_1$ and $\mathcal{B}_2$ are bases for τ_1 and τ_2, respectively. Then,*

$$\mathcal{B}_1 \times \mathcal{B}_2 := \{B_1 \times B_2 : B_1 \in \mathcal{B}_1, B_2 \in \mathcal{B}_2\}$$

is a basis for the product topology of $X_1 \times X_2$.

Proof. We make use of 1.2.4.

Since $\mathcal{B}_1 \subset \tau_1$ and $\mathcal{B}_2 \subset \tau_2$, $\mathcal{B}_1 \times \mathcal{B}_2$ is a subset of the usual basis for the product topology $\mathcal{P}_{X_1 \times X_2}$. Particularly, $\mathcal{B}_1 \times \mathcal{B}_2 \subset \mathcal{P}_{X_1 \times X_2}$.

Fix an open set $U \in \mathcal{P}_{X_1 \times X_2}$ and decompose $U = \cup_{V_1 \times V_2 \in \mathcal{A}}[V_1 \times V_2]$ via a subcollection $\mathcal{A}$ of the usual basis of $\mathcal{P}_{X_1 \times X_2}$. For any $V_1 \times V_2 \in \mathcal{A}$, $V_1 \in \tau_1$ and $V_2 \in \tau_2$, so we may decompose

$$V_1 = \cup_{W_1 \in \mathcal{A}_1} W_1 \text{ and } V_2 = \cup_{W_2 \in \mathcal{A}_2} W_2$$

via some subcollections $\mathcal{A}_1 \subset \mathcal{B}_1$ and $\mathcal{A}_2 \subset \mathcal{B}_2$ of the given bases.

Fix $(x_1, x_2) \in U$. It follows that $(x_1, x_2) \in V_1 \times V_2$ for some $V_1 \times V_2 \in \mathcal{A}$, where $x_1 \in W_1$ and $x_2 \in W_2$ for some $W_1 \in \mathcal{A}_1$ and $\mathcal{W}_2 \in \mathcal{A}_2$ such that $W_1 \subset V_1$ and $W_2 \subset V_2$.

Particularly, $(x_1, x_2) \in W_1 \times W_2 \subset V_1 \times V_2$ with $W_1 \times W_2$ being an element of $\mathcal{B}_1 \times \mathcal{B}_2$ by construction.

The hypotheses of 1.2.4 are satisfied. We conclude that $\mathcal{B}_1 \times \mathcal{B}_2$ is a basis for the product topology. □

Definition 1.4.3 (Projection in Cartesian product)**.** *Suppose X_1, X_2 are sets. For $k \in \{1, 2\}$, the **projection of** $X_1 \times X_2$ **into the** k**-th factor** is the map $\pi_k \colon X_1 \times X_2 \to X_k$ defined by*

$$\forall (x_1, x_2) \in X_1 \times X_2,\ \pi_j(x_1, x_2) := x_k.$$

Lemma 1.4.4 (Images and Preimage of Projections)**.** *Suppose X_1 and X_2 are sets, $V_1 \subset X_1$, and $V_2 \subset X_2$.*

(a) Let $j \in \{1, 2\}$. Then, $\pi_j(V_1 \times V_2) \subset V_j$.

(a.1) If $i \in \{1, 2\} \,/\, \{j\}$ and $V_i \neq \emptyset$, then $\pi_j(V_1 \times V_2) = V_j$.

(b) ${\pi_1}^{-1}(V_1) = V_1 \times X_2$ and ${\pi_2}^{-1}(V_2) = X_1 \times V_2$.

(b.1) Particularly, $V_1 \times V_2 = {\pi_1}^{-1}(V_1) \cap {\pi_2}^{-1}(V_2)$.

Proof. (a) If $V_1 \times V_2 = \emptyset$, then $\pi_j(V_1 \times V_2) = \emptyset \subset V_j$. Hence, we may suppose that $V_1 \times V_2$ is non-empty. For any $(v_1, v_2) \in V_1 \times V_2$, $\pi_j(v_1, v_2) := v_j \in V_j$. That is, $\pi_j(V_1 \times V_2) \subset V_j$ as desired.

(a.1) Suppose $i \in \{1, 2\} \,/\, \{j\}$ and $V_i \neq \emptyset$. By part (a), it remains to show that $\pi_j(V_1 \times V_2) \supset V_j$. If $V_j = \emptyset$, then the containment is trivially true. Thus, we may suppose V_j is non-empty, so that $V_1 \times V_2$ is non-empty as well.

Let $x_i \in V_i$. For any $x_j \in V_j$, $x_j =: \pi_j(x_1, x_2) \in \pi_j(V_1 \times V_2)$. Indeed, $\pi_j(V_1 \times V_2) \supset V_j$.

(b) For any $(x_1, x_2) \in X_1 \times X_2$,

$$\begin{aligned}(x_1, x_2) \in {\pi_1}^{-1}(V_1) &\iff \pi_1(x_1, x_2) \in V_1 \\ &\iff x_1 \in V_1 \text{ and } x_2 \in X_2.\end{aligned}$$

Indeed, ${\pi_1}^{-1}(V_1) = V_1 \times X_2$, where ${\pi_2}^{-1}(V_2) = X_1 \times V_2$ holds via a similar, if not identical, argument.

(b.1) From part (b), we conclude that

$$V_1 \times V_2 = (V_1 \times X_2) \cap (X_1 \times V_2) = {\pi_1}^{-1}(V_1) \cap {\pi_2}^{-1}(V_2).$$

□

Corollary 1.4.5 (Continuity of Projections)**.** *Suppose* (X_1, τ_1)*,* (X_2, τ_2) *are topological spaces. For any* $j \in \{1, 2\}$*, for any open set* $U \in \tau_j$*,* ${\pi_j}^{-1}(U)$ *is open in the product topology.*

Proof. Let $j = 1$ and fix an open set $U \in \tau_1$. By a preceding lemma, ${\pi_j}^{-1}(U) = U \times X_2$, which is an element of the usual basis for the product topology. Particularly, ${\pi_j}^{-1}(U)$ is open in the product topology.

The case for $j = 2$ follows from a similar, if not identical, argument. □

Theorem 1.4.6 (Subbasis for Product Topology via Projections)**.** *Suppose* (X_1, τ_1) *and* (X_2, τ_2) *are topological spaces. Then,*

$$\mathcal{S}_{X_1 \times X_2} := \cup_{j=1}^{2} \left\{ {\pi_j}^{-1}(U_j) : U_j \in \tau_j \right\}$$

is a subbasis of the product topology on $X_1 \times X_2$*.*

Proof. For brevity, denote $\mathcal{S}_{X_1 \times X_2}$ and the usual basis for the product topology on $X_1 \times X_2$ by $\mathcal{S}$ and $\mathcal{B}$, respectively.

Since $X_1 \in \tau_1$ and $\pi_1^{-1}(X_1) = X_1 \times X_2 \in \mathcal{S}$, $\mathcal{S}$ covers $X_1 \times X_2$. By definition, $\mathcal{S}$ is a subbasis for $X_1 \times X_2$. By 1.2.9, $\mathcal{S}$ generates a basis $\mathcal{I}$ of $X_1 \times X_2$ consisting of all finite intersections of elements of $\mathcal{S}$, where the topology $\tau(\mathcal{S})$ of $X_1 \times X_2$ generated by $\mathcal{S}$ coincides with the topology $\tau(\mathcal{I})$ of $X_1 \times X_2$ generated by $\mathcal{I}$ as a basis.

To show that $\mathcal{S}$ is a subbasis for the product topology on $X_1 \times X_2$, it remains to show that the product topology coincide with $\tau(\mathcal{S})$. Since $\tau(\mathcal{S}) = \tau(\mathcal{I})$, it suffices to show that $\mathcal{B}$ coincides with $\mathcal{I}$, for it follows that the product topology is generated by $\mathcal{I}$ and, thus, coincides with $\tau(\mathcal{S})$.

[$\mathcal{B} \subset \mathcal{I}$] For any basis element $B \in \mathcal{B}$, $B = U \times V$ for some $U \in \tau_1$ and $V \in \tau_2$, where

$$B = U \times V = [U \times X_2] \cap [X_1 \times V] = \underbrace{{\pi_1}^{-1}(U)}_{\in \mathcal{S}} \cap \underbrace{{\pi_2}^{-1}(V)}_{\in \mathcal{S}} \in \mathcal{I}.$$

That is, $\mathcal{B} \subset \mathcal{I}$.

[$\mathcal{B} \supset \mathcal{I}$] Fix a basis element $B \in \mathcal{I}$. By the construction of $\mathcal{I}$, $B = \cap_{S \in \mathcal{A}} S$ for some finite subcolleciton $\mathcal{A} \subset \mathcal{S}$. For any $S \in \mathcal{A}$, $S = {\pi_{j_S}}^{-1}(U_S^{j_S})$ for some $j_S \in \{1, 2\}$ and $U_S^{j_S} \in \tau_{j_S}$ by the construction of $\mathcal{S}$. It follows that

$$\begin{aligned} B &= \left[\bigcap_{\substack{S \in \mathcal{A} \\ j_S = 1}} {\pi_1}^{-1}(U_S^1) \right] \cap \left[\bigcap_{\substack{S \in \mathcal{A} \\ j_S = 2}} {\pi_2}^{-1}(U_S^2) \right] = {\pi_1}^{-1} \left(\bigcap_{\substack{S \in \mathcal{A} \\ j_S = 1}} U_S^1 \right) \cap {\pi_2}^{-1} \left(\bigcap_{\substack{S \in \mathcal{A} \\ j_S = 2}} U_S^2 \right) \\ &= \left[\left(\bigcap_{\substack{S \in \mathcal{A} \\ j_S = 1}} U_S^1 \right) \times X_2 \right] \cap \left[X_1 \times \left(\bigcap_{\substack{S \in \mathcal{A} \\ j_S = 2}} U_S^2 \right) \right] = \underbrace{\left(\bigcap_{\substack{S \in \mathcal{A} \\ j_S = 1}} U_S^1 \right)}_{\in \tau_1} \times \underbrace{\left(\bigcap_{\substack{S \in \mathcal{A} \\ j_S = 2}} U_S^2 \right)}_{\in \tau_2}. \end{aligned}$$

It follows that $B \in \mathcal{B}$. That is, $\mathcal{I} \subset \mathcal{B}$.

Indeed, the bases $\mathcal{I}$ and $\mathcal{B}$ coincide. We conclude that $\mathcal{S}$ is a subbasis of the product topology. □

1.5 Subspace Topology

Definition 1.5.1 (Subspace topology)**.** *Suppose (X, τ) is a topological space and $Y \subset X$. Then, the **subspace topology on** Y **(from** (X, τ)**)** is the set $\tau_Y := \{U \cap Y : U \in \tau\}$.*

Unless otherwise specified, we topologize every subset of the underlying set of a topological space by the subspace topology the subset inherits.

Lemma 1.5.1 (Subspace Topology)**.** *Suppose (X, τ) is a topological space and $Y \subset X$. Then, $\tau_Y := \{U \cap Y : U \in \tau\}$ is a topology of Y.*

Proof. Since $X, \emptyset \in \tau$ by the definition of a topology, $Y = X \cap Y, \emptyset = \emptyset \cap Y \in \tau_Y$.

Fix a subcollection $\mathcal{A} \subset \tau_Y$. By definition, every set $V \in \mathcal{A}$ has the form $V = U_V \cap Y$ for some $U_V \in \tau$. It follows that

$$\cup_{V \in \mathcal{A}} V = \cup_{V \in \mathcal{A}} [U_V \cap Y] = \underbrace{[\cup_{V \in \mathcal{A}} U_V]}_{\in \tau} \cap Y \in \tau_Y.$$

Now, fix $n \in \mathbb{N}$ and $V_1, \ldots, V_n \in \tau_Y$. Again, for $j \in \{1, \ldots, n\}$, $V_j = U_j \cap Y$ for some $U_j \in \tau$ by the definition of τ_Y. We observe that

$$\cap_{j=1}^n V_j = \cap_{j=1}^n [U_j \cap Y] = \underbrace{[\cap_{j=1}^n U_j]}_{\in \tau} \cap Y \in \tau_Y.$$

Thus, τ_Y is closed under unions and finite intersections. By definition, τ_Y is a topology on Y. □

Lemma 1.5.2 (Canonically Induced Basis for Subspace Topology)**.** *Suppose (X, τ) is a topological space with a basis $\mathcal{B}$ of τ, and $Y \subset X$. Then, $\mathcal{B}_Y := \{B \cap Y : B \in \mathcal{B}\}$ is a basis for τ_Y.*

Proof. We make use of 1.2.4. For any $B \cap Y \in \mathcal{B}_Y$, $B \in \mathcal{B}$ ensures that it is τ-open ensures and, thus, $B \cap Y \in \tau_Y$ by definition. That is, $\mathcal{B}_Y \subset \tau_Y$.

Fix an open set $V \cap Y \in \tau_Y$ for some $V \in \tau$ and fix $y \in V \cap Y$. Since $y \in V$ and τ has the basis $\mathcal{B}$, there exists a basis element $B \in \mathcal{B}$ such that $y \in B \subset V$. Particularly, $y \in B \cap Y \subset V \cap Y$ with $B \cap Y \in \mathcal{B}_Y$.

The hypotheses of 1.2.4 are satisfied. Indeed, $\mathcal{B}_Y$ is a basis for τ_Y. □

Lemma 1.5.3 (Compatibility of Open Subspace)**.** *Suppose (X, τ) is a topological space and $Y \subset X$ is τ-open. Then, every τ_Y-open subset of Y is τ-open.*

Proof. Fix a τ_Y-open subset $U \subset Y$. By definition, $U = V \cap Y$ for some τ-open subset $V \subset X$. Since Y is τ-open by assumption, U remains τ-open by the closure of τ under finite intersections. □

Theorem 1.5.4 (Compatibility of Product Topology and Subspace Topology)**.** *Suppose (X_1, τ_1) and (X_2, τ_2) are topological spaces with $Y_1 \subset X_1$ and $Y_2 \subset X_2$. Then, the topologies $\mathcal{P}_{Y_1 \times Y_2}$ and $\tau_{Y_1 \times Y_2}$ on $Y_1 \times Y_2$ coincide, where*

(a) $\mathcal{P}_{Y_1 \times Y_2}$ is the product topology on $Y_1 \times Y_2$ induced from the subspace topologies τ_{Y_1}, τ_{Y_2} of Y_1,Y_2, inherited from τ_1 and τ_2, and

(b) $\tau_{Y_1 \times Y_2}$ is the subspace topology on $Y_1 \times Y_2$ inherited from the product topology on $X_1 \times X_2$, induced from τ_1 and τ_2.

Proof. To show the two topologies on $Y_1 \times Y_2$ coincide, it suffices to show they share a basis.

By definition, $\mathcal{P}_{Y_1 \times Y_2}$ has the standard basis

$$\mathcal{B}' := \{V_1 \times V_2 : V_1 \in \tau_{Y_1}, V_2 \in \tau_{Y_2}\},$$

whereas the product topology on $X_1 \times X_2$ has the standard basis

$$\mathcal{B}_0 := \{U_1 \times U_2 : U_1 \in \tau_1, U_2 \in \tau_2\},$$

which canonically induces the basis

$$\mathcal{B} := \{B \cap [Y_1 \times Y_2] : B \in \mathcal{B}_0\}.$$

for the subspace topology $\tau_{Y_1 \times Y_2}$. It remains to show that $\mathcal{B}' = \mathcal{B}$.

[$\mathcal{B}' \subset \mathcal{B}$] Fix $V_1 \times V_2 \in \mathcal{B}'$. For $j \in \{1, 2\}$, $V_j \in \tau_{Y_j}$ ensures $V_j = U_j \cap Y_j$ for some $U_j \in \tau_j$ by the definition of subspace topology. Particularly,

$$V_1 \times V_2 = [U_1 \cap Y_1] \times [U_2 \cap Y_2] = \underbrace{[U_1 \times U_2]}_{\in \mathcal{B}_0} \cap [Y_1 \times Y_2] \in \mathcal{B}.$$

[$\mathcal{B} \subset \mathcal{B}'$] Fix $B \cap [Y_1 \times Y_2]$ for some $B = U_1 \times U_2 \in \mathcal{B}_0$. Via an identical set computation,

$$B \cap [Y_1 \times Y_2] = \underbrace{[U_1 \cap Y_1]}_{\in \tau_{Y_1}} \times \underbrace{[U_2 \cap Y_2]}_{\in \tau_{Y_1}} \in \mathcal{B}'. \qquad \text{(Since } U_1 \in \tau_1, U_2 \in \tau_2\text{)}$$

Indeed, the two bases coincide, which ensures the given topologies $\mathcal{P}_{Y_1 \times Y_2}$ and $\tau_{Y_1 \times Y_2}$ are equal. □

Definition 1.5.2 (Convex set in ordered space)**.**

Theorem 1.5.5 (Compatibility of Order Topology with Convex Subspace)**.**

Example 1.5.6. *Let $[0, 1]$ inherit the subspace topology from that of the standard topology of $\mathbb{R}$. The following are some open sets in the subspace $[0, 1]$.*

$$[0, 1],\ [0, 1),\ (2^{-2}, 2^{-1}),\ [0, 2^{-1}) \cup (2^{-1}, 1]$$

Note that the subspace topology and the order topology of $[0, 1]$ coincide.

Example 1.5.7. *The subspace topology of $\mathbb{Z}$ from that of the standard topology of $\mathbb{R}$ coincides with the discrete topology on $\mathbb{Z}$.*

Exercise 1.5.1 (Compatibility of Subspaces)**.** *Suppose (X, τ) is a topological space and $A \subset Y \subset X$. Then, the subspace topologies on A inherited from (X, τ) and (Y, τ_Y), as a subspace of (X, τ), coincide.*

[Munkres 16.1]

Proof. Here, τ_Y is the subspace topology on Y inherited from (X, τ). Denote $(\tau_Y)_A$ the subspace topology on A inherited from (Y, τ_Y). Similarly, denote τ_A the subspace topology on A inherited from (X, τ_X). It remains to show $(\tau_Y)_A$ and τ_A coincides.

[$\subset$] Fix $U \in (\tau_Y)_A$. By definition of subspace topology, $U = V_Y \cap A$ for some $V_Y \in \tau_Y$, where $V_Y = W \cap Y$ for some $W \in \tau$. That is, $U = W \cap Y \cap A = W \cap A$, since $A \subset Y$ by assumption. Particularly, $U \in \tau_A$ by definition.

[$\supset$] Fix $U \in \tau_A$. By definition, $U = V \cap A$ for some $V \in \tau$. Again, $A \subset Y$ ensures $A = A \cap Y$ and, thus, $U = V \cap A = [V \cap Y] \cap A$. By definition, $V \cap Y \in \tau_Y$, which implies that $U \in (\tau_Y)_A$.

Indeed, the two subspace topologies on A coincide. □

Exercise 1.5.2. ***[Munkres 16.2]***

Exercise 1.5.3. ***[Munkres 16.5]***

Exercise 1.5.4. ***[Munkres 16.7]***

Exercise 1.5.5. ***[Munkres 16.9]***

Exercise 1.5.6. ***[Munkres 16.10]***

1.6 Closed Sets and Limit Points

1.6.1 Closed Sets

Theorem 1.6.1 (Closure of Closed Sets)**.** *Suppose (X, τ) is a topological space. Then,*

(a) $X, \emptyset$ are closed sets, and

(b) the collection of closed sets is closed under intersections (i.c.c) and finite unions (f.u.c.c).

Proof. (a) Since $X, \emptyset$ are open, their respective complements $\emptyset, X$ are closed by definition.

(b) Fix a collection $\mathcal{C}$ of closed sets. By definition, $\cap_{C\in\mathcal{C}}C$ is closed since its complement

$$X/\left[\cap_{C\in\mathcal{C}}C\right] = \cup_{C\in\mathcal{C}}[X/C]$$

is the union of open sets by definition and, thus, open.

Now, fix $n \in \mathbb{N}$ and closed sets $C_1, \ldots, C_n$. Similarly, $\cup_{j=1}^{n} C_j$ is closed since its complement

$$X/\cup_{j=1}^{n} C_j = \cap_{j=1}^{n}[X/C_j]$$

is, by definition, the intersection of finitely many open sets and, thus, open. □

Example 1.6.2. *(1) All subsets of a given set is closed in the discrete space of the given set.*

(2) $K := \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$ is not closed in the standard topology of $\mathbb{R}$.

Proof. Suppose, to the contrary, that K is closed. By definition, $\mathbb{R}/K$ is open and contains 0. Since the standard topology of $\mathbb{R}$ is generated by all open intervals of $\mathbb{R}$, there is an open interval (a, b) with $0 \in (a, b) \subset \mathbb{R}/K$ for some $a, b \in \mathbb{R}$ with $a < b$.

Particularly, $b > 0$. Consider $N := \left\lceil \frac{1}{b} \right\rceil + 1 \in \mathbb{N}$, so that $\frac{1}{N} \in (0, b) \subset (a, b) \subset \mathbb{R}/K$. However, $N \in \mathbb{N}$ ensures that $\frac{1}{N} \in K$, which is a contradiction. Indeed, K is not closed. □

(3) Let X be a set and consider the finite complement topology

$$\tau_C := \{X/A : A \text{ is a finite set}\} \cup \{\emptyset\}.$$

The space has as its closed sets all finite subsets of X and the underlying set X.

Lemma 1.6.3 (Product Preserves Closedness)**.** *Suppose (X_1, τ_1) and (X_2, τ_2) are topological spaces with subsets $A_1 \subset X_1$ and $A_2 \subset X_2$. If A_1 and A_2 are closed in the respective topological spaces, then $A_1 \times A_2$ is closed in the product topology.*

Proof. Suppose A_1 and A_2 are closed in the respective topologies, so their complements are open in the respective topologies by definition. It follows that $(X_1/A_1) \times X_2$ and $X_1 \times (X_2/A_2)$ are standard basis elements and, thus, open sets for the product topology. Lastly,

$$\begin{aligned}(X_1 \times X_2)/(A_1 \times A_2) =& (X_1 \times X_2)/([A_1 \times X_2] \cap [X_1 \times A_2]) \\ =& [(X_1 \times X_2)/(A_1 \times X_2)] \cup [(X_1 \times X_2)/(X_1 \times A_2)] \\ =& [(X_1/A_1) \times X_2] \cup [X_1 \times (X_2/A_2)]\end{aligned}$$

is open in the product topology. By definition, $A_1 \times A_2$ is closed in the product topology. □

Lemma 1.6.4 (Compatibility of Closed Set and Subspace Topology)**.** *Suppose (X, τ) is a topological space and $Y \subset X$. Then, for any $A \subset Y$, A is closed in $(Y, \tau_Y) \iff A = C \cap Y$ for some closed set C in (X, τ).*

Proof. Fix a subset $A \subset Y$. By definition, A is closed in $(Y, \tau_Y) \iff$

$$Y/A \text{ is open in } (Y, \tau_Y) \iff Y/A = U \cap Y \text{ for some } U \in \tau.$$

[$\Longrightarrow$ **]** Suppose A is closed in (Y, τ_Y), so $Y/A = U \cap Y$ for some $U \in \tau$. Consider the closed set $U := X/U$ in (X, τ). We observe that

$$A = Y/[Y/A] = Y/[U \cap Y] = [Y/U] \cup \emptyset = [X/U] \cap Y = C \cap Y,$$

[$\Longleftarrow$ **]** Suppose $A = C \cap Y$ for some closed set C in (X, τ) and consider the open set $U := X/C \in \tau$. We observe that $X/A = [X/C] \cup [X/Y]$, so that

$$Y/A = Y \cap [X/A] = [X/C] \cap Y = U \cap Y.$$

By the above equivalence, A is closed in (Y, τ_Y). □

1.6.2 Closure and interior

Definition 1.6.1 (Interior and Closure)**.** *Suppose (X, τ) is a topological space and $E \subset X$. Then,*

(a) the interior int E *of E is the union of all open sets in (X, τ) contained in E.*

(b) the closure $\overline{E}$ of E is the intersection of all closed sets in (X, τ) containing E.

In discussions involving multiple topological spaces, we write $\text{int}_\tau(E)$ *or* $\text{int}_X(E)$ *($\text{cl}_\tau(E)$ or $\text{cl}_X(E)$) to specify that the interior (closure) of E is defined relative to the topological space (X, τ).*

Lemma 1.6.5 (Extremality of Interior and Closure)**.** *Suppose (X, τ) is a topological space and $A \subset X$. Then,*

(a) int *A is the maximal open set in (X, τ) contained in A.*

(a.1) Particularly, A is open $\iff A =$ int *A.*

(b) $\overline{A}$ is the minimal closed set in (X, τ) containing in A.

(a.1) Particularly, A is closed $\iff A = \overline{A}$.

Proof. (a) Since topologies are closed under unions, int A is an open set contained in A by definition. For any open set U contained in A, $U \subset$ int A by the definition of interior. That is, int A is the maximal open set contained in A.

(a.1) **[** $\Longrightarrow$ **]** Suppose A is open. By definition, int $A \subset A$. Since A is an open set contained in A, the maximality of int A ensures that $A \subset$ int A. Indeed, $A =$ int A.

[$\Longleftarrow$ **]** Suppose $A =$ int A. Since int A is open, A is open.

(b) & (b.1) These implications follow from arguments symmetric to that presented in parts (a) and (a.1). □

Theorem 1.6.6 (Compatibility of Closure and Interior with Subspace Topology)**.** *Suppose* (X, τ) *is a topological space and* $Y \subset X$*. Then, for any* $A \subset Y$*,*

(a) $\text{cl}_Y A = [\text{cl}_X A] \cap Y$*, and*

(b) $\text{int}_Y A = [\text{int}_X A] \cap Y$*.*

Proof. Fix $A \subset Y$.

(a) Since $\text{cl}_Y A$ is closed in the subspace (Y, τ_Y), $\text{cl}_Y A = C \cap Y$ for some closed set C of (X, τ). Conversely, $\text{cl}_X A$ is closed in (X, τ) ensures that $[\text{cl}_X A] \cap Y$ is closed in (Y, τ_Y).

By definition, $[\text{cl}_X A] \cap Y \supset A \cap Y = A$. Particularly, $[\text{cl}_X A] \cap Y$ is a closed set of (Y, τ_Y) containing A. By the minimality of $\text{cl}_Y A$, $[\text{cl}_X A] \cap Y \supset \text{cl}_Y A$.

Similarly, $C \cap Y = \text{cl}_Y A \supset A$ ensures C is a closed set of (X, τ) containing A. By the minimality of $\text{cl}_X A$, $\text{cl}_X A \subset C$, so that $[\text{cl}_X A] \cap Y \subset \text{cl}_Y A$. Indeed, $\text{cl}_Y A = [\text{cl}_X A] \cap Y$.

(b) This implication follows from arguments symmetric to that presented in part (a). □

Lemma 1.6.7 (Characterization of Living in interior)**.** *Suppose* (X, τ) *is a topological space and* $A \subset X$*. Then,* $\forall x \in X$*,* $x \in \text{int } A \iff$ *there is a neighborhood* $U \in \tau$ *of* x *contained in* A*.*

Proof. Fix $x \in X$.

[$\implies$] Suppose $x \in \text{int } A$ and consider $U := \text{int } A$. By definition, $U \subset A$, where U is a neighborhood of x by assumption.

[$\impliedby$] Suppose there is a neighborhood $U \in \tau$ of x contained in A. By the maximality of interiors, $x \in U \subset \text{int } A$. □

Theorem 1.6.8 (Characterization of Living in Closure)**.** *Suppose* (X, τ) *is a topological space and* $A \subset X$*.*

(a) For any $x \in X$*,* $x \in \overline{A} \iff$ *every neighborhood* $U \in \tau$ *of* x *intersects* A *non-trivially.*

(b) Suppose $\mathcal{B}$ *is a basis for* τ*. Then,* $\forall x \in X$*,* $x \in \overline{A} \iff$ *every basis element* $B \in \mathcal{B}$ *containing* x *intersects* A *non-trivially.*

Proof. Fix $x \in X$.

(a) [$\implies$] Suppose $x \in \overline{A}$. Suppose, to the contrary, that there is a neighborhood $U \in \tau$ of x such that $U \cap A = \emptyset$. The disjointness of U and A ensures that X/U is a closed set containing A.

The minimality of closures then implies that $\overline{A} \subset X/U$. However, $x \in \overline{A}$ implies $x \in X/U$, which contradicts the assumption that U is a neighborhood of x.

Thus, every neighborhood U of x in (X, τ) intersects A non-trivially.

[$\impliedby$] Suppose every neighborhood $U \in \tau$ of x intersects A non-trivially. Suppose, to the contrary, that $x \in X/\overline{A}$, which is a neighborhood of x. By assumption, $X/\overline{A}$ intersects A non-trivially.

Particularly, there exists $y \in [X/\overline{A}] \cap A$. However, $\overline{A} \supset A \implies X/\overline{A} \subset X/A$, so that $y \in [X/A] \cap A$, which is a contradiction. Thus, $x \in \overline{A}$.

(b) By part (a), it suffices to show the following equivalence.

Every neighborhood $U \in \tau$ of x intersects A non-trivially $\iff$ every basis element $B \in \mathcal{B}$ containing x intersects A non-trivially.

Since a basis of a topology is a subset of the topology, the implication [$\Longrightarrow$] holds immediately.

[$\Longleftarrow$] Suppose every basis element $B \in \mathcal{B}$ containing x intersects A non-trivially.

Fix a neighborhood $U \in \tau$ of x. Since $\mathcal{B}$ is a basis for τ, there is a basis element $B \in \mathcal{B}$ such that $x \in B \subset U$. By assumption, B intersects A non-trivially. Since U contains B, U intersects A non-trivially as well. This completes the proof. □

Corollary 1.6.9 (Characterization of Denseness)**.** *Suppose (X, τ) is a topological space and $X \subset X$ is non-empty. Then, $\overline{A} = X \iff$ every non-empty open set $U \in \tau$ intersects A non-trivially.*

Proof. **[$\Longrightarrow$]** Suppose $\overline{A} = X$. Fix a non-empty open set $U \in \tau$ and consider $x \in U$. Since $x \in X = \overline{A}$, the preceding theorem ensures that the neighborhood U of x intersects A non-trivially.

[$\Longleftarrow$] Suppose every non-empty open set $U \in \tau$ intersects A non-trivially. Suppose, to the contrary, that $X/\overline{A}$ is non-empty. Since $\overline{A}$ is closed, $X/\overline{A}$ is a non-empty open set.

By assumption, $X/\overline{A}$ intersects A non-trivially. However, $X/\overline{A} \subset X/A$ implies that X/A and A have a non-empty intersection, which is a contradiction. Indeed, $X = \overline{A}$. □

1.6.3 Closure and Limit Point

Definition 1.6.2 (Limit point)**.** *Suppose (X, τ) is a topological space and $A \subset X$.*

*(a) A **limit point (accumulation/cluster point) of** $A \subset X$ is a point $x \in X$ contained in $\overline{A/\{x\}}$.*

(b) The collection of limit points of $A \subset X$ in (X, τ) is denoted by A'.

Remark 1.6.1. *As a reminder, a limit point x of a subset A was defined in class via non-trivial intersections of any neighborhood of x with $A/\{x\}$.*

Remark 1.6.2. *As the term suggests, we somehow should be able to relate limit points with limits of convergent sequences.*

Theorem 1.6.10 (Characterization of Closure via Limit Point)**.** *Suppose (X, τ) is a topological space, $A \subset X$, and A' is the set of limit points of A. Then,*

$$\overline{A} = A \cup A'.$$

Proof. For any limit point x of A, $x \in \overline{A/\{x\}} \subset \overline{A}$ by definition. That is, $A' \subset \overline{A}$, so that $A \cup A' \subset \overline{A}$.

Fix $x \in \overline{A}$. If $x \in A$, then $x \in A \cup A'$. Hence, we suppose that $x \in X/A$. Since $A/\{x\} = A$, $x \in \overline{A} = \overline{A/\{x\}}$. That is, x is a limit point of A by definition, so $x \in A'$. Particularly, $\overline{A} \subset A \cup A'$, so that $\overline{A} = A \cup A'$. □

Corollary 1.6.11 (Characterization of Closed Set Via Limit Point)**.** *Suppose (X, τ) is a topological space and $A \subset X$. Then, A is closed $\iff$ A contains all of its limit points.*

Proof. Recall that A is closed $\iff A = \overline{A} = A \cup A'$ by the preceding characterizations.

If A is closed, then $A' \subset A$, so A contains all of its limit points. If A contains all of its limit points, then $\overline{A} = A \cup A' \subset A$ ensures that $A = \overline{A}$, which implies A is closed. □

Definition 1.6.3 (Boundary)**.** *Suppose (X, τ) is a topological space and $A \subset X$.*

(a) The boundary of A is the set $\partial A := \overline{A} \cap \overline{X/A}$.

(b) A boundary point of A is an element of the boundary of A.

Proposition 1.6.12 (Properties of Boundary)**.** *Suppose (X, τ) is a topological space and $A \subset X$.*

(a) The interior and boundary of A are disjoint.

(a.1) Particularly, $\overline{A} = \text{int } A \sqcup \partial A$.

(b) The boundary of A is empty $\iff$ A is open and closed.

(c) A is open $\iff \partial A = \overline{A}/A$.

Proof. (a) Suppose, to the contrary, that there exists $x \in \text{int } A \cap \partial A$. Since $x \in \partial A$, $x \in \overline{X/\text{int } A}$ as $\text{int } A \subset A$ by definition. Thus, any neighborhood of x intersects $X/\text{int } A$ non-trivially.

Particularly, $\text{int } A$ is a neighborhood of x by assumption, so $\text{int } A$ intersects $X/\text{int } A$ non-trivially, which is a contradiction. Indeed, the interior and boundary of A are disjoint.

(a.1) By part (a), the interior and boundary of A are disjoint. By definition, $\overline{A} \supset \text{int } A \sqcup \partial A$, so it remains to show $\overline{A} \subset \text{int } A \sqcup \partial A$.

Fix $x \in \overline{A}$. If $x \in \text{int } A$, then $x \in \text{int } A \sqcup \partial A$. Thus, we suppose $x \notin \text{int } A$. It follows that, for any neighborhood U of x, $U \not\subset A$,

> Recall that $x \in \text{int } A \iff$ there is a neighborhood of x contained in A.

which ensures that $U \cap [X/A] \neq \emptyset$. Particularly, $x \in \overline{X/A}$ and, thus, $x \in \partial A \subset \text{int } A \sqcup \partial A$. Indeed, $\overline{A} \subset \text{int } A \sqcup \partial A$, so the equality holds.

(b) This implication follows immediately from (a.1)

[$\implies$ **]** Suppose $\partial A = \emptyset$. It follows that $\overline{A} = \text{int } A \cup \partial A = \text{int } A$, which implies $A = \overline{A} = \text{int } A$ for we have $\text{int } A \subset A \subset \overline{A}$. Indeed, A is closed and open.

[$\impliedby$ **]** Suppose A is closed and open, so that A and X/A are both closed and coincide with their respective closures. Particularly,

$$\partial A := \overline{A} \cap \overline{X/A} = A \cap [X/A] = \emptyset.$$

(c) **[** $\implies$ **]** Suppose A is open, so that X/A is closed and coincides with its closure. It follows that

$$\partial A := \overline{A} \cap \overline{X/A} = \overline{A} \cap [X/A] = \overline{A}/A.$$

[$\impliedby$ **]** Suppose that $\partial A = \overline{A}/A$. By (a),

$$\text{int } A = \overline{A}/\partial A = \overline{A}/[\overline{A}/A] = A.$$

Indeed, A is open. □

1.6.4 Hausdorff Space

Remark 1.6.3 (Singletons Need not be Closed)**.** *Despite how nice a world it would be, it need not be true that singletons are closed in every topological space.*

As a trivial counterexample, the indiscrete space with an underlying set consisting of at least 2 elements has non-closed singletons.

As a more interesting counterexample, the finite topology τ on $\{a, b, c\}$ defined by

$$\tau := \{\{a, b, c\}, \{a, b\}, \{b, c\}, \{b\}, \emptyset\}$$

has a non-closed singleton $\{b\}$.

Remark 1.6.4 (Limits Need not be Unique)**.** *Similarly, it need not be true that every convergent sequence, which we define later, in a topological space has a unique limit.*

For instance, sequences in the indiscrete space with an underlying set containing more than 1 element converge to every element of X.

For a more interesting example, consider $\{a, b, c\}$ with the topology $\tau := \{\{a, b, c\}, \{a, b\}, \{c\}, \emptyset\}$. Then, every sequence in $\{a, b, c\}$ consisting of terms a and b has limits a and b.

Remark 1.6.5. *To remediate the above inconveniences, we consider a type of very well-behaved topological spaces, namely the Hausdorff spaces.*

Definition 1.6.4 (Hausdorff space)**.** *A **Hausdorff space (T_2 space)** is a topological space (X, τ) in which any distinct points in X can be separated by disjoint open sets; that is, for any distinct $x, y \in X$, there exist disjoint neighborhoods $U_x, W_y \in \tau$ of x and y, respectively.*

Definition 1.6.5 (T_1 space)**.** *Suppose (X, τ) is a topological space. Then, (X, τ) is T_1 if all finite subsets of X are closed.*

Exercise 1.6.1 (Characterization of T_1 Axiom)**.** *Suppose (X, τ) is a topological space. Then, (X, τ) is $T_1 \iff$ for any distinct point $x, y \in X$, there exist neighborhoods $V_x, V_y \in \tau$ of x and y, respectively, such that $x \notin V_y$ and $y \notin V_x$. **[Munkres 17.15]***

Proof. **[** $\implies$ **]** Suppose (X, τ) is T_1 and fix distinct points $x, y \in X$. By definition of T_1, $\{x\}$ and $\{y\}$ are closed sets. Particularly, $V_x := X/\{y\}$ and $V_y := X/\{x\}$ are open sets.

Since $x \neq y$, $x \in X/\{y\} =: V_x$ and $y \notin V_x$ by construction. That is, V_x is a neighborhood of x not containing y. Similarly, V_y is a neighborhood of y not containing x.

[$\impliedby$ **]** Suppose, for any distinct point $x, y \in X$, that there exist neighborhoods $V_x, V_y \in \tau$ of x and y, respectively, such that $x \notin V_y$ and $y \notin V_x$.

Fix $x \in X$. By assumption, every $y \in X/\{x\}$ has a neighborhood V_y not containing x and, thus, $V_y \subset X/\{x\}$. Particularly, $X/\{x\} = \cup_{y \in X/\{x\}} V_y$ is open. By definition, $\{x\}$ is closed.

Lastly, every finite subset of X is the finite union of singletons in X, which we showed are closed. Since the collection of all closed sets is closed under finite union, every finite subset of X is closed. That is, (X, τ) is T_1 by definition. □

Remark 1.6.6. *As a reminder, a T_1 space was defined in class via the separation property above.*

Lemma 1.6.13 (Hausdorff Spaces are T_1)**.** *Suppose (X, τ) is a topological space. If (X, τ) is Hausdorff, then (X, τ) is T_1.*

Proof. Suppose (X, τ) is Hausdorff and fix $x \in X$. For any $y \in X/\{x\}$, there exist disjoint neighborhoods $V_y, U_y \in \tau$ of x and y, respectively, where the disjointness of the neighborhoods ensures that $U_y \subset X/\{x\}$. Particularly, $X/\{x\} = \cup_{y \in X/\{x\}} U_y$, so $X/\{x\}$ is open. By definition, $\{x\}$ is closed.

Since the collection of closed sets is closed under finite unions, the above ensures that finite subsets of X are closed. That is, (X, τ) is T_1 by definition. □

Lemma 1.6.14. *$\mathbb{R}$ equipped with the standard topology is Hausdorff.*

Proof. Fix distinct points $x, y \in \mathbb{R}$. Suppose, without loss of generality, that $x < y$ and consider the midpoint $m := \frac{x+y}{2}$.

We observe $x \in I_x := (x - 1, m)$ and $y \in I_y := (m, y + 1)$. Particularly, I_x and I_y are disjoint neighborhoods of x and y, respectively. By definition, $\mathbb{R}$ with the standard topology is Hausdorff. □

Remark 1.6.7. *A common question on the comprehensive exams is to give examples of topological spaces satisfying a certain separation property but not another.*

For instance, one may be asked to provide a topological space that is T_1 but not T_2.

Lemma 1.6.15 (Infinite Finite Complement Space is T_1 but not T_2)**.** *Suppose X is a set with infinitely many elements. Then, X equipped with the finite complement topology is T_1 but not Hausdorff.*

Proof. Denote the finite complement topology on X by τ_C.

By the definition of τ_C, $A \subset X$ is closed $\iff$ X/A is open $\iff$ A is finite or $X/A = \emptyset$. We observe that $\emptyset$ is trivially closed, and every non-empty finite subset of X is closed by the above equivalence. That is, (X, τ_C) is T_1 by definition.

By the definition of τ_C, $A \subset X$ is open $\iff$ X/A is finite or A is empty. Thus, for any non-empty open set U, we may enumerate $X/U = \{u_1, \ldots, u_n\}$ for some $n \in \mathbb{N}$, so that $U = X/[X/U] = X/\{u_1, \ldots, u_n\}$.

Particularly, given two non-empty open sets $U = X/\{u_1, \ldots, u_n\}$ and $V = X/\{v_1, \ldots, v_m\}$ for some $n, m \in \mathbb{N}$, their intersection

$$U \cap V = X/\{u_1, \ldots, u_n, v_1, \ldots, v_m\}$$

remains a set with infinitely many elements and, therefore, non-empty.

Thus, given two distinct elements $x, y \in X$, there are no disjoint neighborhoods U_x and U_y of x and y, respectively. That is, (X, τ) is not Hausdorff. □

Example 1.6.16 (Line with 2 Origins is not Hausdorff)**.**

Theorem 1.6.17 (Subspaces and Product Spaces Inherit Hausdorffness)**.** *Suppose (X, τ_X) is a Hausdorff space.*

(a) If $A \subset X$, then the subspace (A, τ_A) is Hausdorff.

(b) If (Y, τ_Y) is a Hausdorff space, then the product space $(X \times Y, \mathcal{P}_{X \times Y})$ is Hausdorff.

Proof. (a) Suppose $A \subset X$ and fix distinct elements $a, b \in A$. Since $a, b \in X$ and (X, τ_X) is Hausdorff, there are disjoint neighborhoods $U_x, U_y \in \tau_X$ of x and y, respectively.

It follows that $U_x \cap A$ and $U_y \cap A$ are disjoint neighborhoods in the subspace (A, τ_A) of x and y, respectively. By definition, the subspace (A, τ_A) is Hausdorff.

(b) Suppose (Y, τ_Y) is a Hausdorff space and fix distinct elements $(x_1, y_1), (x_2, y_2) \in X \times Y$. Since the elements are distinct, $x_1 \neq x_2$ or $y_1 \neq y_2$. We suppose, without loss of generality, that $x_1 \neq x_2$.

Since (X, τ_X) is a Hausdorff, there are disjoint neighborhoods $U_1, U_2 \in \tau_X$ of x_1 and x_2, respectively. It follows that the sets $V_1 := U_1 \times Y, V_2 := U_2 \times Y \in \mathcal{P}_{X \times Y}$ are disjoint neighborhoods of (x_1, y_1) and (x_2, y_2), respectively.

Since $x_1 \in U_1$ and $y_1 \in Y$, $(x_1, y_1) \in V_1$. Similarly, $(x_2, y_2) \in V_2$. Lastly,

$$V_1 \cap V_2 = [U_1 \times Y] \cap [U_2 \times Y] = [U_1 \cap U_2] \times [Y \cap Y] = \emptyset \times Y = \emptyset.$$

By definition, the product space $(X \times Y, \mathcal{P}_{X \times Y})$ is Hausdorff. □

1.6.5 Convergence and Limit

Definition 1.6.6 (Convergence of sequence, limit)**.** *Suppose (X, τ) is a topological space and $(x_n)_{n \in \mathbb{N}}$ is a sequence of elements of X.*

*(a) The sequence $(x_n)_{n \in \mathbb{N}}$ **converges to** $x_0 \in X$ **in** (X, τ) and we write $x_n \to x_0$ **in** (X, τ) if for every neighborhood U of x_0, there exists $N \in \mathbb{N}$ such that $\forall n \in \mathbb{N}$, $n > N \implies x_n \in U$.*

*(b) A **limit of** $(x_n)_{n \in \mathbb{N}}$ **in** (X, τ) is a point $x_0 \in X$ to which $(x_n)_{n \in \mathbb{N}}$ converges, if any.*

Lemma 1.6.18 (Characterization of Convergence)**.** *Suppose (X, τ) is a topological space and $x, x_1, x_2, \cdots \in X$. Then, $(x_n)_{n \in \mathbb{N}}$ converges to x in (X, τ) $\iff$ for every neighborhood U of x, the collection $\{n \in \mathbb{N} : x_n \in X/U\}$ is finite.*

Proof. **[** $\implies$ **]** Suppose $(x_n)_{n \in \mathbb{N}}$ converges to x in (X, τ). Fix a neighborhood U of x, so there exists $N \in \mathbb{N}$ such that $\forall n \in \mathbb{N}$ $n > N \implies x_n \in U$, by the definition of convergence.

We observe $\{n \in \mathbb{N} : x_n \in X/U\} \subset \{0, 1, \ldots, N\}$, for if there is some $n \in \mathbb{N} / \{0, 1, \ldots, N\}$ with $x_n \in X/U$, then $n > N$ ensures $x_n \in U$, which is a contradiction. Thus, $\{n \in \mathbb{N} : x_n \in X/U\}$ is finite, for it is a subset of a finite set.

[$\impliedby$ **]** Suppose for every neighborhood U of x, the collection $\{n \in \mathbb{N} : x_n \in X/U\}$ is finite. Fix a neighborhood U of x, so $\{n \in \mathbb{N} : x_n \in X/U\}$ is finite by assumption.

Consider $N := \max\{n \in \mathbb{N} : x_n \in X/U\} \in \mathbb{N}$. For any $n \in \mathbb{N}$ with $n > N$, $x_n \in U$, for otherwise $x_n \in X/U$ implies that $N \geq n > N$, which is a contradiction. By definition, $(x_n)_{n \in \mathbb{N}}$ converges to x in (X, τ). □

Lemma 1.6.19. *Suppose (X, τ) is a topological space and $x, x_1, x_2, \cdots \in X$.*

(a) If $(x_n)_{n \in \mathbb{N}}$ converges to x in (X, τ), then for every neighborhood U of x, the collection $\{x_n : n \in \mathbb{N}\} \cap [X/U]$ is finite.

(b) The converse of (a) need not hold.

Proof. (a) Suppose $(x_n)_{n\in\mathbb{N}}$ converges to x in (X,τ). Fix a neighborhood U of x. By the preceding lemma, $K := \{n \in \mathbb{N} : x_n \in X/U\}$ is finite and so is $\{x_k : k \in K\}$. Since

$$\{x_n : n \in \mathbb{N}\} \cap [X/U] \subset \{x_k : k \in K\}$$

and $\{x_k : k \in K\}$ is finite, $\{x_n : n \in \mathbb{N}\} \cap [X/U]$ is finite.

(b) Consider $X := \{0,1\}$ equipped with the discrete topology τ on X. Let x, and $\forall j \in \mathbb{N}\ x_j := 1$. For any neighborhood U of x, $\{x_n : n \in \mathbb{N}\} \cap [X/U]$ is finite since

$$\{x_n : n \in \mathbb{N}\} \cap [X/U] \subset X := \{0,1\}.$$

However, $(x_n)_{n\in\mathbb{N}}$ does not converge to x in (X,τ), since for the neighborhood $U := \{0\}$ of x $\forall j \in \mathbb{N}\ x_n \notin U$ by construction. □

Theorem 1.6.20 (Uniqueness of Limit in Hausdorff Space)**.** *Suppose (X,τ) is a Hausdorff space and $x, x', x_1, x_2, \cdots \in X$. If $(x_n)_{n\in\mathbb{N}}$ converges to x and x' in (X,τ), then $x = x'$.*

Proof. Suppose $(x_n)_{n\in\mathbb{N}}$ converges to x and x' in (X,τ). Suppose, to the contrary, that x and x' are distinct. Since (X,τ) is Hausdorff, there are disjoint neighborhoods $V_x, V_{x'}$ of x and x', respectively.

By the convergence of $(x_n)_{n\in\mathbb{N}}$, there exists $N, N' \in \mathbb{N}$ such that $\forall n \in \mathbb{N}$

$$n > N \implies x_n \in V_x \text{ and } n > N' \implies V_{x'}.$$

Particularly, $x_M \in V_x \cap V_{x'}$ for $M := \max\{N, N'\} + 1 \in \mathbb{N}$, which is a contradicts the disjointnes of V_x and $V_{x'}$. Indeed, $x = x'$. □

Theorem 1.6.21 (Characterization of Limit Point in T_1 Space via Infinite Intersection)**.** *Suppose (X,τ) is a T_1 topological space and $A \subset X$. Then, $\forall x \in X$, x is limit point of $A \iff$ every neighborhood $U \in \tau$ of x intersects A at infinitely many points.*

Proof. **[INCOMPLETE]** □

1.6.6 Exercises

Exercise 1.6.2 (Compatibility of Closure and Product Space)**.** *Suppose (X,τ_X) and (Y,τ_Y) are topological spaces with subsets. If $A \subset X$ and $B \subset Y$, then*

$$\mathrm{cl}_{X\times Y}(A \times B) = \mathrm{cl}_X(A) \times \mathrm{cl}_Y(B).$$

[Munkres 17.9]

Proof. Suppose $A \subset X$ and $B \subset Y$. Since $\mathrm{cl}_X(A)$ and $\mathrm{cl}_Y(B)$ are closed in the respective topological spaces, Exercise 17.3 (Munkres 100) ensures that $\mathrm{cl}_X(A) \times \mathrm{cl}_Y(B)$ is closed in the product space $(X \times Y, \mathcal{P}_{X\times Y})$.

Since $\mathrm{cl}_X(A) \supset A$ and $\mathrm{cl}_Y(B) \supset B$, $\mathrm{cl}_X(A) \times \mathrm{cl}_Y(B)$ is a closed in $(X \times Y, \mathcal{P}_{X\times Y})$ containing $A \times B$. By the minimality of $\mathrm{cl}_{X\times Y}(A \times B)$, $\mathrm{cl}_{X\times Y}(A \times B) \subset \mathrm{cl}_X(A) \times \mathrm{cl}_Y(B)$.

Fix $(x,y) \in \mathrm{cl}_X(A) \times \mathrm{cl}_Y(B)$ and it remains to show that $(x,y) \in \mathrm{cl}_{X\times Y}(A \times B)$. Fix a neighborhood $U \in \mathcal{P}_{X\times Y}$ of (x,y). By the definition of product topology, there is a standard basis element $V_x \times V_y$ of $\mathcal{P}_{X\times}$, for some $V_1 \in \tau_X$ and $V_2 \in \tau_Y$, such that $(x,y) \in V_x \times V_y \subset U$.

Since $x \in \mathrm{cl}_X(A)$ and $x \in V_1$, A intersects V_x non-trivially. Similarly, B intersects V_y non-trivially. Particularly,

$$[A \cap V_x] \times [B \cap V_y] = [V_x \times V_y] \times [A \times B]$$

is non-empty and, thus, so is $U \cap [A \times B]$. By the characterization of elements in closure, $(x, y) \in \mathrm{cl}_{X\times Y}[A \times B]$. Indeed, $\mathrm{cl}_{X\times Y}(A \times B) \supset \mathrm{cl}_X(A) \times \mathrm{cl}_Y(B)$, so the equality holds. □

Exercise 1.6.3 (Characterization Hausdorff Space via Closed Diagonal)**.** *Suppose (X, τ) is a topological space. Then (X, τ) is Hausdorff $\iff$ the diagonal $\Delta := \{(x, x) : x \in X\}$ of X is closed in the product space $(X^2, \mathcal{P}_{X^2})$.* ***[Munkres 17.13]***

Proof. **[** $\implies$ **]** Suppose (X, τ) is Hausdorff. To show that Δ is closed, we prove that $\Delta = \overline{\Delta}$. Fix $(x, y) \in \overline{\Delta}$ and it remains to show that $(x, y) \in \Delta$.

Suppose, to the contrary, that $(x, y) \notin \Delta$, so that x and y are distinct. Since (X, τ) is Hasudindorff, there are disjoint neighborhoods $V_x, V_y \in \tau$ of x and y, respectively.

Since $V_x \times V_y$ is a standard basis element and, thus, an open set of the product space, $V_x \times V_y$ is a neighborhood of (x, y). Since $(x, y) \in \overline{\Delta}$ by assumption, $V_x \times V_y$ intersects Δ non-trivially.

Particularly, there is some $(x', y') \in [V_x \times V_y] \cap \Delta$, where $(x', y') \in \Delta$ ensures that $x' = y'$. Lastly, $(x', y') \in V_x \times V_y$ implies that V_x and V_y are not disjoint, which is a contradiction.

Indeed, $(x, y) \in \Delta$, so that $\Delta = \overline{\Delta}$ is closed in the product space.

[$\impliedby$ **]** Suppose Δ is closed in the product space $(X^2, \mathcal{P}_{X^2})$. Fix distinct points $x, y \in X$, so $[X^2]/\Delta \in \mathcal{P}_{X^2}$ is a neighborhood of (x, y).

Particularly, there is a standard basis element $V_1 \times V_2$ of $\mathcal{P}_{X^2}$ for some $V_x, V_y \in \tau$ such that $(x, y) \in V_x \times V_y \subset [X^2]/\Delta$. It follows that V_x and V_y are disjoint, for otherwise if $\exists x' \in V_x \cap V_y$, then $(x', x') \in V_x \times V_y \subset [X^2]/\Delta$, which is a contradiction as $(x', x') \in \Delta$.

As a result, V_x and V_y are disjoint neighborhoods of x and y, respectively. By definition, (X, τ) is Hasudorff. □

Exercise 1.6.4 (Munkres 17.14)**.** *In the finite complement topology on $\mathbb{R}$, to what point or points does the sequence $(x_n)_{n\in\mathbb{N}}$ defined by $x_n := \frac{1}{n}$ $\forall n \in \mathbb{N}$ converge?*

Proof. For brevity, denote the finite complement topology of $\mathbb{R}$ by τ. We show that the sequence converges, in $(\mathbb{R}, \tau)$, to all real numbers.

Fix $x \in \mathbb{R}$ and a neighborhood U of x in τ. Recall that

$$\tau = \{U \subset \mathbb{R} : \mathbb{R}/U \text{ is finite or } U = \emptyset\},$$

so U being non-empty ensures that it has a finite complement, which we may enumerate as $\mathbb{R}/U := \{s_1, \ldots, s_n\}$ for some $n \in \mathbb{N}$. Since $U = \mathbb{R}/[\mathbb{R}/U]$, we obtain $U = \mathbb{R}/\{s_1, \ldots, s_n\}$.

Consider $\delta := \min\{s_j : j \in \{1, \ldots, n\}, s_j > 0\} > 0$, which ensures that $(0, \delta) \subset U$.

> To see this, we fix $\alpha \in (0, \delta)$ and show that $\alpha \neq s_j$, $\forall j \in \{1, \ldots, n\}$. Fix $j \in \{1, \ldots, n\}$. If $s_j \leq 0$, then $\alpha \neq s_j$ since $\alpha > 0$ by construction. If $s_j > 0$, then $s_j \geq \delta > \alpha$ by construction, which ensures $\alpha \neq s_j$. Indeed, $(0, \delta) \subset U$.

Lastly, consider $N \in \mathbb{N}$ with $\frac{1}{N} < \delta$; say $N := \left\lceil \frac{1}{\delta} \right\rceil + 1$, where $\lceil \cdot \rceil$ denotes the usual ceiling function on $\mathbb{R}$. It follows that $\forall n \in \mathbb{N}$, $n > N$ ensures $\frac{1}{n} < \frac{1}{N} < \delta$ and, thus, $x_n := \frac{1}{n} \in (0, \delta) \subset U$.

By definition, $(x_n)_{n\in\mathbb{N}}$ converges to x in $(\mathbb{R}, \tau)$ as desired. □

Exercise 1.6.5 (Kuratowski)**.** ***[Munkres 17.21]***

Proof. **[Outline for Attempt 2]** Consider the idempotence of taking closure and the invertibility of complementation. Argue that to generate sets distinct from $A, X/A$, and $\overline{A}$, it is necessary to consider sets generated strictly by alternating between taking closures and complementations.

From attempt #1, we obtained the duality int $A = X/\overline{X/A}$ and $\overline{A} = X/[\text{int } (X/A)]$.

> *Proof.* By definition, int $A \subset A$, so that $X/\text{int } A \supset X/A$. Since $X/\text{int } A$ is a closed set containing X/A, the minimality of $\overline{X/A}$ ensures that
>
> $$X/\text{int } A \supset \overline{X/A} \supset X/A \implies \text{int } A \subset X/\overline{X/A} \subset A.$$
>
> Since $X/\overline{X/A}$ is an open set residing within A, the maximality of int A ensures that int $A \supset X/\overline{X/A}$. That is, int $A = X/\overline{X/A}$ as desired.
>
> Consider $B := X/A$. The first part ensures that $X/\text{int } B = \overline{X/B} = \overline{A}$. That is,
>
> $$\overline{A} = X/[\text{int } (X/A)]$$
>
> □

From the first equality, most sets of interest will be generated by alternating strictly between taking closures and interiors. The respective maximality and minimality of interiors and closures then show that int $(\overline{\text{int } (\overline{\text{int } A})}) = \text{int } (\overline{\text{int } A})$.

Proof. We observe that $\overline{\text{int } A} \supset \text{int } A$ ensures

$$\text{int } (\overline{\text{int } A}) \supset \text{int } A,$$

since int $(\overline{\text{int } A})$ is the maximal open set contained in $\overline{\text{int } A}$. By applying the above containment to $A' := \overline{\text{int } A}$, we obtain

$$\text{int } (\overline{\text{int } (\overline{\text{int } A})}) \supset \text{int } (\overline{\text{int } A}).$$

Since $\overline{\text{int } A}$ is a closed set containing int $(\overline{\text{int } A})$, the minimality of $\overline{\text{int } (\overline{\text{int } A})}$ ensures

$$\overline{\text{int } (\overline{\text{int } A})} \subset \overline{\text{int } A}.$$

By taking the interior of the above sets, we obtain

$$\text{int } (\overline{\text{int } (\overline{\text{int } A})}) \subset \text{int } (\overline{\text{int } A}).$$

Indeed, int $(\overline{\text{int } (\overline{\text{int } A})}) = \text{int } (\overline{\text{int } A})$. □

Combine the previous observations to conclude that there are no more than 14 distinct sets generated by taking closure and complementation □

1.7 Continuous Functions

1.7.1 Elementary Definitions and Examples

Definition 1.7.1 (Continuous map)**.** *Suppose (X, τ_X) and (Y, τ_Y) are topological spaces.*

(a) A continuous function from (X, τ_X) to (Y, τ_Y) is a function $f\colon X \to Y$ satisfying

$$\forall W \in \tau_Y,\ f^{-1}(W) \in \tau_X.$$

(b) We write $f\colon (X, \tau_X) \to (Y, \tau_Y)$ is continuous if f is a continnuous function from (X, τ_X) to (Y, τ_Y). If the topologies are clear, we write more succinctly that $f\colon X \to Y$ is continuous or f is continuous.

Example 1.7.1 (Some Continuous Function)**.** *Suppose (X, τ_X) and (Y, τ_Y) are topological spaces. Then,*

(a) the identity map $I\colon (X, \tau_X) \to (X, \tau_X)$ is continuous;

(b) if $A \subset X$, the inclusion map $\iota\colon (A, \tau_A) \to (X, \tau_X)$ is continuous;

(c) every constant map $C\colon (X, \tau_X) \to (Y, \tau_Y)$ is continuous;

(d) the canonical projection maps $\pi_1\colon X \times Y \to X$ and $\pi_2\colon X \times Y \to Y$ are continuous.

Proof. We only prove (b) and (c), since (a) is immediate and we proved (d) previously.

(b) Fix $A \subset X$. Note the inclusion map is defined by $\forall x \in A$, $\iota(x) := x$. For any $U \subset X$,

$$\iota^{-1}(U) := \{x \in A : x \in U\} = U \cap A.$$

Particularly, for any open set $U \in \tau_X$, $\iota^{-1}(U) = U \cap A \in \tau_A$. By definition, ι is continuous.

(c) Fix a constant map $C\colon X \to Y$, say with value $y_0 \in Y$. That is, C is defined by $\forall x \in X$, $C(x) := y_0$. For any $U \subset Y$,

$$C^{-1}(U) = \begin{cases} \emptyset & \text{if } y_0 \notin U \\ X & \text{if } y_0 \in U \end{cases}.$$

Particularly, $\forall U \in \tau_Y$, $C^{-1}(U) \in \tau_X$. By definition, the constant map is closed. □

Example 1.7.2 (Identity Map Need Not be Continuous)**.** *Suppose X is a set with at least two elements. Then, the identity map $I\colon (X, \{\emptyset, X\}) \to (X, \mathcal{P}(X))$ is not continuous.*

Proof. Let $a \in X$. Since X has at least two elements by assumption, $\{a\} \neq X$ and $\{a\} \in \mathcal{P}(X)$. It follows that $I^{-1}(\{a\}) = \{a\} \notin \{\emptyset, X\}$. Particularly, I is not continuous. □

Example 1.7.3 (Continuity of Maps from Discrete Space)**.** *Suppose (X, τ) is a topological space. Then, every $f\colon (X, \mathcal{P}(X)) \to (X, \tau)$ is continuous.*

Lemma 1.7.4 (Identifying Discrete Topology)**.** *Suppose (X, τ) is a topological space. If, for every topology τ' of X, every map $f\colon (X, \tau) \to (X, \tau')$ is continuous, then $\tau = \mathcal{P}(X)$.*

Proof. Suppose, for every topology τ' of X, every map $f\colon (X, \tau) \to (X, \tau')$ is continuous. Consider the topology $\tau' := \mathcal{P}(X)$ and the identity map $I\colon (X, \tau) \to (X, \tau')$, which is continuous by assumption.

For any $U \subset X$, $U \in \tau'$ ensures that $I^{-1}(U) = U \in \tau$ by the continuity of I. That is, $\mathcal{P}(X) \subset \tau$, which ensures τ is the discrete topology. □

1.7.2 Elementary Properties

Lemma 1.7.5 (Continuity on (sub)Basis Implies Continuity)**.** *Suppose (X, τ_X) and (Y, τ_Y) are topological spaces and $f\colon X \to Y$. If $\mathcal{B}$ is a basis or subabsis for τ_Y and $\forall B \in \mathcal{B}\ f^{-1}(B) \in \tau_X$, then f is continuous.*

Proof. Suppose $\mathcal{B}$ is a basis or subabsis for τ_Y and $\forall U \in \tau_Y\ f^{-1}(U) \in \tau_X$.

[Case 1] Suppose $\mathcal{B}$ is a basis for τ_Y. For any open set $U \in \tau_Y$, we may decompose $U = \cup_{B\in\mathcal{A}} B$ for some subcollection $\mathcal{A}$ of $\mathcal{B}$, which ensures that

$$f^{-1}(U) = \cup_{B\in\mathcal{A}} \underbrace{f^{-1}(B)}_{\in\tau_X} \in \tau_X$$

by assumption. Indeed, f is continuous.

[Case 2] Suppose $\mathcal{B}$ is a subbasis for τ_Y. It follows that τ_Y has a basis $\mathcal{B}'$ consisting of all finite intersections of elements of $\mathcal{B}$. Particularly, for any $\cap_{j=1}^n B_j \in \mathcal{B}'$,

$$f^{-1}(\cap_{j=1}^n B_j) = \cap_{j=1}^n \underbrace{f^{-1}(B_j)}_{\in\tau_X} \in \tau_X$$

by assumption. That is, we reduced this case to the previous. By Case 1, f is continuous. □

Definition 1.7.2 (Continuous at a point)**.** *Suppose (X, τ_X) and (Y, τ_Y) are topological spaces, $f\colon X \to Y$, and $x_0 \in X$. Then, f is continuous at x_0 if, for every neighborhood $W \in \tau_Y$ of $f(x_0)$, there is a neighborhood $U \in \tau_X$ of x_0 such that $f(U) \subset W$.*

Theorem 1.7.6 (Characterization of Continuity)**.** *Suppose (X, τ_X) and (Y, τ_Y) are topological spaces and $f\colon X \to Y$. Then, the following statements are equivalent.*

(a) f is continuous.

(b) For any $A \subset X$, $f(\overline{A}) \subset \overline{f(A)}$.

(c) For every $x \in X$, f is continuous at x.

(d) For every closed set C of (Y, τ_Y), $f^{-1}(C)$ is closed in (X, τ_X).

Proof. We show that $(a) \implies (b) \implies (c) \implies (d) \implies (a)$.

$[(a) \implies (b)]$ Suppose f is continuous and fix $A \subset X$. Suppose, to the contrary, that there exists $y \in f(\overline{A}) \cap [Y/\overline{f(A)}]$.

Since $y \in f(\overline{A})$, $y = f(x)$ for some $x \in \overline{A}$. Since closures are closed, $Y/\overline{f(A)}$ is a neighborhood of y, where $f^{-1}(Y/\overline{f(A)}) = X/f^{-1}(\overline{f(A)})$ is open in (X, τ_X) by the continuity of f.

Particularly, $X/f^{-1}(\overline{f(A)})$ is a neighborhood of x, which ensures that A intersects $X/f^{-1}(\overline{f(A)})$ non-trivially, for $x \in \overline{A}$ by assumption.

That is, there exists $a \in A$ such that $f(a) \notin \overline{f(A)}$, which is a contradiction since $f(a) \in f(A) \subset \overline{f(A)}$ by definition. Indeed, $f(\overline{A}) \subset \overline{f(A)}$.

Alternative proof: Fix $A \subset X$ and $x \in \overline{A}$. It remains to show that $f(x) \in \overline{f(A)}$.

Fix a neighborhood $V \in \tau_Y$ of $f(x)$. The continuity of f ensures x has the neighborhood $f^{-1}(V)$. Since $x \in \overline{A}$, $A \cap f^{-1}(V)$ is non-empty.

Particularly, there is some $a \in A$ with $f(a) \in V$. That is, $f(A) \cap V$ is non-trivial. As a result, $f(x) \in \overline{f(A)}$. That is, we showed $f(\overline{A}) \subset \overline{f(A)}$.

$[(b) \implies (c)]$ Suppose for any $A \subset X$, $f(\overline{A}) \subset \overline{f(A)}$. Fix $x \in X$ and a neighborhood $V \in \tau_Y$ of $f(x)$, and consider $A := f^{-1}(Y/V)$. By assumption,

$$f(\overline{A}) \subset \overline{f(A)} \subset \overline{Y/V} = Y/V, \qquad \text{(Since } \forall S \subset Y \ f(f^{-1}(S)) \subset S \text{ and } Y/V \text{ is closed)}$$

which implies that $\overline{A} \subset f^{-1}(Y/V) =: A$, so that A is closed.

Particularly, $X/A \in \tau_X$ is a neighborhood of x since $X/A = X/[X/f^{-1}(V)] = f^{-1}(V)$. Lastly,

$$f(X/A) = f(f^{-1}(V)) \subset V.$$

Alternatively, $\overline{A} \subset f^{-1}(Y/V) = X/f^{-1}(V)$ ensures that $x \in f^{-1}(V) \subset X/\overline{A}$. By the characterization of living in the closure, there exists a neighborhood $U \in \tau_X$ of x that intersects A trivially. That is, $x \in U \subset X/A = f^{-1}(V)$ with $f(U) \subset V$.

Alternative proof: Fix a closed set $B \subset Y$ and consider $A := f^{-1}(B)$.
Fix $x \in \overline{A}$. The assumption ensures that $f(x) \in \overline{f(A)} \subset \overline{f(f^{-1}(B))} \subset \overline{B} = B$, where the equality holds since B is closed by assumption.

That is, $x \in f^{-1}(B) =: A$ so that $\overline{A} \subset A$ and, thus, $\overline{A} = A$. Particularly, $A := f^{-1}(B)$ is closed as desired.

$[(c) \implies (d)]$ Suppose for every $x \in X$, f is continuous at x. Fix a closed set C in (Y, τ_Y). To show $f^{-1}(C)$ is closed, we show it contains its closure.

It suffices to show f maps $\overline{f^{-1}(C)}$ into $\overline{C} = C$. Fix $x \in \overline{f^{-1}(C)}$ and a neighborhood $V \in \tau_Y$ of $f(x)$. The continuity of f at x ensures there exists a neighborhood $U \in \tau_X$ of x such that $f(U) \subset V$, where $x \in \overline{f^{-1}(C)}$ ensures U intersects $f^{-1}(C)$ non-trivially.

Particularly, there exists $u \in U$ with $f(u) \in C \cap f(U)$. That is, C intersects V non-trivially since $f(U) \subset V$. Thus, $f(x) \in \overline{C} = C$, since C is closed. Particularly, $x \in f^{-1}(C)$.

That is, $\overline{f^{-1}(C)} \subset f^{-1}(C)$ as desired. Indeed, $\overline{f^{-1}(C)} = f^{-1}(C)$, so $f^{-1}(C)$ is closed.

$[(d) \implies (a)]$ Suppose for every closed set C of (Y, τ_Y), $f^{-1}(C)$ is closed in (X, τ_X).

For any open set $V \in \tau_Y$, its complement is closed and, thus, $X/f^{-1}(V) = f^{-1}(Y/V)$ is closed by assumption, which ensures that $f^{-1}(V) \in \tau_X$. By definition, f is continuous. □

Remark 1.7.1. *Statement (b) in the above theorem states precisely that, for every $A \subset X$, f maps A and the limit point of A into $f(A)$ and the limit points of $f(A)$.*

Corollary 1.7.7 (Characterization of Real Maps via ϵ-δ)**.** *Suppose $f\colon \mathbb{R} \to \mathbb{R}$. Then, f is continuous (relative to the standard topology of $\mathbb{R}$) $\iff$ for every $x_0 \in X$,*

$$\forall \epsilon > 0,\ \exists \delta > 0 \text{ such that } \forall x \in \mathbb{R},\ |x - x_0| < \delta \implies |f(x) - f(x_0)| < \epsilon.$$

1.7.3 Homeomorphism

Definition 1.7.3 (Homeomorphism)**.** *Suppose (X, τ_X) and (Y, τ_Y) ar topological spaces.*

(a) A homeomorphism from (X, τ_X) to (Y, τ_Y) is a continuous bijection $f\colon (X, \tau_X) \to (Y, \tau_Y)$ with a continuous inverse.

(b) The topological spaces (X, τ_X) and (Y, τ_Y) are homeomorphic and we write $(X, \tau_X) \cong (Y, \tau_Y)$ if there is a homeomorphism between the two topological spaces.

Example 1.7.8. *$\mathbb{R}$ equipped with the standard topology is homeomorphic to its subspace $(-\frac{\pi}{2}, \frac{\pi}{2})$ via the homeomorphism $f\colon \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$ defined by*

$$f(x) := \arctan(x)$$

Example 1.7.9. *Consider $f\colon [0, 1) \to S^1$ defined by*

$$f(t) := (\cos(2\pi t), \sin(2\pi t)),\ \forall t \in [0, 1).$$

The map f is a continuous bijection. Yet, its inverse F is not continuous, since $F^{-1}([0, \frac{1}{2}))$ is not open, for we have a bad point on S^1 at $(1, 0)$.

1.7.4 Constructing Continuous Functions

Theorem 1.7.10 (Constructing Continuous Functions)**.** *Suppose (X, τ_X), (Y, τ_Y), and (Z, τ_Z) are topological spaces.*

(a) Suppose $A \subset X$. Then, the inclusion map $i\colon A \to X$ defined by $i(a) := a\ \forall a \in A$ is continuous.

(b) Suppose $f\colon X \to Y$ and $g\colon Y \to Z$ are continuous. Then, $g \circ f\colon X \to Z$ is continuous.

(c) Suppose $f\colon X \to Y$ is continuous and $A \subset X$. Then, $f|_A\colon A \to Y$ is continuous.

(d) Suppose $f\colon X \to Y$ and $\mathcal{A} \subset \tau_X$ is an open cover of X such that $\forall A \in \mathcal{A}$, $f|_A\colon A \to Y$ is continuous. Then, f is continuous.

Proof. (a) For any $U \in \tau_X$, $i^{-1}(U) = U \cap A \in \tau_A$ by the definition of subspace topology. By definition, i is continuous.

(b) Fix $U \in \tau_Z$. We observe that $[g \circ f]^{-1}(U) = f^{-1}(g^{-1}(U))$. Since g is continuous, $g^{-1}(U) \in \tau_Y$. Since f is continuous, $f^{-1}(g^{-1}(U)) = [g \circ f]^{-1}(U) \in \tau_X$. That is, $g \circ f$ is continuous.

(c) Consider the inclusion map $i\colon A \to X$. We observe that $f|_A = f \circ i$, which is the composition of continuous functions. By part (b), $f|_A$ is continuous.

(d) Fix $V \in \tau_Y$. For any $A \in \mathcal{A}$, $f|_A$ is continuous by assumption, so ${f|_A}^{-1}(V)$ is open in the subspace (A, τ_A), which ensures that ${f|_A}^{-1}(V) = U_A \cap A$ for some $U_A \in \tau_X$ so ${f|_A}^{-1}(V) \in \tau_X$. Lastly, we compute

$$\begin{aligned} f^{-1}(V) =& f^{-1}(V) \cap X = f^{-1}(V) \cap \cup_{A \in \mathcal{A}} A \\ =& \cup_{A \in \mathcal{A}} [f^{-1}(V) \cap A] = \cup_{A \in \mathcal{A}} [{f|_A}^{-1}(V)], \end{aligned}$$

which is the union of open sets and, thus, an open set of (X, τ_X). By definition, f is continuous. □

Lemma 1.7.11 (Product Preserves Continuity)**.** *Suppose (X_1, τ_1), (X_2, τ_2), $(Y_1, \mathcal{T}_1)$, and $(Y_2, \mathcal{T}_2)$ are topological spaces with continuous maps $f_1 \colon (X_1, \tau_1) \to (Y_1, \mathcal{T}_1)$ and $f_2 \colon (X_2, \tau_2) \to (Y_2, \mathcal{T}_2)$. Then, the product map $f_1 \times f_2 \colon X_1 \times X_2 \to Y_1 \times Y_2$ defined by*

$$[f_1 \times f_2](x_1, x_2) := (f_1(x_1), f_2(x_2)),\ \forall (x_1, x_2) \in X_1 \times X_2,$$

is continuous.

Proof. For brevity, denote $f_1 \times f_2$ by p.

For any $V_1 \subset Y_1$ and $V_2 \subset Y_2$, $p^{-1}(V_1 \times V_2) = {f_1}^{-1}(V_1) \times {f_2}^{-1}(V_2)$ by definition.

Fix $V_1 \subset Y_1$ and $V_2 \subset Y_2$. For any $(x_1, x_2) \in X_1 \times X_2$,

$$\begin{aligned}(x_1, x_2) \in p^{-1}(V_1 \times V_2) &\iff p(x_1, x_2) \in V_1 \times V_2 \\ &\iff f_1(x_1) \in V_1 \text{ and } f_2(x_2) \in V_2 \\ &\iff x_1 \in {f_1}^{-1}(V_1) \text{ and } x_2 \in {f_2}^{-1}(V_2) \\ &\iff (x_1, x_2) \in {f_1}^{-1}(V_1) \times {f_2}^{-1}(V_2).\end{aligned}$$

Particularly, for any standard basis element $V_1 \times V_2$ of the product space $(Y_1 \times Y_2, \mathcal{P}_{Y_1 \times Y_2})$, $V_1 \in \mathcal{T}_1$ and $V_2 \in \mathcal{T}_2$, so the continuity of f_1 and f_2 ensures that $p^{-1}(V_1 \times V_2)$ remains a standard basis element and, thus an open set of the product space $(X_1 \times X_2, \mathcal{P}_{X_1 \times X_2})$.

By 1.7.5, p is continuous. □

Theorem 1.7.12 (Pasting Lemma)**.** *Suppose (X, τ_X) and (Y, τ_Y) are topological spaces and $\{A, B\}$ is a closed cover of X. If $f \colon A \to Y$ and $g \colon B \to Y$ are continuous maps with $f = g$ on $A \cap B$, then the map $h \colon X \to Y$ defined by*

$$h(x) := \begin{cases} f(x) & \text{if } x \in A \\ g(x) & \text{if } x \in B \end{cases},\ \forall x \in X,$$

is continuous.

Proof. Fix a closed set C of (Y, τ_Y). By the continuity of f and g, $f^{-1}(C)$ and $g^{-1}(C)$ are closed in the respective subspaces. Particularly, $f^{-1}(C) = C_f \cap A$ and $g^{-1}(C) = C_g \cap B$ for some closed sets C_f, C_g of (X, τ_X). As a result,

$$\begin{aligned}h^{-1}(C) =& h^{-1}(C) \cap X = [h^{-1}(C) \cap A] \cup [h^{-1}(C) \cap B] && \text{(Since } \{A, B\} \text{ covers } X) \\ =& f^{-1}(C) \cup g^{-1}(C) && \text{(By the construction of } h) \\ =& [C_f \cap A] \cup [C_g \cap B],\end{aligned}$$

which remains a closed set of (X, τ_X) by assumption. Indeed, h is continuous by the characterization of continuity via closed sets. □

Remark 1.7.2. *The pasting lemma is convenient in constructing paths, homotopies, and, thus, the fundamental groups.*

1.7.5 Exercises

Exercise 1.7.1 (Joint Continuity Implies Separate Continuity)**.** *Let $F\colon X \times Y \to Z$. We say that F is **continuous in each variable separately** if for each y' in Y, the map $h_{y'}\colon X \to Z$ defined by*

$$h_{y'}(x) := F(x, y'),\ \forall x \in X$$

is continuous and for each x' in X, the map $k_{x'}\colon y \to Z$ defined by

$$k_{x'}(y) := F(x', y),\ \forall y \in Y$$

is continuous. Show that if F is continuous, then F is continuous in each variable separately.

[Munkres §18 exercise 11]

Proof. Suppose F is continuous.

For every $(x, y) \in X \times Y$, F is continuous at (x, y), by Theorem 18.1 (Munkres 104), ensures that for any neighborhood $V \in \tau_Z$ of $F(x, y)$, there are neighborhoods $U_x \in \tau_X$ and $U_y \in \tau_Y$ of x and y, respectively, such that $F(U_x \times U_y) \subset V$.

> Fix $(x, y) \in X \times Y$ and fix a neighborhood $V \in \tau_Z$ of $F(x, y)$. By Theorem 18.1, F is continuous at (x, y) ensures (x, y) has a neighborhood $U \in \mathcal{P}_{X\times Y}$ with $F(U) \subset V$.
>
> By the definition of product topology, (x, y) has a neighborhood $U_x \times U_y \in \mathcal{P}_{X\times Y}$ contained in U. Particularly, x and y respectively have the neighborhoods $U_x \in \tau_X$ and $U_y \in \tau_Y$ with $F(U_x \times U_y) \subset F(U) \subset V$.

The above shows, for each $y' \in Y$, that $h_{y'}$ is continuous at every $x \in X$ and, thus, continuous by Theorem 18.1 (Munkres 104).

> Fix $y' \in Y$. Moreover, fix $x \in X$ and a neighborhood $V \in \tau_Z$ of $h_{y'}(x)$.
>
> Since $h_{y'}(x) := F(x, y')$, V is a neighborhood of $F(x, y')$ ensures there are neighborhoods $U_x \in \tau_X$ and $U_{y'} \in \tau_Y$ of x and y', respectively, with $F(U_x \times U_{y'}) \subset V$. Particularly,
>
> $$h_{y'}(U_x) = F(U_x \times \{y'\}) \subset F(U_x \times U_{y'}) \subset V.$$
>
> That is, x has a neighborhood $U_x \in \tau_X$ with $h_{y'}(U_x) \subset V$, so that $h_{y'}$ is continuous at x. By Theorem 18.1 (Munkres 104), $h_{y'}$ is continuous.

Via a similar, if not identical, argument, for each $x' \in X$, $k_{x'}$ is continuous.

By definition, F is continuous in each variable separately. □

Definition 1.7.4 (Locally finite)**.** *Suppose (X, τ) is a topological space. A collection $\mathcal{A}$ of subsets of X is **locally finite** if each $x \in X$ has a neighborhood $U_x \in \tau$ intersecting only finitely many elements of $\mathcal{A}$ non-trivially.*

Exercise 1.7.2 (Continuity via Closed Cover and Local Continuity)**.** *Suppose (X, τ_X) and (Y, τ_Y) are topological spaces with $f\colon X \to Y$. Suppose $\mathcal{A}$ is a τ_X-closed cover of X such that $\forall A \in \mathcal{A}$, $f|_A\colon (A, \tau_A) \to (Y, \tau_Y)$ is continuous.*

(a) If $\mathcal{A}$ is finite, then f is continuous.

(b) If $\mathcal{A}$ is locally finite, then f is continuous.

[Munkres 18.9 (a) & (c); appeared on Spring 2025 comprehensive exam]

Proof. (a) Suppose $\mathcal{A}$ is finite and enumerate $\mathcal{A} = \{A_1, \dots, A_n\}$ for some $n \in \mathbb{N}$. Fix a closed subset C of (Y, τ_Y). Since $\mathcal{A}$ covers X by assumption,

$$\begin{aligned} f^{-1}(C) =& f^{-1}(C) \cap X = \cup_{j=1}^{n} [f^{-1}(C) \cap A_j] \\ =& \cup_{j=1}^{n} [f|_{A_j}]^{-1}(C). \qquad \text{(By the definition of preimage)} \end{aligned}$$

For each $j \in \{1, \dots, n\}$, the continuity of $f|_{A_j}$ ensures $[f|_{A_j}]^{-1}(C)$ remains closed in the subspace (A_j, τ_{A_j}) and, therefore, in (X, τ_X) as A_j is closed by assumption.

Particularly, $f^{-1}(C)$ is the finite union of closed sets of (X, τ_X) and, thus, a closed set of (X, τ_X). By the characterization of continuity, f is continuous.

(b) Suppose $\mathcal{A}$ is locally finite. To show the desired continuity, it suffices to fix $x \in X$ and show that f is continuous at x.

By the local finiteness of $\mathcal{A}$, $x \in X$ has a neighborhood $U_x \in \tau$ intersecting only finitely many elements of $\mathcal{A}$, say $A_1, \dots, A_n \in \mathcal{A}$ for some $n \in \mathbb{N}$, non-trivially. Particularly, $\mathcal{A}$ covers X ensures that $U_x \subset \cup_{j=1}^{n} A_j$.

Consider $X' := \cup_{j=1}^{n} A_j$ and $F\colon X' \to Y$ defined by $F := f|_{X'}$. By part (a), the compatibility of subspace topologies (1.5.1) and the continuity of $f|_{A_1}, \dots, f|_{A_n}$ ensure that F remains continuous relative to the subspace $(X', \tau_{X'})$ of (X, τ_X) and (Y, τ_Y). Particularly, F is continuous at x.

> Consider the subspace $(X', \tau_{X'})$ of (X, τ_X) and fix $j \in \{1, \dots, n\}$.
>
> Since A_j is a closed subset of (X, τ_X), $A_j = A_j \cap X'$ remains a closed subset of $(X', \tau_{X'})$. By the compatibility of subspaces (1.5.1), the subspace topologies $(\tau_{X'})_{A_j}$ and τ_{A_j} on A_j respectively inherited from $\tau_{X'}$ and τ_X coincide.
>
> Particularly, for any $V \in \tau_Y$, $\left[F|_{A_j}\right]^{-1}(V) = \left[f|_{A_j}\right]^{-1}(V) \in \tau_{A_j}$ by the continuity of $f|_{A_j}$, so that $\left[F|_{A_j}\right]^{-1}(V) \in (\tau_{X'})_{A_j}$. That is, $F|_{A_j}\colon (A_j, (\tau_{X'})_{A_j}) \to (Y, \tau_Y)$ is continuous.
>
> Particularly, $\mathcal{A}' := \{A_1, \dots, A_n\}$ is a finite $\tau_{X'}$-closed cover of X' such that $\forall A \in \mathcal{A}'$, $F|_A\colon (A, (\tau_{X'})_A) \to (Y, \tau_Y)$ is continuous.
>
> That is, the hypotheses of part (a) are satisfied for $F\colon (X', \tau_{X'}) \to (Y, \tau_Y)$ with the $\tau_{X'}$-closed cover $\mathcal{A}'$, so that F is continuous.

Fix a neighborhood $V \in \tau_Y$ of $f(x)$. Since $F(x) := f(x)$, V remains a neighborhood of $F(x)$, so there exists $U_F \in \tau_X$ such that $x \in U_F \cap X'$ with $F(U_F \cap X') = f(U_F \cap X') \subset V$.

Lastly, x has the neighborhood $U_F \cap U_x \in \tau_X$, where $U_x \subset X'$ ensures that $U_F \cap U_x \subset U_F \cap X'$ and, thus,

$$f(U_F \cap U_x) \subset f(U_F \cap X') \subset V.$$

By definition, f is continuous at x. By the characterization of continuity, f is continuous. □

Example 1.7.13 (Munkres 18.9 (b))**.** *We provide an example in which the statement of [Munkres 18.9 (a)] fails, given that we remove the finiteness of the closed cover $\mathcal{A}$.* ***(Appeared on Spring 2025 comprehensive exam)***

Consider $X := \mathbb{N}$ and $Y := \{0, 1\}$ equipped with the finite complement topology τ and the discrete topology $\mathcal{P}(Y)$, respectively. Define $f \colon X \to Y$ by

$$f(x) := \begin{cases} 1 & \text{if } x \text{ is odd} \\ 2 & \text{if } x \text{ is even} \end{cases}, \ \forall x \in X.$$

By definition, $C \subset X$ is closed $\iff$ X/C is open $\iff$ C is finite or $C = X$. We observe that $\{1\}$ is closed in $(Y, \mathcal{P}(Y))$, yet $f^{-1}(\{1\}) = \{2n + 1 : n \in \mathbb{W}\} \neq X$ and is not finite.

That is, $f^{-1}(\{1\})$ is not closed in (X, τ). Particularly, f is not continuous by the characterization of continuity.

Consider the countable cover $\mathcal{A} := \{\{n\} : n \in \mathbb{N}\}$ of X. Since singletons are finite, $\mathcal{A}$ consists of closed subsets of X. Fix $\{n\} \in \mathcal{A}$ and it remains to show $f|_{\{n\}}$ is continuous.

Recall that $\mathcal{P}(Y) = \{\emptyset, Y, \{1\}, \{2\}\}$. We see that ${f|_{\{n\}}}^{-1}(\emptyset) = \emptyset$ and ${f|_{\{n\}}}^{-1}(Y) = \{n\}$ are open in the subspace $(\{n\}, \tau_{\{n\}})$, where

$${f|_{\{n\}}}^{-1}(\{1\}) = f^{-1}(\{1\}) \cap \{n\} = \begin{cases} \{n\} & \text{if } n \text{ is odd} \\ \emptyset & \text{if } n \text{ is even} \end{cases}$$

and, via a similar computation, ${f|_{\{n\}}}^{-1}(\{2\})$ also remain open in the subspace. By definition, $f|_{\{n\}}$ is continuous.

Exercise 1.7.3 (Continuous Extension to Closure is Unique)**.** *Suppose (X, τ_X) and (Y, τ_Y) are topological spaces, $A \subset X$, (Y, τ_Y) is Hausdorff, and $f \colon A \to Y$ is continuous. If f has a continuous extension $g \colon \overline{A} \to Y$, then g is unique.*

[Munkres 18.13]

Proof. Suppose f has continuous extensions $g, h \colon \overline{A} \to Y$. It remains to show $g = h$.

Suppose, to the contrary, that $g \neq h$, so there exists $x \in \overline{A}$ such that $g(x) \neq h(x)$. Since g and h extends f, g and h coincide with f on A and, thus, coincide on A, which ensures that $x \in \overline{A}/A$.

Since (Y, τ_Y) is Hausdorff, there exist disjoint neighborhoods $G, H \in \tau_Y$ of $g(x)$ and $h(x)$, respectively. By the continuity of g and h, x has the neighborhoods $g^{-1}(G), h^{-1}(H) \in \tau_{\overline{A}}$.

By the definition of subspace topology, $g^{-1}(G) = U_G \cap \overline{A}$ and $h^{-1}(H) = U_H \cap \overline{A}$ for some $U_G, U_H \in \tau_X$. That is, x has the neighborhood $U_G \cap U_H \in \tau_X$, so $U_G \cap U_H \cap A$ is non-empty, say with element y, since $x \in \overline{A}$.

Particularly, $y \in A$ ensures $g(y) = f(y) = h(y)$, where $y \in U_G \cap \overline{A} = g^{-1}(G)$ implies that $g(y) = h(y) \in G$. Similarly, $h(y) \in H$ since $y \in U_H \cap \overline{A} = h^{-1}(H)$. That is, we showed $h(y) \in G \cap H$, which contradicts the disjointness of G and H.

We conclude that $g = h$, so that the continuous extension of f to $\overline{A}$ is unique. □

1.7.6 *Continuity of Vector Operations

Remark 1.7.3. *The following characterizations were given in Rudin's Functional Analysis. Although seemingly unfamiliar, they follow immediately from the continuity theorem. We derive them here for some practice.*

Lemma 1.7.14 (Continuous Maps from Product Space)**.** *Let $(X_1, \tau_1), \ldots, (X_3, \tau_3)$ be topological spaces and $\Lambda\colon X_1 \times X_2 \to X_3$. Then, Λ is continuous $\iff$ for every $(x_1, x_2) \in X_1 \times X_2$, for every neighborhood $U \in \tau_3$ of $\Lambda(x_1, x_2)$, there are neighborhoods $V_1 \in \tau_1$ of x_1 and $V_2 \in \tau_2$ of x_2 such that $\Lambda(V_1 \times V_2) \subset U$.*

Proof. We observe the chain of equivalences: Λ is continuous

$\iff$ for every $(x_1, x_2) \in X_1 \times X_2$, Λ is continuous at (x_1, x_2)

$\iff$ for every $(x_1, x_2) \in X_1 \times X_2$, for every neighborhood $U \in \tau_3$ of $\Lambda(x_1, x_2)$, there is a neighborhood $V \in \mathcal{P}_{X_1 \times X_2}$ of (x_1, x_2) such that $\Lambda(V) \subset U$.

$\iff$ for every $(x_1, x_2) \in X_1 \times X_2$, for every neighborhood $U \in \tau_3$ of $\Lambda(x_1, x_2)$, there are neighborhoods $V_1 \in \tau_1$ of x_1 and $V_2 \in \tau_2$ of x_2 such that $\Lambda(V_1 \times V_2) \subset U$.

Notice that the last equivalence holds since the product topology $\mathcal{P}_{X_1 \times X_2}$ is, by definition, generated by the basis $\{V_1 \times V_2 : V_1 \in \tau_1, V_2 \in \tau_2\}$. □

Proposition 1.7.15 (Characterization of Continuous Vector Addition)**.** *Suppose X is a vector space with a topology τ. Then, the vector addition $A\colon X \times X \to X$ of X is continuous $\iff$ for any $x, y \in X$, for any neighborhood $U \in \tau$ of $x + y$, there are neighborhoods $V_x, V_y \in \tau$ of x and y such that $V_x + V_y \subset U$.*

Proof. Since $A(V_x \times V_y) := V_x + V_y$ for any $V_x, V_y \subset X$, this proposition follows immediately from the preceding lemma.

Fix $V_x, V_y \subset X$. For any $(x, y) \in V_x \times V_y$, $A(x, y) := x + y \in V_x + V_y$. Particularly, for any $v \in X$,

$$v \in A(V_x \times V_y) \iff v = A(x, y) \text{ for some } x \in V_x, y \in V_y$$
$$\iff v = x + y \text{ for some } x \in V_x, y \in V_y \iff v \in V_x + V_y.$$

□

Proposition 1.7.16 (Characterization of Continuous Scalar-Vector Multiplication)**.** *Suppose X is a vector space, over field $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$, with a topology τ. Then, the scalar-vector multiplication $M\colon \mathbb{F} \times X \to X$ of X is continuous $\iff$ for any $(\alpha, x) \in \mathbb{F} \times X$, for any neighborhood $U \in \tau$ of αx, there are neighborhoods $V_x \in \tau$ of x and $B_\delta(\alpha)$ of α for some $\delta > 0$ such that*

$$\forall \beta \in B_\delta(\alpha),\ \beta V_x \subset U.$$

Proof. Denote by $\tau_{\mathbb{F}}$ the standard topology on $\mathbb{F}$. For any $\delta > 0$,$\alpha \in \mathbb{F}$, $B_\delta(\alpha)$ is the open ball

$$B_\delta(\alpha) := \{\beta \in \mathbb{F} : |\beta - \alpha| < \delta\}.$$

By a preceding lemma, M is continuous $\iff$ for every $(\alpha, x) \in \mathbb{F} \times X$, for every neighborhood $U \in \tau$ of αx, there are neighborhoods $V_x \in \tau$ of x and $B_\alpha \in \tau_{\mathbb{F}}$ of α such that $M(B_\alpha \times V_x) \subset U$,

which is equivalent to for every $(\alpha, x) \in \mathbb{F} \times X$, for every neighborhood $U \in \tau$ of αx, there are neighborhoods $V_x \in \tau$ of x and $B_\delta(\alpha)$ of α for some $\delta > 0$ such that $\forall \beta \in B_\delta(\alpha),\ \beta V_x \subset U$.

We verify the last equivalence, which is very straightforward.

[$\Longrightarrow$] Suppose for every $(\alpha, x) \in \mathbb{F} \times X$, for every neighborhood $U \in \tau$ of αx, there are neighborhoods $V_x \in \tau$ of x and $B_\alpha \in \tau_{\mathbb{F}}$ of α such that $M(B_\alpha \times V_x) \subset U$.

Fix $(\alpha, x) \in \mathbb{F} \times X$ and a neighborhood $U \in \tau$ of αx, so there are neighborhoods $V_x \in \tau$ of x and $B_\alpha \in \tau_{\mathbb{F}}$ of α such that $M(B_\alpha \times V_x) \subset U$. Since $\alpha \in B_\alpha$, there exists $\delta > 0$ such that $\alpha \in B_\delta(\alpha) \subset B_\alpha$ by the definition of $\tau_{\mathbb{F}}$. Particularly, $\forall \beta \in B_\delta(\alpha)$,

$$\begin{aligned} \beta \in B_\alpha &\implies \forall x \in V_x,\ (\beta, x) \in B_\alpha \times V_x \\ &\implies \forall x \in V_x,\ M(\beta, x) = \beta x \in U \qquad \text{(By assumption)} \\ &\implies \beta V_x \subset U. \end{aligned}$$

[$\Longleftarrow$] Suppose for every $(\alpha, x) \in \mathbb{F} \times X$, for every neighborhood $U \in \tau$ of αx, there are neighborhoods $V_x \in \tau$ of x and $B_\delta(\alpha)$ of α for some $\delta > 0$ with $\forall \beta \in B_\delta(\alpha),\ \beta V_x \subset U$.

Fix $(\alpha, x) \in \mathbb{F} \times X$ and a neighborhood $U \in \tau$ of αx, so there are neighborhoods $V_x \in \tau$ of x and $B_\delta(\alpha)$ of α for some $\delta > 0$ with $\forall \beta \in B_\delta(\alpha),\ \beta V_x \subset U$.

By the definition of $\tau_{\mathbb{F}}$, $B_\delta(\alpha)$ is a neighborhood of α. Now, for any $(\beta, x) \in B_\delta(\alpha) \times V_x$, $\beta V_x \subset U$ ensures that $M(\beta, x) = \beta x \in U$. That is, $M(B_\delta(\alpha) \times V_x) \subset U$ as desired.

□

1.7.7 *Topological Group

Definition 1.7.5 (Group)**.** *A group is an ordered pair $(G, *)$ consisting of a set G and a map $*\colon G \times G \to G$ satisfying*

(a.1) associativity ($\forall a, b, c \in G$, $(a * b) * c = a * (b * c)$),

(a.2) there exists an identity element $e_G \in G$ such that $\forall a \in G$, $a * e_G = e_G * a = a$, and

(a.3) for every $a \in G$, there is an inverse element $a^{-1} \in G$ of a satisfying

$$a * a^{-1} = a^{-1} * a = e_G.$$

If no ambiguity may arise, we

(a) refer to a group $(G, *)$ by its underlying set G;

(b) denote the group operation of a group by $*$;

(b) denote, for any $a, b \in G$, $a * b$ more succinctly by $a \cdot b$ or ab;

(c) denote an identity element of $(G, *)$ by e or 1.

Definition 1.7.6 (Topological group)**.** *A topological group is a group G equipped with a topology τ such that*

(a) (G, τ) is T_1,

(b) the group operation $*\colon G \times G \to G$ of G is continuous, and

(c) the inverse operation $\mathrm{Inv}\colon G \to G$ defined by $\mathrm{Inv}(g) := g^{-1}$, $\forall g \in G$, is continuous.

Exercise 1.7.4 (Characterization of Topological Group)**.** *Suppose G is a group with a topology τ such that (G, τ) is T_1. Then, G is a topological group $\iff$ the map $\varphi\colon G \times G \to G$ defined by*

$$\varphi(x, y) := xy^{-1}, \ \forall (x, y) \in G \times G$$

is continuous. ***[Munkres *22.1]***

Proof. **[** $\Longrightarrow$ **]** Suppose (G, τ) is a topological group. By definition, the group operations $*$ and Inv are continuous. Consider the identity map Id on G, which is continuous relative to (G, τ).

Since product maps preserve continuity (1.7.11), the map $\mathrm{Id} \times \mathrm{Inv}\colon G \times G \to G \times G$ defined by

$$[\mathrm{Id} \times \mathrm{Inv}](x, y) := (x, y^{-1})$$

remains continuous relative to the product space $(G \times G, \mathcal{P}_{G \times G})$. Since composition preserves continuity (1.7.10, the map $* \circ [\mathrm{Id} \times \mathrm{Inv}]$ remains continuous. Since φ is precisely the composition $* \circ [\mathrm{Id} \times \mathrm{Inv}]$, φ is continuous.

[$\Longleftarrow$ **]** Suppose φ is continuous. By 1.7.1, φ is separately continuous. Particularly, the map $\varphi_e\colon G \to G$ defined by $\varphi_e(y) := \varphi(e, y)$ is continuous. That is, Inv is continuous since

$$\forall y \in G, \ \varphi_e(y) = ey^{-1} = y^{-1} =: \mathrm{Inv}(y).$$

As a result, the product map $\mathrm{Id} \times \mathrm{Inv}\colon G \times G \to G$ again remains continuous by 1.7.11. Since composition preserves continuity (1.7.10), we conclude that $*$ is continuous as

$$\forall x, y \in G, \ [\varphi \circ [\mathrm{Id} \times \mathrm{Inv}]](x, y) = \varphi(x, y^{-1}) = x[y^{-1}]^{-1} = xy$$

ensures that $* = \varphi \circ [\mathrm{Id} \times \mathrm{Inv}]$. By definition, (G, τ) is a topological group. □

Exercise 1.7.5 (Topological Subgroup)**.** *Suppose (G,τ) is a topological group with a subgroup $H \subset G$. Then, $\overline{H}$ is a subgroup of G, and (H,τ_H) and $(\overline{H},\tau_{\overline{H}})$ are topological groups.*

[Munkres *22.3]

Proof. By a preceding characterization of topological group, the map $\varphi\colon G\times G\to G$ defined by

$$\varphi(x,y) := xy^{-1},\ \forall x,y\in G$$

is continuous.

$[(H,\tau_H)$ is a topological group] Via the above characterization, it remains to prove that the restriction $\varphi'\colon H\times H\to H$ of φ to $H\times H$ remains continuous, relative to the product topology on $H\times H$ induced by the subspace topology of H.

Since H is a subgroup of G by assumption, H is closed under the group operations $*$ and Inv of G. Particularly, φ' is well-defined. For any open set $U\in\tau$, we compute

$$\begin{aligned}\varphi'^{-1}(U\cap H) =&\varphi^{-1}(U\cap H)\cap[H\times H]\\ =&\varphi^{-1}(U)\cap\varphi^{-1}(H)\cap[H\times H]\\ =&\varphi^{-1}(U)\cap\varphi'^{-1}(H)=\varphi^{-1}(U)\cap[H\times H],\end{aligned}$$

where the continuity of φ ensures that $\varphi^{-1}(U)$ is open relative to the product topology of $G\times G$. That is, φ' is continuous relative to the subspace topology of $H\times H$ by definition and, thus, the product topology of $H\times H$ by 1.5.4.

Indeed, (H,τ_H) is a topological group by the mentioned characterization.

$[(\overline{H},\tau_{\overline{H}})$ is a topological group] For this implication, it suffices to show that $\overline{H}$ is a subgroup of G and apply the preceding part. To achieve so, it suffices to show that φ maps $\overline{H}\times\overline{H}$ into $\overline{H}$.

Fix $(h_1,h_2)\in\overline{H}\times\overline{H}$ and a neighborhood $V\in\tau$ of $\varphi(h_1,h_2)$. The continuity of φ ensures that φ is continuous at (h_1,h_2), so there exist neighborhoods $U_1,U_2\in\tau$ of h_1 and h_2, respectively, such that $\varphi(U_1\times U_2)\subset V$.

Particularly, for $j\in\{1,2\}$, $h_j\in\overline{H}$ ensures that $U_j\cap H$ is non-empty, say with an element g_j. Since H is a subgroup by assumption, $\varphi(g_1,g_2)$ remains elements of H. Particularly,

$$\varphi([U_1\cap H]\times[U_2\times H])\subset\varphi(U_1\times U_2)\subset V$$

ensures that $\varphi(g_1,g_2)\in U\cap H$, so that $V\cap H$ is non-empty. As a result, $\varphi(h_1,h_2)\in\overline{H}$,. That is, $\overline{H}$ is closed under φ, which ensures that $\overline{H}$ is a subgroup of G and, thus, a topological group by the preceding part.

> **$[\overline{H}$ is a subgroup]** Since H is a subgroup of G, $e_G\in H\subset\overline{H}$. For any $h\in\overline{H}$, $(e_G,h)\in\overline{H}\times\overline{H}$ ensures $\varphi(e_G,h)=e_Gh^{-1}=h^{-1}\in\overline{H}$. For any $h_1,h_2\in\overline{H}$, $h_1,{h_2}^{-1}\in\overline{H}$ ensures that $\varphi(h_1,{h_2}^{-1})=h_1h_2\in\overline{H}$.
>
> That is, $\overline{H}$ is closed under Inv and $*$. Indeed, $\overline{H}$ is a subgroup of G.

□

Exercise 1.7.6 (Topological Group is Homogeneous)**.** *Suppose (G,τ) is a topological group.*

(a) Then, for any $a \in G$, the maps $f_a, g_a \colon G \to G$ defined by

$$f_a(x) := ax \text{ and } g_a(x) := xa, \ \forall x \in G$$

are homeomorphism.

(a.1) Particularly, for any $x, y \in G$, there is a homeomorphism $\psi \colon G \to G$ with $\psi(x) = y$.

[Munkres *22.4]

Proof. Fix $a \in G$. The maps f_a, g_a have the respective inverses ${f_a}^{-1}, {g_a}^{-1} \colon G \to G$ given by

$$ {f_a}^{-1}(y) := a^{-1}y \text{ and } {g_a}^{-1}(y) := ya^{-1}, \ \forall y \in G.$$

For any $y, x \in G$,

$$\begin{aligned} f_a({f_a}^{-1}(y)) :=& a{f_a}^{-1}(y) = aa^{-1}y = y, \\ {f_a}^{-1}(f_a(x)) :=& a^{-1}f_a(x) = a^{-1}ax = x, \\ g_a({g_a}^{-1}(y)) :=& {g_a}^{-1}(y)a = ya^{-1}a = y, \text{ and} \\ {g_a}^{-1}(g_a(x)) :=& g_a(x)a^{-1} = xaa^{-1} = x. \end{aligned}$$

Indeed, ${f_a}^{-1}$ and ${g_a}^{-1}$ are the respective inverses of f_a, g_a.

Particularly, f_a and g_a are bijective.

Since (G, τ) is a topological group, the inverse operation Inv of G is continuous, where a preceding characterization ensures $\varphi \colon G \times G \to G$ defined by $\varphi(x, y) := xy^{-1}$, $\forall x, y \in G$, is continuous.

Particularly, φ is separately continuous by 1.7.1, so, $\forall b \in G$, the maps ${}_b\varphi, \varphi_b \colon G \to G$ defined by

$${}_b\varphi(x) := \varphi(b, x) := bx^{-1} \text{ and } \varphi_b(x) := \varphi(x, b) := xb^{-1}, \ \forall x \in G,$$

are continuous. Since

$$\begin{aligned} f_a = {}_a\varphi \circ \mathrm{Inv}, \ g_a = \varphi_{a^{-1}} \circ \mathrm{Id}, \\ {f_a}^{-1} = {}_{a^{-1}}\varphi \circ \mathrm{Inv}, \text{ and } {g_a}^{-1} = \varphi_a \circ \mathrm{Id} \end{aligned}$$

are compositions of continuous maps and, thus, continuous by 1.7.10, we conclude f_a and g_a are homeomorphisms by definition.

(a.1) Fix $x, y \in G$ and consider $a := yx^{-1}$. By the preceding part, f_a is a homeomorphism from G onto G, where $f_a(x) := ax = yx^{-1}x = y$ as desired. □

Definition 1.7.7 (Product and inverse of set)**.** *Suppose G is a group. For any $U, V \subset G$,*

(a) $U \cdot V := \{uv : u \in U, v \in V\}$;

(a.1) if $x \in G$, $U \cdot x := U \cdot \{x\}$ and $x \cdot U := \{x\} \cdot U$;

(b) $U^{-1} := \left\{u^{-1} : u \in U\right\}$.

Definition 1.7.8 (Symmetric subset of a group)**.** *Suppose G is a group. Then, a symmetric subset of G is a subset $V \subset G$ such that $V = V^{-1}$.*

Lemma 1.7.17 (Product and Inverse Inherits Openness)**.** *Suppose (G, τ) is a topological group. If $U \subset G$ is open, then*

(a) U^{-1} is open, and

(b) for any $V \subset G$, $U \cdot V$ and $V \cdot U$ are open.

Proof. Fix an open set $U \subset G$. By definition, Inv is continuous, so $\text{Inv}^{-1}(U)$ remains open. Particularly, $\text{Inv}^{-1}(U) = U^{-1}$ ensures that U^{-1} is open.

For any $x \in G$,

$$x \in \text{Inv}^{-1}(U) \iff x^{-1} \in U \iff x \in U^{-1}.$$

Fix $a \in G$. A preceding exercise ensures that the map $g_a \colon G \to G$ defined by

$$g_a(x) := xa, \ \forall x \in G,$$

is a homeomorphism. Since $U \cdot a = g_a(U) = G^{-1}(U)$, $U \cdot a$ remains open by the continuity of the inverse G of g_a. Particularly, $U \cdot V = \cup_{a \in V}[U \cdot a]$ implies $U \cdot V$ is a union of open sets and, thus, an open set of (G, τ). Via a symmetric argument, $V \cdot U$ is open as well. □

Theorem 1.7.18 (Separation Properties of Topological Group)**.** *Suppose (G, τ) is a topological group. Then, the following statements hold.*

(a) If U is a neighborhood of e, then there exists a symmetric neighborhood V of e such that $V \cdot V \subset U$.

(b) For distinct $x, y \in G$, there exists a (symmetric) neighborhood V of e such that $V \cdot x$ and $V \cdot y$ are disjoint.

(b.1) Particularly, (G, τ) is Hausdorff.

(c) For any closed subset $A \subset G$ and $x \in G/A$, there exists a (symmetric) neighborhood V of e such that $V \cdot A$ and $V \cdot x$ are disjoint.

(c.1) Particularly, (G, τ) is regular.

[Munkres *22.7 (a-c)]

Proof. (a) Fix a neighborhood U of e. By definition, $*$ is continuous and, thus, continuous at (e, e). Since $e * e = e$, U is a neighborhood of $e * e$, there exist neighborhoods V_1, V_2 of e such that $V_1 \cdot V_2 \subset U$.

Consider $V := (V_1 \cap V_2) \cap (V_1 \cap V_2)^{-1}$, which remains a neighborhood of e by a preceding lemma. By construction, $V \subset V_1$ and $V \subset V_2$ ensures $V \cdot V \subset V_1 \cdot V_2 \subset U$. Hence, it remains to show V is symmetric.

For any $x \in G$, we observe

$$\begin{aligned} x \in V &\iff x \in V_1 \cap V_2 \text{ and } x \in (V_1 \cap V_2)^{-1} \\ &\iff x^{-1} \in (V_1 \cap V_2)^{-1} \text{ and } x^{-1} \in (V_1 \cap V_2) \\ &\iff x^{-1} \in V \iff x \in V^{-1}. \end{aligned}$$

Indeed, V is symmetric by definition.

(b) Fix distinct elements $x, y \in G$. Since (G, τ) is T_1 by definition, there are neighborhoods U_x and U_y of x and y, respectively, such that $U_x \subset X/\{y\}$ and $U_y \subset X/\{x\}$.

By a preceding lemma, $U_x \cdot x^{-1}$ and $U_y \cdot y^{-1}$ are neighborhoods of e, which yields the neighborhood $U_e := [U_x \cdot x^{-1}] \cap [U_y \cdot y^{-1}]$ of e.

By part (a), there is a symmetric neighborhood V of e such that $V \cdot V \subset U_e$. Particularly, $e \in V$ ensures that $V \cdot x$ and $V \cdot y$ are neighborhood of x and y, respectively.

Suppose, to the contrary, that $[V \cdot x] \cap [V \cdot y]$ is non-empty, say with an element z. It follows that $z = v_1 x = v_2 y$ for some $v_1, v_2 \in V$. Particularly,

$$\begin{aligned} x = {v_1}^{-1} v_2 y \in & [V \cdot V] \cdot y && \text{(Since } V \text{ is symmetric)} \\ \subset & U_e \cdot y \subset [U_y \cdot y^{-1}] \cdot y && \text{(By construction)} \\ = & U_y \subset X/\{x\}, && \text{(By construction)} \end{aligned}$$

which is a contradiction. Indeed, $V \cdot x$ and $V \cdot y$ are disjoint neighborhoods of x and y, respectively. By definition, (G, τ) is Hausdorff.

(c) Fix a closed set $A \subset G$ and a point $x \in G/A$. It remains to show there are disjoint neighborhoods of A and x.

Since G/A is open by definition, $[G/A] \cdot x^{-1}$ is a neighborhood of e. Let V be a symmetric neighborhood of e with $V \cdot V \subset [G/A] \cdot x^{-1}$.

Since V is a neighborhood of e, $V \cdot A$ and $V \cdot x$ are neighborhoods of A and x, respectively. Suppose, to the contrary, that $[V \cdot A] \cap [V \cdot x]$ is non-empty, say with an element z. It follows that $z = v_1 a = v_2 x$ for some $v_1, v_2 \in V$ and $a \in A$. Particularly,

$$a = {v_1}^{-1} v_2 x \in [V \cdot V] \cdot x \subset ([G/A] \cdot x^{-1}) \cdot x = G/A, \qquad \text{(Since } V \text{ is symmetric)}$$

which is a contradiction. Indeed, $V \cdot A$ and $V \cdot x$ are disjoint. By definition, (G, τ) is regular. □

Remark 1.7.4. *The arguments presented in this section may be applied almost verbatim to yield analogous separation properties for topological vector spaces.*

1.8 *Initial Topology

Remark 1.8.1. *An initial topology is a type of topology on a set X induced by a collection of map(s) on X with topologized codomains.*

Examples of such include the subspace topology induced by the inclusion map and the product topology induced by the canonical projections, which we studied.

As we will prove, an initial topology induced by a collection of maps is the smallest topology making these maps continuous.

We study initial topologies very briefly to gain some familiarity, in the hope that I can make some sense of weak topologies of topological vector spaces and topologies on vector spaces induced by a separating family of semi-norms.

Lemma 1.8.1. *Suppose X is a set and $\{(Y_\alpha, \tau_\alpha)\}_{\alpha\in I}$ is a collection of topological spaces with a collection of maps $\mathcal{F} := \left\{f_\alpha \colon X \to Y_\alpha\right\}_{\alpha\in I}$ for some index set I. Then,*

$$\mathcal{S}(\mathcal{F}) := \cup_{\alpha\in I} \left\{{f_\alpha}^{-1}(U_\alpha) : U_\alpha \in \tau_\alpha\right\}$$

is a subbasis of X

Proof. Let $\alpha \in I$. Since $Y_\alpha \in \tau_\alpha$, $X = {f_\alpha}^{-1}(Y_\alpha) \in \mathcal{S}(\mathcal{F})$. Particularly, $\mathcal{S}(\mathcal{F})$ covers X. By definition, $\mathcal{S}(\mathcal{F})$ is a subbasis of X. □

Definition 1.8.1 (Initial topology)**.** *Suppose X is a set and $\{(Y_\alpha, \tau_\alpha)\}_{\alpha\in I}$ is a collection of topological spaces with a collection of maps $\mathcal{F} := \left\{f_\alpha \colon X \to Y_\alpha\right\}_{\alpha\in I}$ for some index set I. Then, the **($\mathcal{F}$-topology) initial topology** $\tau(\mathcal{F})$ **on** X **induced by** $\mathcal{F}$ is the topology on X generated by the subbasis of X*

$$\mathcal{S}(\mathcal{F}) := \cup_{\alpha\in I} \left\{{f_\alpha}^{-1}(U_\alpha) : U_\alpha \in \tau_\alpha\right\}.$$

Lemma 1.8.2 (Minimality of Initial Topology)**.** *Suppose X is a set and $\{(Y_\alpha, \tau_\alpha)\}_{\alpha\in I}$ is a collection of topological spaces with a collection of maps $\mathcal{F} := \left\{f_\alpha \colon X \to Y_\alpha\right\}_{\alpha\in I}$ for some index set I. Then, the initial topology $\tau(\mathcal{F})$ is the smallest topology on X relative to which, $\forall\alpha \in I$, f_α is continuous.*

Proof. By definition, $\tau(\mathcal{F})$ is the topology generated by the subbasis $\mathcal{S}(\mathcal{F})$. Particularly, $\mathcal{S}(\mathcal{F}) \subset \tau(\mathcal{F})$ by 1.2.9.

For any $\alpha \in I$, for any open set $U_\alpha \in \tau_\alpha$, ${f_\alpha}^{-1}(U_\alpha) \in \mathcal{S}(\mathcal{F})$ ensures that ${f_\alpha}^{-1}(U_\alpha) \in \tau(\mathcal{F})$, so that f_α is continuous relative to $(X, \tau(\mathcal{F}))$ and (Y, τ_α) by definition.

Now, fix a topology τ on X such that, for any $\alpha \in I$, $f_\alpha \colon (X, \tau) \to (Y, \tau_\alpha)$ is continuous. By construction, $\mathcal{S}(\mathcal{F}) \subset \tau$. By the minimality of $\tau(\mathcal{F})$ as the smallest topology on X containing $\mathcal{S}(\mathcal{F})$ (1.2.1), $\tau \supset \tau(\mathcal{F})$. That is, $\tau(\mathcal{F})$ is the smallest topology on X relative to which, $\forall\alpha \in I$, f_α is continuous. □

Remark 1.8.2. *Notice that the following proposition is simply a generalization of 1.4.3, for we may write the Cartesian products of basis elements as the (finite) intersection of preimages of the canonical projections.*

Proposition 1.8.3 (Canonically Inherited Basis)**.** *Suppose X is a set and $\{(Y_\alpha, \tau_\alpha)\}_{\alpha\in I}$ is a collection of topological spaces with a collection of maps $\mathcal{F} := \left\{f_\alpha \colon X \to Y_\alpha\right\}_{\alpha\in I}$ for some index set I. If, $\forall\alpha \in I$, $\mathcal{B}_\alpha$ is a basis for τ_α, then $\tau(\mathcal{F})$ has the basis*

$$\left\{\cap_{j=1}^{n} {f_{\alpha_j}}^{-1}(B_{\alpha_j}) : n \in \mathbb{N}, \alpha_1, \ldots, \alpha_n \in I, \text{ and } \forall j \in \{1, \ldots, n\}, B_{\alpha_j} \in \mathcal{B}_{\alpha_j}\right\}$$

and the subbasis $\left\{{f_\alpha}^{-1}(B_\alpha) : \alpha \in I, B_\alpha \in \mathcal{B}_\alpha\right\}$.

Proof. Suppose, $\forall \alpha \in I$, $\mathcal{B}_\alpha$ is a basis for τ_α and denote by $\mathcal{B}'$ the collection

$$\left\{ \cap_{j=1}^{n} {f_{\alpha_j}}^{-1}(B_{\alpha_j}) : n \in \mathbb{N}, \alpha_1, \ldots, \alpha_n \in I, \text{ and } \forall j \in \{1, \ldots, n\}, B_{\alpha_j} \in \mathcal{B}_{\alpha_j} \right\}.$$

To show $\mathcal{B}'$ is a basis for $\tau(\mathcal{F})$, we make use of 1.2.4.

By definition, $\tau(\mathcal{F})$ is the topology generated by the subbasis $\mathcal{S}(\mathcal{F})$. Particularly, $\mathcal{S}(\mathcal{F}) \subset \tau(\mathcal{F})$ by 1.2.9. For any $\cap_{j=1}^{n} {f_{\alpha_j}}^{-1}(B_{\alpha_j}) \in \mathcal{B}'$, $\forall j \in \{1, \ldots, n\}$, the basis element B_{α_j} belongs to τ_{α_j} and, thus, $\mathcal{S}(\mathcal{F})$, which ensures that $\cap_{j=1}^{n} {f_{\alpha_j}}^{-1}(B_{\alpha_j}) \in \tau(\mathcal{F})$ by the closure of $\tau(\mathcal{F})$ under finite intersections. That is, $\mathcal{B}'$ is a collection of open sets of $\tau(\mathcal{F})$.

Now, fix an open set $U \in \tau(\mathcal{F})$ and fix $x \in U$. By the definition of a topology generated by a subbasis, there exists an intersection I_0 of finitely many elements of $\mathcal{S}(\mathcal{F})$, say $I_0 := \cap_{j=1}^{n} {f_{\alpha_j}}^{-1}(U_{\alpha_j})$, such that $x \in I_0 \subset U$.

For each $j \in \{1, \ldots, n\}$, $f_{\alpha_j}(x) \in U_{\alpha_j}$ ensures that there is a basis element $B_{\alpha_j} \in \mathcal{B}_{\alpha_j}$ with $f_{\alpha_j}(x) \in B_{\alpha_j} \subset U_{\alpha_j}$, so that $x \in {f_{\alpha_j}}^{-1}(B_{\alpha_j}) \subset {f_{\alpha_j}}^{-1}(U_{\alpha_j})$. Particularly,

$$x \in \underbrace{\cap_{j=1}^{n} {f_{\alpha_j}}^{-1}(B_{\alpha_j})}_{\in \mathcal{B}'} \subset I_0 \subset U.$$

Indeed, $\mathcal{B}'$ is a basis for $\tau(\mathcal{F})$ by 1.2.4. Lastly, consider the collection

$$\mathcal{S}' := \left\{ {f_\alpha}^{-1}(B_\alpha) : \alpha \in I, B_\alpha \in \mathcal{B}_\alpha \right\},$$

which is a subbasis of X.

> Let $\alpha \in I$. Since $\mathcal{B}_\alpha$ is a basis for τ_α by assumption, $\mathcal{B}_\alpha$ covers Y_α by defintion. That is, $\cup_{B \in \mathcal{B}_\alpha} B = Y_\alpha$. Particularly,
>
> $$X = {f_\alpha}^{-1}(Y_\alpha) = {f_\alpha}^{-1}\left(\cup_{B \in \mathcal{B}_\alpha} B\right) = \cup_{B \in \mathcal{B}_\alpha} {f_\alpha}^{-1}(B).$$
>
> Since $\left\{ {f_\alpha}^{-1}(B) : B \in \mathcal{B}_\alpha \right\} \subset \mathcal{S}'$, $\mathcal{S}'$ covers X, so $\mathcal{S}'$ is a subbasis for X by definition.

Since the basis on X induced by $\mathcal{S}'$ consisting of every finite intersection of elements of $\mathcal{S}'$ coincides with the basis $\mathcal{B}'$ of $\tau(\mathcal{F})$, $\mathcal{S}'$ is a subbasis for $\tau(\mathcal{F})$ by 1.2.9. □

Corollary 1.8.4 (Canonically Inherited Subbasis)**.** *Suppose X is a set and $\{(Y_\alpha, \tau_\alpha)\}_{\alpha \in I}$ is a collection of topological spaces with a collection of maps $\mathcal{F} := \left\{ f_\alpha : X \to Y_\alpha \right\}_{\alpha \in I}$ for some index set I. If, $\forall \alpha \in I$, $\mathcal{S}_\alpha$ is a subbasis for τ_α, then*

$$\left\{ {f_\alpha}^{-1}(S_\alpha) : \alpha \in I, S_\alpha \in \mathcal{S}_\alpha \right\}$$

is a subbasis for the initial topology $\tau(\mathcal{F})$.

Proof. Suppose, $\forall \alpha \in I$, $\mathcal{S}_\alpha$ is a subbasis for τ_α and denote by $\mathcal{S}$ the collection

$$\left\{ {f_\alpha}^{-1}(S_\alpha) : \alpha \in I, S_\alpha \in \mathcal{S}_\alpha \right\}.$$

For each $\alpha \in I$, $\mathcal{S}_\alpha$ is a subbasis for τ_α ensures that τ_α has a basis $\mathcal{B}_\alpha$, consisting of every finite intersections of elements of $\mathcal{S}_\alpha$ by 1.2.9. By the preceding proposition, $\tau(\mathcal{F})$ has the subbasis

$$\mathcal{S}' := \left\{ {f_\alpha}^{-1}(B_\alpha) : \alpha \in I, B_\alpha \in \mathcal{B}_\alpha \right\}.$$

It remains to show the topologies $\tau(\mathcal{S})$ and $\tau(\mathcal{S}')$ on X coincide, for we have $\tau(\mathcal{F}) = \tau(\mathcal{S}')$.

For any ${f_\alpha}^{-1}(S_\alpha) \in \mathcal{S}$, S_α belongs to $\mathcal{S}_\alpha$ and, thus, $\mathcal{B}_\alpha$, which implies that ${f_\alpha}^{-1}(S_\alpha) \in \mathcal{S}'$. Particularly, $\mathcal{S} \subset \mathcal{S}' \subset \tau(\mathcal{S}')$, so that $\tau(\mathcal{S}) \subset \tau(\mathcal{S}')$ by the minimality of $\tau(\mathcal{S})$ (1.2.1).

To show the remaining containment, we consider the bases $\mathcal{B}$ and $\mathcal{B}'$ for $\tau(\mathcal{S})$ and $\tau(\mathcal{S}')$, from (1.2.9), induced by $\mathcal{S}$ and $\mathcal{S}'$, respectively consisting of every finite intersections of elements of $\mathcal{S}$ and $\mathcal{S}'$.

It remains to show that $\mathcal{B} \supset \mathcal{B}'$, for it follows from the minimality of $\tau(\mathcal{B}')$ (1.2.1) that

$$\tau(\mathcal{S}) = \tau(\mathcal{B}) \supset \tau(\mathcal{B}') = \tau(\mathcal{S}').$$

For each $\alpha \in I$, by the construction of $\mathcal{B}_\alpha$, every basis element $B \in \mathcal{B}_\alpha$ has the form

$$B = \bigcap_{j=1}^{n(B)} [S_j^B]$$

for some $n(B) \in \mathbb{N}$ and $S_1^B, \ldots, S_{n(B)}^B \in \mathcal{S}_\alpha$. It follows that every element of $\mathcal{S}'$ has the form

$${f_\alpha}^{-1}(B) = \bigcap_{j=1}^{n(B)} \underbrace{[{f_\alpha}^{-1}(S_j^B)]}_{\in \mathcal{S}}$$

for some $\alpha \in I$ and $B \in \mathcal{B}_\alpha$. That is, every element of $\mathcal{S}'$ is a finite intersection of elements of $\mathcal{S}$.

Since every finite intersection of finite intersections of elements of $\mathcal{S}$ remains a finite intersection of elements of $\mathcal{S}$, every basis element of $\mathcal{B}'$ is a finite intersection of elements of $\mathcal{S}$ and, thus, a basis element of $\mathcal{B}$.

Indeed, $\mathcal{B}' \subset \mathcal{B}$ as claimed, so that $\tau(\mathcal{S}) \subset \tau(\mathcal{S}')$. Thus, $\tau(\mathcal{S}) = \tau(\mathcal{S}') = \tau(\mathcal{F})$. That is, $\mathcal{S}$ is a subbasis for the initial topology $\tau(\mathcal{F})$ by definition. □

Lemma 1.8.5 (Identifying Initial Topology)**.** *Suppose X is a set and $\{(Y_\alpha, \tau_\alpha)\}_{\alpha \in I}$ is a collection of topological spaces with a collection of maps $\mathcal{F} := \{f_\alpha \colon X \to Y_\alpha\}_{\alpha \in I}$ for some index set I. If τ is a topology on X such that*

(a) $\tau \subset \tau(\mathcal{F})$ and

(b) $\forall \alpha \in I$, f_α is continuous relative to (X, τ) and (Y_α, τ_α),

then τ is the initial topology on X induced by $\mathcal{F}$.

Proof. Suppose τ is a topology on X such that $\tau \subset \tau(\mathcal{F})$ and $\forall \alpha \in I$, f_α is continuous relative to (X, τ) and (Y_α, τ_α). Since $\tau(\mathcal{F})$ is the minimal topology on X relative to which every $f_\alpha \in \mathcal{F}$ is continuous, the assumption ensures that $\tau \supset \tau(\mathcal{F})$ and, therefore, $\tau = \tau(\mathcal{F})$. □

Corollary 1.8.6 (Identifying Initial Topology via Generator)**.** *Let X be a set and $\{(Y_\alpha, \tau_\alpha)\}_{\alpha \in I}$ be a collection of topological spaces with a collection of maps $\mathcal{F} := \{f_\alpha \colon X \to Y_\alpha\}_{\alpha \in I}$ for some index set I. If τ is a topology on X*

(a) generated by a set, basis, or subbasis contained in $\tau(\mathcal{F})$, and

(b) $\forall \alpha \in I$, f_α is continuous relative to (X, τ) and (Y_α, τ_α),

then τ is the initial topology on X induced by $\mathcal{F}$.

Remark 1.8.3. *The above corollary shows that the topology defined, in Theorem 1.37 of Rudin's Functional Analysis, on a vector space X with a separating family $\mathcal{P}$ of semi-norms on X is, in*

disguise, the initial topology on X induced by the collection $\{p_y : p \in \mathcal{P}, y \in X\}$ of translations of semi-norms, where for every $(p, y) \in \mathcal{P} \times X$, $p_y \colon X \to \mathbb{R}$ is defined by

$$\forall x \in X,\ p_y(x) := p(x - y).$$

Remark 1.8.4. *The above corollary shows that we may generate the initial topology $\tau(\mathcal{F})$ via a nicer subcollection $\mathcal{C}$ of itself, so long as every $f_\alpha \in \mathcal{F}$ remains continuous relative to $\tau(\mathcal{C})$.*

Remark 1.8.5. *By considering underlying sets, topological spaces, and maps with additional structures, we may obtain immensely nicer, more concrete bases and subbases for the initial topologies that the maps induce.*

For instance, we may consider topological vector spaces with initial topologies induced by linear functionals or semi-norms.

1.9 Product Topology (Revisited)

Definition 1.9.1 ((General) Cartesian product)**.** *Suppose $\{X_\alpha\}_{\alpha\in J}$ is a collection of non-empty sets for some index set J. Then,*

(a) the (general) Cartesian product of $\{X_\alpha\}_{\alpha\in J}$ is the collection

$$\prod_{\alpha\in J} X_\alpha := \left\{\tilde{x}\colon J \to \cup_{\alpha\in J} X_\alpha : \forall \alpha \in J,\ \tilde{x}(\alpha) \in X_\alpha\right\};$$

(a.1) if $\tilde{x} \in \prod_{\alpha\in J} X_\alpha$, we denote, $\forall \alpha \in J$, $\tilde{x}_\alpha := \tilde{x}(\alpha)$ and $(\tilde{x}_\alpha)_{\alpha\in J} := \tilde{x}$;

(a.2) if X is a set, then we denote more succinctly $\prod_{\alpha\in J} X$ by X^J.

(b) for every $\beta \in J$, the projection of $\prod_{\alpha\in J} X_\alpha$ into the β-factor is the map $\pi_\beta\colon \prod_{\alpha\in J} X_\alpha \to X_\beta$ defined by

$$\pi_\beta(\tilde{x}) := \tilde{x}(\beta),\ \forall \tilde{x} \in \prod_{\alpha\in J} X_\alpha.$$

Definition 1.9.2 ((General) Cartesian product of subsets)**.** *Suppose $\{X_\alpha\}_{\alpha\in J}$ is a collection of non-empty sets for some index set J. If $\forall \alpha \in J$, $U_\alpha \subset X_\alpha$, then the Cartesian product of $\{U_\alpha\}_{\alpha\in J}$ relative to $\{X_\alpha\}_{\alpha\in J}$ is the subset*

$$\prod_{\alpha\in J} U_\alpha := \left\{\tilde{x}\colon J \to \cup_{\alpha\in J} X_\alpha : \forall \alpha \in J,\ \tilde{x}(\alpha) \in U_\alpha\right\}$$

of $\prod_{\alpha\in J} X_\alpha$.

Unless otherwise specified, for any collection $\{U_\alpha\}_{\alpha\in J}$ such that $\forall \alpha \in J$, $U_\alpha \subset X_\alpha$, $\prod_{\alpha\in J} U_\alpha$ is defined as above relative to $\{X_\alpha\}_\alpha$ and as a subset of $\prod_{\alpha\in J} X_\alpha$.

Remark 1.9.1. *I adopt the convention that two functions are equal if and only if they share the same assignment, domain, and codomain. Respect to this convention, two functions with different codomain are thus distinct.*

Particularly, given a collection $\{X_\alpha\}_{\alpha\in J}$ of non-empty sets for some index set J and some collection $\{U_\alpha\}_{\alpha\in J}$ such that $\forall \alpha \in J$, $U_\alpha \subset X_\alpha$, a function $\tilde{y} \in \prod_{\alpha\in J} X_\alpha$ cannot be an element of $\left\{\tilde{x}\colon J \to \cup_{\alpha\in J} U_\alpha : \forall \alpha \in J,\ \tilde{x}(\alpha) \in U_\alpha\right\}$ and vice-versa, due to the distinctness of the codomains.

To remediate the inconvenience and maintain adherence to the mentioned convention, we adopted the preceding definition, which is convenient since every product of subsets is immediately a subset of $\prod_{\alpha\in J} X_\alpha$ by definition.

Definition 1.9.3. *Suppose $\{X_\alpha\}_{\alpha\in J}$ is a collection of non-empty sets for some index set J and $\{U_\alpha\}_{\alpha\in J}$ is a collection satisfying $\forall \alpha \in J$, $U_\alpha \subset X_\alpha$.*

(a) Let $n \in \mathbb{N}$. For any $\beta_1, \ldots, \beta_n \in J$, if $\forall j \in \{1, \ldots, n\}$, $V_{\beta_j} \subset X_{\beta_j}$, then

$$\left[\prod_{\alpha\in J/\{\beta_1,\ldots,\beta_n\}} U_\alpha\right] \times \left[\prod_{j=1}^{n} V_{\beta_j}\right] := \prod_{\alpha\in J} U'_\alpha,$$

where $\forall \alpha \in J$,

$$U'_\alpha := \begin{cases} U_\alpha & \text{if } \alpha \notin \{\beta_1, \ldots, \beta_n\} \\ V_\alpha & \text{if } \alpha \in \{\beta_1, \ldots, \beta_n\} \end{cases}.$$

(b) Let $n \in \mathbb{N}$ and $\tilde{x} \in \prod_{\alpha\in J} X_\alpha$. For any $\beta_1, \ldots, \beta_n \in J$, if $\forall j \in \{1, \ldots, n\}$, $x_{\beta_j} \in X_{\beta_j}$, then

$$(\tilde{x}_\alpha)_{\alpha\in J/\{\beta_1,\ldots,\beta_n\}} \times (\tilde{x}_{\beta_j})_{j=1}^n := (\tilde{x}'_\alpha)_{\alpha\in J},$$

where $\forall \alpha \in J$

$$\tilde{x}'_\alpha := \begin{cases} \tilde{x}_\alpha & \text{if } \alpha \notin \{\beta_1, \ldots, \beta_n\} \\ x_\alpha & \text{if } \alpha \in \{\beta_1 \ldots, \beta_n\} \end{cases}.$$

Remark 1.9.2. *Several arguments from the preceding section on finite product spaces apply almost verbatim to theorems in this section.*

Additionally, implications regarding initial topologies may be directly applied to yield some of the results in this section concerning the product topology.

We, however, provide explicit proofs to gain some familiarity with the general Cartesian products.

Example 1.9.1. *(a) Let X be a non-empty set. The Cartesian product $\prod_{n\in\mathbb{N}} X$ consists of all sequences of elements in X.*

(b) The Cartesian product $\prod_{\alpha\in\mathbb{R}} \mathbb{R}$ consists of all real-valued maps defined on $\mathbb{R}$.

Definition 1.9.4 ($\mathbb{R}^\omega$)**.** *We denote $\mathbb{R}^\omega := \mathbb{R}^{\mathbb{N}}$, the collection of all real-valued sequences.*

Lemma 1.9.2 (Images and Preimage of Projections)**.** *Suppose $\{X_\alpha\}_{\alpha\in J}$ is a collection of non-empty sets for some index set J and $\{V_\alpha\}_{\alpha\in J}$ is a collection satisfying $\forall \alpha \in J$, $V_\alpha \subset X_\alpha$.*

(a) Let $\beta \in J$. Then, $\pi_\beta(\prod_{\alpha\in J} V_\alpha) \subset V_\beta$.

(a.1) If $\forall \alpha \in J/\{\beta\}$ $V_\alpha \neq \emptyset$, then $\pi_\beta(\prod_{\alpha\in J} V_\alpha) = V_\beta$.

(b) Let $\beta \in J$. Then, ${\pi_\beta}^{-1}(V_\beta) = \left[\prod_{\alpha\in J/\{\beta\}} X_\alpha\right] \times V_\beta$.

(b.1) Particularly, $\prod_{\alpha\in J} V_\alpha = \cap_{\alpha\in J} {\pi_\alpha}^{-1}(V_\alpha)$.

Proof. (a) If $\prod_{\alpha\in J} V_\alpha = \emptyset$, then $\pi_\beta(\prod_{\alpha\in J} V_\alpha) = \emptyset \subset V_\beta$. Hence, we may suppose that $\prod_{\alpha\in J} V_\alpha$ is non-empty. For any $(v_\alpha)_{\alpha\in J} \in \prod_{\alpha\in J} V_\alpha$, $\pi_\beta((v_\alpha)_{\alpha\in J}) := v_\beta \in V_\beta$. That is, $\pi_\beta(\prod_{\alpha\in J} V_\alpha) \subset V_\beta$.

(a.1) Suppose $\forall \alpha \in J/\{\beta\}$, V_α is non-empty, say with an element x_α. By part (a), it remains to show that $\pi_\beta(\prod_{\alpha\in J} V_\alpha) \supset V_\beta$. If $V_\beta = \emptyset$, then the containment holds trivially. Thus, we suppose V_β is non-empty.

Fix $x_\beta \in V_\beta$. By construction, $x_\beta =: \pi_\beta((x_\alpha)_{\alpha\in J}) \in \pi_\beta(\prod_{\alpha\in J} V_\alpha)$. Indeed, $\pi_\beta(\prod_{\alpha\in J} V_\alpha) \supset V_\beta$.

(b) For any $(x_\alpha)_{\alpha\in J} \in \prod_{\alpha\in J} X_\alpha$,

$$\begin{aligned}(x_\alpha)_{\alpha\in J} \in {\pi_\beta}^{-1}(V_\beta) &\iff \pi_\beta((x_\alpha)_{\alpha\in J}) \in V_\beta \\ &\iff \forall \alpha \in J,\ x_\alpha \in X_\alpha \text{ and } x_\beta \in V_\beta.\end{aligned}$$

Indeed, ${\pi_\beta}^{-1}(V_\beta) = \left[\prod_{\alpha\in J/\{\beta\}} X_\alpha\right] \times V_\beta$.

(b.1) From part (b), we conclude that

$$\prod_{\alpha\in J} V_\alpha = \cap_{\alpha\in J} {\pi_\alpha}^{-1}(V_\alpha).$$

□

Lemma 1.9.3. *Suppose $\{X_\alpha\}_{\alpha\in J}$ is a collection of non-empty sets for some index set J. If $\{U_\alpha\}_{\alpha\in J}$ and $\{V_\alpha\}_{\alpha\in J}$ are collections satisfying $\forall\alpha\in J$, $U_\alpha, V_\alpha \subset X_\alpha$, then*

$$\left[\prod_{\alpha\in J} U_\alpha\right] \cap \left[\prod_{\alpha\in J} V_\alpha\right] = \prod_{\alpha\in J}[U_\alpha \cap V_\alpha].$$

Proof. Suppose $\{U_\alpha\}_{\alpha\in J}$ and $\{V_\alpha\}_{\alpha\in J}$ are collections of subsets such that $\forall\alpha\in J$, $U_\alpha, V_\alpha \subset X_\alpha$.

For any $\tilde{x}\in\prod_{\alpha\in J} X_\alpha$,

$$\begin{aligned}\tilde{x}\in\prod_{\alpha\in J}[U_\alpha\cap V_\alpha] &\iff \forall\alpha\in J,\ \tilde{x}_\alpha\in U_\alpha \text{ and } \tilde{x}_\alpha\in V_\alpha\\ &\iff \tilde{x}\in\prod_{\alpha\in J} U_\alpha\cap\prod_{\alpha\in J} V_\alpha.\end{aligned}$$

Indeed, the two sets of interest coincide. □

Lemma 1.9.4 (Basis for Product)**.** *Suppose $\{(X_\alpha,\tau_\alpha)\}_{\alpha\in J}$ is a collection of topological spaces for some index set J. Then,*

$$\mathcal{B}_{box} := \left\{\prod_{\alpha\in J} U_\alpha : \forall\alpha\in J,\ U_\alpha\in\tau_\alpha\right\}.$$

is a basis for $\prod_{\alpha\in J} X_\alpha$.

Proof. For any $\alpha\in J$, (X_α,τ_α) is a topological space ensure that $X_\alpha\in\tau_\alpha$. Particularly, $\prod_{\alpha\in J} X_\alpha\in\mathcal{B}_{box}$, so $\mathcal{B}_{box}$ covers the product $\prod_{\alpha\in J} X_\alpha$.

Fix $\tilde{x}\in\prod_{\alpha\in J} X_\alpha$ and suppose that $x\in B_1\cap B_2$ for some $B_1, B_2\in\mathcal{B}_{box}$, say $B_1=\prod_{\alpha\in J} U_\alpha$ and $B_2=\prod_{\alpha\in J} V_\alpha$. Since $\prod_{\alpha\in J} U_\alpha\cap\prod_{\alpha\in J} V_\alpha=\prod_{\alpha\in J}[U_\alpha\cap V_\alpha]\in\mathcal{B}_{box}$, $B_3:=\prod_{\alpha\in J}[U_\alpha\cap V_\alpha]\in\mathcal{B}_{box}$ satisfies $\tilde{x}\in B_3\subset B_1\cap B_2$.

By definition, $\mathcal{B}_{box}$ is a basis for $\prod_{\alpha\in J} X_\alpha$. □

Lemma 1.9.5 (Subbasis for Product)**.** *Suppose $\{(X_\alpha,\tau_\alpha)\}_{\alpha\in J}$ is a collection of topological spaces for some index set J. Then,*

$$\mathcal{S}_{prod} := \cup_{\alpha\in J}\left\{{\pi_\alpha}^{-1}(U_\alpha) : U_\alpha\in\tau_\alpha\right\}.$$

is a subbasis for $\prod_{\alpha\in J} X_\alpha$.

Proof. Let $\alpha'\in J$. Since $(X_{\alpha'},\tau_{\alpha'})$ is a topological space by assumption, $X_{\alpha'}\in\tau_{\alpha'}$ ensures that $\prod_{\alpha\in J} X_\alpha = {\pi_{\alpha'}}^{-1}(X_{\alpha'})\in\mathcal{S}_{prod}$. Particularly, $\mathcal{S}_{prod}$ is a cover and, thus, a subbasis for $\prod_{\alpha\in J} X_\alpha$. □

Definition 1.9.5 (Box topology)**.** *Suppose $\{(X_\alpha,\tau_\alpha)\}_{\alpha\in J}$ is a collection of topological spaces for some index set J. Then, the box topology on $\prod_{\alpha\in J} X_\alpha$ is the topology generated by the basis*

$$\mathcal{B}_{box} := \left\{\prod_{\alpha\in J} U_\alpha : \forall\alpha\in J,\ U_\alpha\in\tau_\alpha\right\}.$$

Definition 1.9.6 (Product topology)**.** *Suppose $\{(X_\alpha,\tau_\alpha)\}_{\alpha\in J}$ is a collection of topological spaces for some index set J. Then, the product topology on $\prod_{\alpha\in J} X_\alpha$ is the topology generated by the subbasis*

$$\mathcal{S}_{prod} := \cup_{\alpha\in J}\left\{{\pi_\alpha}^{-1}(U_\alpha) : U_\alpha\in\tau_\alpha\right\}.$$

Unless otherwise specified, we endow $\prod_{\alpha\in J} X_\alpha$ with its product topology.

Theorem 1.9.6 (Characterization of Product Topology)**.** *Suppose $\{(X_\alpha, \tau_\alpha)\}_{\alpha\in J}$ is a collection of topological spaces for some index set J. Then, the product topology on $\prod_{\alpha\in J} X_\alpha$ is generated by the basis*

$$\mathcal{B}_{prod} := \left\{ \prod_{\alpha\in J} U_\alpha : \forall \alpha \in J,\ U_\alpha \in \tau_\alpha \text{ and } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha \in J \right\}.$$

Proof. Since the product topology has the subbasis

$$\mathcal{S}_{prod} := \cup_{\alpha\in J} \left\{ {\pi_\alpha}^{-1}(U_\alpha) : U_\alpha \in \tau_\alpha \right\},$$

it is generated by the basis $\mathcal{I}$ consisting of all finite intersections of elements of $\mathcal{S}_{prod}$. Hence, it remains to show that $\mathcal{B}_{prod}$ and $\mathcal{I}$ coincide.

For any $U \subset \prod_{\alpha\in J} X_\alpha$, $U \in \mathcal{B}_{prod} \iff U$ has the form $U = \left[\prod_{\alpha\in J/\{\beta_1,\ldots,\beta_n\}} X_\alpha\right] \times \prod_{j=1}^n V_{\beta_j}$ for some $n \in \mathbb{N}$, $\beta_1, \ldots, \beta_n \in J$ and $V_{\beta_1} \in \tau_{\beta_1}, \ldots, V_{\beta_n} \in \tau_{\beta_n} \iff U \in \mathcal{I}$, since

$$\left[\prod_{\alpha\in J/\{\beta_1,\ldots,\beta_n\}} X_\alpha\right] \times \prod_{j=1}^n V_{\beta_j} = \bigcap_{j=1}^n \underbrace{{\pi_{\beta_j}}^{-1}(V_{\beta_j})}_{\in \mathcal{S}_{prod}} \in \mathcal{I}.$$

Indeed, the two collections coincide, and the product topology is generated by the basis $\mathcal{B}_{prod}$. □

Remark 1.9.3. *As a reminder, the product topology was defined in lecture via the basis above.*

Lemma 1.9.7 (Box Topology is Finer Than Product Topology)**.** *Suppose $\{(X_\alpha, \tau_\alpha)\}_{\alpha\in J}$ is a collection of topological spaces for some index set J. Then, the box topology on $\prod_{\alpha\in J} X_\alpha$ is finer than the product topology on $\prod_{\alpha\in J} X_\alpha$.*

Proof. Since the product topology is generated by a basis that is contained in the standard basis of the box topology, the product topology is contained in the box topology.

That is, the box topology is finer than the product topology. □

Lemma 1.9.8 (Box and Product Topologies on Finite Product Coincide)**.** *Suppose $n \in \mathbb{N}$ and $(X_1, \tau_1), \ldots, (X_n, \tau_n)$ are topological spaces. Then, the box topology and the product topology on $\prod_{j=1}^n X_j$ coincide.*

Proof. Since the standard bases of the box topology and product topology on $\prod_{j=1}^n X_j$ coincide by construction, the two topologies coincide. □

Example 1.9.9 (Box Topology may be Strictly Finer)**.** *Equip $\mathbb{R}$ with its standard topology and consider the product $\mathbb{R}^\omega$ and its subset $(-1,1)^{\mathbb{N}}$.*

Since $(-1,1)^{\mathbb{N}}$ is a standard basis element for the box topology, it is open relative to the box topology of $\mathbb{R}^\omega$. However, it is not open relative to the product topology of $\mathbb{R}^\omega$.

Suppose, to the contrary, that $(-1,1)^{\mathbb{N}}$ is open in the product topology. Since the product topology is generated by its standard basis, there is a basis element $\prod_{k\in\mathbb{N}} U_k$ with $\tilde{0} \in \prod_{k\in\mathbb{N}} U_k \subset (-1,1)^{\mathbb{N}}$ satisfying $U_k = \mathbb{R}$ for all but finitely many $k \in \mathbb{N}$ with $\forall k \in \mathbb{N}$, $\tilde{0}_k = 0 \in U_k$.

Especially, $\mathbb{N}$ is infinite ensures there is some $N \in \mathbb{N}$ with $U_N = \mathbb{R}$. Thus, $\tilde{y}\colon \mathbb{R}^\omega \to \mathbb{R}$ defined by

$$\tilde{y}(k) := \begin{cases} 0 & \text{if } k \neq N \\ 2 & \text{if } k = N \end{cases}$$

lives in $\prod_{k\in\mathbb{N}} U_k \subset (-1,1)^{\mathbb{N}}$, which yields the contradiction $2 \in (-1,1)$ since $\tilde{y}(N) = 2 \in U_N$. Indeed, $(-1,1)^{\mathbb{N}}$ is not open in the product topology.

Theorem 1.9.10 (Canonically Inherited Basis)**.** *Suppose $\{(X_\alpha, \tau_\alpha)\}_{\alpha\in J}$ is a collection of topological spaces for some index set J and $\forall\alpha \in J$, (X_α, τ_α) has a basis $\mathcal{B}_\alpha$. Then, the box topology and the product topology are respectively generated by the bases*

$$\mathcal{B}'_{box} := \left\{ \prod_{\alpha\in J} B_\alpha : \forall\alpha \in J,\ B_\alpha \in \mathcal{B}_\alpha \right\} \text{ and}$$

$$\mathcal{B}'_{prod} := \left\{ \prod_{\alpha\in J} B_\alpha : B_\alpha \in \mathcal{B}_\alpha \text{ for only finitely many } \alpha \text{ and } B_\alpha = X_\alpha \text{ otherwise} \right\}.$$

Proof. We make use of 1.2.4

Since basis elements are open sets, $\mathcal{B}'_{box}$ remains a collection of standard basis elements and, thus, open sets of the box topology. Similarly, $\mathcal{B}'_{prod}$ remains a subcollection of the product topology.

Fix an open set U in the box topology and fix $x \in U$. By the standard basis of the box topology, there exists a collection $\{U_\alpha\}_{\alpha\in J}$ satisfying $\forall\alpha \in J\ U_\alpha \in \tau_\alpha$ with $x \in \prod_{\alpha\in J} U_\alpha \subset U$.

Particularly, for any $\alpha \in J$, τ_α has the basis $\mathcal{B}_\alpha$ and $x_\alpha \in U_\alpha$ ensures there is some $B_\alpha \in \mathcal{B}_\alpha$ such that $x_\alpha \in B_\alpha \in U_\alpha$. That is, $\prod_{\alpha\in J} B_\alpha \in \mathcal{B}'_{box}$ satisfies

$$x \in \prod_{\alpha\in J} B_\alpha \subset \prod_{\alpha\in J} U_\alpha \subset U,$$

which allows us to conclude that $\mathcal{B}'_{box}$ is a basis for the box topology by 1.2.4.

Fix an open set U in the product topology and fix $x \in U$. By the standard basis of the product topology, there exists a basis element $B = \left[\prod_{\alpha\in J/\{\beta_1,\ldots,\beta_n\}} X_\alpha\right] \times \prod_{j=1}^n U_{\beta_j}$ satisfying $x \in B \subset U$, for some $n \in \mathbb{N}$, $\beta_1, \ldots, \beta_n \in J$, and $U_{\beta_1} \in \tau_{\beta_1}, \ldots, U_{\beta_n} \in \tau_{\beta_n}$.

Thus, for $\beta \in \{\beta_1, \ldots, \beta_n\}$, there is some basis element $B_\beta \in \mathcal{B}_\beta$ with $x_\beta \in B_\beta \subset U_\beta$. That is,

$$B' := \left[\prod_{\alpha\in J/\{\beta_1,\ldots,\beta_n\}} X_\alpha\right] \times \prod_{j=1}^n B_{\beta_j} \in \mathcal{B}'_{prod}$$

satisfies $x \in B' \subset B \subset U$, which implies, via 1.2.4, the product topology has $\mathcal{B}'_{prod}$ as a basis. □

Theorem 1.9.11 (Product Preserves Hausdorffness)**.** *Suppose $\{(X_\alpha, \tau_\alpha)\}_{\alpha\in J}$ is a collection of Hausdorff spaces for some index set J. Then, $\prod_{\alpha\in J} X_\alpha$ is Hausdorff relative to the product topology and the box topology.*

Proof. Denote by τ_{box} and τ_{prod} the box topology and the product topology, respectively. Since τ_{box} is finer than τ_{prod}, it suffices to show the τ_{prod} is Hausdorff.

Fix distinct elements $\tilde{x}, \tilde{y} \in \prod_{\alpha\in J} X_\alpha$. Since $\tilde{x} \neq \tilde{y}$, $\tilde{x}_\beta \neq \tilde{y}_\beta$ for some $\beta \in \alpha$. Since (X_β, τ_β) is Hausdorff by assumption, there are disjoint neighborhoods U_β, V_β of $\tilde{x}_\beta$ and $\tilde{y}_\beta$, respectively.

Particularly, $\tilde{x} \in {\pi_\beta}^{-1}(U_\beta)$ and $\tilde{y} \in {\pi_\beta}^{-1}(V_\beta)$, where ${\pi_\beta}^{-1}(U_\beta)$ and ${\pi_\beta}^{-1}(V_\beta)$ are standard subbasis elements of τ_{prod} and, thus, open sets of τ_{prod}. Moreover, the preimages are disjoint by construction.

> Suppose, to the contrary, that the preimages intersects non-trivially, say with an element $\tilde{z}$. It follows that $\tilde{z}_\beta \in U_\beta \cap V_\beta$, which contradicts the disjointness of U_β and V_β.

That is, we showed $\tilde{x}$ and $\tilde{y}$ have disjoint neighborhoods in τ_{prod} and, hence, in τ_{box}, since $\tau_{prod} \subset \tau_{box}$. By definition, $\prod_{\alpha \in J} X_\alpha$ is Hausdorff relative to both τ_{prod} and τ_{box}. □

Theorem 1.9.12 (Compatibility of Subspace with Product)**.**

Proof. **[INCOMPLETE]** □

Theorem 1.9.13 (Compatibility of Closure with Product)**.**

Proof. **[INCOMPLETE]** □

Lemma 1.9.14 (Projections are Continuous)**.** *Suppose $\{(X_\alpha, \tau_\alpha)\}_{\alpha \in J}$ is a collection of topological spaces for some index set J. Then, $\forall \beta \in J$, the projection map $\pi_\beta \colon \prod_{\alpha \in J} X_\alpha \to X_\beta$ is continuous relative to both the product topology and box topology.*

Proof. Fix $\beta \in J$. Again, by the fineness of the box topology, it suffices to show that π_β is continuous relative to the product topology.

For any open set $U_\beta \in \tau_\beta$, the preimage ${\pi_\beta}^{-1}(U_\beta)$ is a standard subbasis element and, thus, a open set of the product topology and, hence, the box topology. That is, π_β is continuous relative to both the product topology and the box topology by definition. □

Theorem 1.9.15 (Characterization of Continuous Map into Product Space)**.** *Suppose (Y, τ_Y) is a topological space, $\{(X_\alpha, \tau_\alpha)\}_{\alpha \in J}$ is a collection of topological spaces for some index set J, and $f \colon Y \to \prod_{\alpha \in J} X_\alpha$. Then, f is continuous relative to the product topology $\iff$ $\forall \alpha \in J$, the composition $\pi_\alpha \circ f \colon Y \to X_\alpha$ is continuous.*

Proof. [$\implies$] Suppose f is continuous relative to the product topology. For every $\alpha \in J$, $\pi_\alpha \circ f$ is the composition of continuous functions and, thus, continuous by 1.7.10.

[$\impliedby$] Suppose, $\forall \alpha \in J$, the composition $\pi_\alpha \circ f \colon Y \to X_\alpha$ is continuous. In light of 1.7.5, it suffices to show each subbasis elements of the product topology has an open preimage under f.

Fix a standard subbasis element ${\pi_\alpha}^{-1}(U_\alpha)$ of the product topology. We compute that

$$f^{-1}({\pi_\alpha}^{-1}(U_\alpha)) = [\pi_\alpha \circ f]^{-1}(U_\alpha),$$

which remains an open set of τ_Y by assumption. Indeed, f is continuous by 1.7.5. □

Example 1.9.16. *We provide an example in which the preceding theorem fails if $\prod_{\alpha \in J} X_\alpha$ is equipped with the box topology instead.*

Equip $\mathbb{R}$ with its standard topology. Consider $\mathbb{R}^\omega$ equipped with the box topology and the map $f \colon \mathbb{R} \to \mathbb{R}^\omega$ defined by $\forall t \in \mathbb{R},\ f(t) := (t)_{n \in \mathbb{N}}$.

By construction, for every $n \in \mathbb{N}$, $\pi_n \circ f = \mathrm{Id}$, which is continuous relative to $(\mathbb{R}, \tau_{\mathbb{R}})$ and $(\mathbb{R}, \tau_{\mathbb{R}})$. However, the open set $U := \prod_{n \in \mathbb{N}} (-\frac{1}{n}, \frac{1}{n})$ in the box topology has the preimage under f

$$f^{-1}(U) := \left\{ t \in \mathbb{R} : (t)_{n \in \mathbb{N}} \in \prod_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n} \right) \right\} = \cap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n} \right) = \{0\},$$

which is not open in the standard topology of $\mathbb{R}$. That is, f is not continuous.

Remark 1.9.4 (Related to the comprehensive exam)**.** *Due to the nature and abstractness of the product topology and box topology on general Cartesian products, problems in the comprehensive exams concerning those topics tended to be less difficult.*

That is, those topics are great opportunities to score points in the comprehensive exam.

2 Metric Spaces

Definition 2.0.1 (Metric)**.** *(a) Suppose X is a set. A metric on X is a map $d\colon X \times X \to \mathbb{R}$ satisfying the following properties.*

(a.1) non-negativity: $\forall x, y \in X,\ d(x, y) \geq 0$

(a.2) symmetry: $\forall x, y \in X,\ d(x, y) = d(y, x)$

(a.3) vanishing on and only on reflexive pairs: $\forall x, y \in X$,

$$d(x, x) = 0 \text{ and if } d(x, y) = 0, \text{ then } x = y$$

(a.4) triangle inequality:

$$\forall x, y, z \in X,\ d(x, z) \leq d(x, y) + d(y, z)$$

(b) A metric space is an ordered pair (X, d) consisting of a set X and a metric d on X.

Definition 2.0.2 (Balls)**.** *Suppose (X, d) is a metric space. Then,*

(a) an open ball is a set $B_\epsilon(x) := \{y \in X : d(x, y) < \epsilon\}$ for some radius $\epsilon > 0$ and center $x \in X$ and

(b) a closed ball is a set $\overline{B}_\epsilon(x) := \{y \in X : d(x, y) \leq \epsilon\}$ for some radius $\epsilon > 0$ and center $x \in X$.

In discussions involving multiple metrics, we write $B^d_\epsilon(x)$ ($\overline{B^d}_\epsilon(x)$) to indicate that the open (closed) ball is defined relative to the metric d.

Lemma 2.0.1 (Metric Topology)**.** *Suppose (X, d) is a metric space. Then,*

$$\mathcal{B}_d := \{B_\epsilon(x) : x \in X, \epsilon > 0\}$$

is a basis for X.

Proof. For any $x \in X$, $x \in B_1(x) \in \mathcal{B}_d$. That is, $\mathcal{B}_d$ covers X.

Fix $x \in X$ and suppose that $x \in B_1 \cap B_2$ for some for some $B_1, B_2 \in \mathcal{B}_d$, say $B_1 = B_{\epsilon_1}(x_1)$ and $B_2 = B_{\epsilon_2}(x_2)$. For each $j \in \{1, 2\}$, $d(x, x_j) < \epsilon_j$ by definition ensures that

$$\epsilon := \min_{j \in \{1,2\}} [\epsilon_j - d(x, x_j)] > 0.$$

Particularly, for each $j \in \{1, 2\}$, $\forall y \in B_3 := B_\epsilon(x)$

$$\begin{aligned} d(y, x_j) \leq& d(y, x) + d(x, x_j) < \epsilon + d(x, x_j) \\ \leq& \epsilon_j - d(x, x_j) + d(x, x_j) = \epsilon_j. \end{aligned}$$

That is, $x \in B_3 \subset B_1 \cap B_2$ with $B_3 \in \mathcal{B}_d$. By definition, $\mathcal{B}_d$ is a basis for X. □

Definition 2.0.3 (Metric topology)**.** *Suppose (X, d) is a metric space. Then, the metric topology on X (induced from d) is the topology generated by the basis*

$$\mathcal{B}_d := \{B_\epsilon(x) : \epsilon > 0,\ x \in X\}.$$

Unless otherwise specified, we endow the underlying set X of a metric space (X, d) with the metric topology induced by d and denote by (X, d) the corresponding topological space.

Definition 2.0.4 (Compatible metric, metrizability)**.** *Suppose (X, τ) is a topological space.*

(a) A compatible metric with τ is a metric on X whose induced metric topology coincides with τ.

(b) (X, τ) is metrizable if there is a metric on X that is compatible with τ.

Lemma 2.0.2 (Characterization of Open Set in Metric Space)**.** *Suppose (X, d) is a metric space. Then, $U \subset X$ is open $\iff$ $\forall x \in U$, there exists $\epsilon > 0$ such that $x \in B_\epsilon(x) \subset U$.*

Proof. Fix $U \subset X$.

[$\implies$] Suppose U is open and fix $x \in U$. Since the metric topology is generated by the basis consisting of open balls, there exists some open ball $B_\epsilon(y)$ for some $\epsilon > 0$ and $y \in X$ with $x \in B_\epsilon(y) \subset U$.

By definition, $d(x, y) < \epsilon$ ensures $\delta := \epsilon - d(x, y) > 0$. Particularly, for any $x' \in B_\delta(x)$,

$$d(x', y) \leq d(x', x) + d(x, y) < \delta + d(x, y) \leq \epsilon.$$

That is, $x \in B_\delta(x) \subset B_\epsilon(y) \subset U$.

[$\impliedby$] Suppose $\forall x \in U$, there exists $\epsilon_x > 0$ such that $x \in B_{\epsilon_x}(x) \subset U$. By assumption,

$$U = \cup_{x \in X} B_{\epsilon_x}(x),$$

so that U is the union of standard basis elements and, thus, open sets of the metric topology. Particularly, U is open by the closure of topologies under unions. □

Definition 2.0.5 (Standard Metric of $\mathbb{R}^n$)**.** *Let $n \in \mathbb{N}$. The standard (Euclidean) metric on $\mathbb{R}^n$ is the metric $d\colon \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ defined by*

$$d(x, y) := \sqrt{\sum_{j=1}^{n} (x_j - y_j)^2}, \ \forall x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in \mathbb{R}^n.$$

Lemma 2.0.3 (Compatibility of Standard Metric and Topology of $\mathbb{R}$)**.** *The standard metric of $\mathbb{R}$ is compatible with the standard topology of $\mathbb{R}$.*

Proof. It remains to show that the standard topology of $\mathbb{R}$ coincides with the topology induced by the standard metric of $\mathbb{R}$. It suffices to show the topologies on $\mathbb{R}$ share a common basis.

By definition, every open ball relative to the standard metric of $\mathbb{R}$ has the form $B_\epsilon(x)$ for some $\epsilon > 0$ and $x \in \mathbb{R}$, where

$$B_\epsilon(x) := \{y \in \mathbb{R} : |x - y| < \epsilon\} = (x - \epsilon, x + \epsilon)$$

ensures every open ball is an open interval of $\mathbb{R}$.

Conversely, every (finite) open interval of $\mathbb{R}$ has the form (a, b) for some $a, b \in \mathbb{R}$ with $a < b$, where by considering $x := \frac{a+b}{2}$ and $\epsilon := \frac{b-a}{2}$ we obtain

$$(a, b) = (x - \epsilon, x + \epsilon) = B_\epsilon(x),$$

so every open interval is an open ball relative to the standard metric of $\mathbb{R}$.

That is, we showed the standard bases for the two topologies of interest coincide, so the two topologies also coincide. Indeed, the standard metric of $\mathbb{R}$ is compatible with the standard topology of $\mathbb{R}$ by definition. □

Lemma 2.0.4 (Discrete Metric)**.** *Suppose X is a set. Then,*

(a) $d\colon X \times X \to \mathbb{R}$ defined by

$$d(x,y); = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

is a metric on X, and

(b) d is compatible with the discrete topology of X.

Proof. Fix $x, y, z \in X$.

By definition, $d(X \times X) \subset \{0, 1\}$. That is, d is non-negative. By construction, $d(x, x) := 0$ and if $d(x, y) = 0$, then $x = y$. That is, d vanishes on and only on reflexive pairs.

Moreover, if $x \neq y$, then $d(x, y) = 0 = d(y, x)$, whereas if $x = y$, then $d(x, y) = 0 = d(y, x)$. That is, d is symmetric.

Lastly, if $x = z$, then $d(x, z) = 0 \leq d(x, y) + d(y, z)$ since d is non-negative. If $x \neq z$, then

$$d(x,z) \leq d(x,y) + d(y,z) = \begin{cases} d(x,z) & \text{if } x = y \\ 1 + d(y,z) & \text{if } x \neq y \end{cases}.$$

Indeed, d satisfies the triangle inequality. By definition, d is a metric on X.

For any $x \in X$, $B_1(x) = \{x\}$ since for any $y \in X$

$$d(x, y) < 1 \iff d(x, y) = 0 \iff x = y.$$

Particularly, every singletons of X is open in the metric topology, as the metric topology is by definition generated by the open balls.

Thus, the metric topology contains and, thus, coincides with the discrete topology of X. By definition, d is compatible with the discrete topology of X. □

Definition 2.0.6 (Discrete Metric)**.** *Suppose X is a set. Then, the discrete metric of X is the metric $d\colon X \times X \to \mathbb{R}$ defined by*

$$d(x,y); = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}.$$

Example 2.0.5 (Closure of Open Balls Need Not be the Closed Ball)**.** *The equality*

$$\overline{B_\epsilon(x)} = \overline{B}_\epsilon(x)$$

need not hold for every metric space.

Consider the metric space (X, d) consisting of a set X with at least two elements and the discrete metric d of X. Let $x \in X$ and $\epsilon := 1$. Since d induces the discrete topology of X,

$$\overline{B_\epsilon(x)} = \overline{\{x\}} = \{x\}$$

as every set in the discrete topology is closed. However,

$$\overline{B}_\epsilon(x) = \{y \in X : d(x, y) \leq 1\} = X$$

as $d(X \times X) \subset \{0, 1\}$ by definition.

Theorem 2.0.6 (Metric Spaces are Hausdorff)**.** *Every metric space is Hausdorff.*

Proof. Fix a metric space (X, d) and fix distinct $x, y \in X$. By definition, $d(x, y) > 0$ ensures $\delta := \frac{d(x,y)}{2}$ remains positive.

Thus, x and y respectively have disjoint neighborhoods $B_\delta(x)$ and $B_\delta(y)$ in the metric topology.

> Suppose, to the contrary, that z lives in the intersection of the open balls. Particularly,
>
> $$d(x, y) \leq d(x, z) + d(z, y) < 2\delta = d(x, y),$$
>
> which is a contradiction. Indeed, the two open balls are disjoint

By definition, (X, d) is Hausdorff. □

Corollary 2.0.7 (Infinite Finite-Complement Space is not Metrizable)**.** *Suppose X is a set with infinitely many elements. Then, X equipped with the finite-complement topology is not metrizable.*

Proof. Denote τ_C the finite-complement topology of X and recall that (X, τ_C) is not Hasudorff.

If (X, τ) is metrizable, then (X, τ) is Hasudorff by the preceding theorem, which is a contradiction. Indeed, (X, τ) is not metrizable. □

Lemma 2.0.8 (Uniform Metric)**.** *Suppose (X, d) is a metric space. Then, $\overline{d}\colon X \times X \to \mathbb{R}$ defined by $\forall x, y \in X,\ \overline{d}(x, y) := \min\{d(x, y), 1\}$, is a metric of X.*

Proof. Since d and 1 are non-negative, $\overline{d}$ remains non-negative.

Fix $x, y, z \in X$. By definition, d is symmetric ensures

$$\overline{d}(x, y) := \min\{d(x, y), 1\} = \min\{d(y, x), 1\} =: \overline{d}(y, x),$$

so $\overline{d}$ is symmetric.

If $\overline{d}(x, y) = 0$, then $\overline{d}(x, y) \in \{d(x, y), 1\}$ ensures that $d(x, y) = \overline{d}(x, y) = 0$ and, therefore, $x = y$. Moreover, $d(x, x) = 0$ ensures $\overline{d}(x, x) := 0$. Indeed, $\overline{d}$ vanishes on and only on reflexive pairs.

Lastly, we verify that $\overline{d}$ satisfies the triangle inequality.

If $\overline{d}(x, y) + \overline{d}(y, z) \geq 1$, then $\overline{d}(x, z) \leq 1 \leq \overline{d}(x, y) + \overline{d}(y, z)$. Thus, we suppose that $\overline{d}(x, y) + \overline{d}(y, z) < 1$, so the non-negativity of $\overline{d}$ ensures that $\overline{d}(x, y), \overline{d}(y, z) < 1$ and, thus,

$$\overline{d}(x, y) = d(x, y) \text{ and } \overline{d}(y, z) = d(y, z).$$

Since d satisfies the triangle inequality, we conclude

$$\overline{d}(x, z) \leq d(x, z) \leq d(x, y) + d(y, z) = \overline{d}(x, y) + \overline{d}(y, z).$$

Indeed, $\overline{d}$ satisfies the triangle inequality. By definition, $\overline{d}$ is a metric on X. □

Definition 2.0.7 (Uniform metric)**.** *Suppose (X, d) is a metric space. Then, the uniform (standard bounded) metric (induced from d) is the metric $\overline{d}$ of X defined by*

$$\forall x, y \in X,\ \overline{d}(x, y) := \min\{d(x, y), 1\}.$$

Lemma 2.0.9. *Suppose (X, d) is a metric space, $\epsilon > 0$, and $x \in X$. If $\epsilon' < \min\{\epsilon, 1\}$, then $B^{\overline{d}}_{\epsilon'}(x) \subset B^d_\epsilon(x)$.*

Proof. Suppose $\epsilon' < \min\{\epsilon, 1\}$ and fix $y \in B^{\overline{d}}_{\epsilon'}(x)$. Since $\epsilon' < 1$, $\overline{d}(x, y) = d(x, y)$, whereas $\epsilon' < \epsilon$ ensures that $\epsilon > \epsilon' > \overline{d}(x, y) = d(x, y)$, which implies $y \in B^d_\epsilon(x)$ by definition. Indeed, $B^{\overline{d}}_{\epsilon'}(x) \subset B^d_\epsilon(x)$. □

Example 2.0.10 ($\overline{d}$-Balls Need not be Smaller)**.** *The containment $B^{\overline{d}}_\epsilon(x) \subset B^d_\epsilon(x)$ need not holds for every metric space (X, d), $\epsilon > 0$ and $x \in X$.*

Indeed, equip $\mathbb{R}$ with its standard metric d and consider $x := 0$ and $\epsilon := 2$.

We observe that $d(10, 0) := 10$ ensures that $10 \notin B^d_\epsilon(x)$ and $\overline{d}(10, 0) = 1$. However, $10 \in B^{\overline{d}}_\epsilon(x)$ since $\overline{10, 0} < 2$.

Theorem 2.0.11 (Compatibility of Uniform Metric)**.** *Suppose (X, d) is a metric space. Then, the uniform metric $\overline{d}$ is compatible with the metric topology of X induced by d.*

Proof. It suffices to show the metric topologies on X share the basis

$$\overline{\mathcal{B}} := \left\{ B^d_\epsilon(x) : \epsilon \in (0, 1),\ x \in X \right\},$$

which we achieve via 1.2.4.

By definition, $\overline{\mathcal{B}}$ is a collection of the standard basis and, therefore, open sets of the d-metric topology. For any $B^d_\epsilon(x) \in \overline{\mathcal{B}}$, $B^d_\epsilon(x) = B^{\overline{d}}_\epsilon(x)$,

> For any $y \in X$,
>
> $$y \in B^d_\epsilon(x) \iff d(x, y) < \epsilon \iff \overline{d}(x, y) < \epsilon \iff B^{\overline{d}}_\epsilon(x)$$
>
> since $\epsilon < 1$ ensures $\overline{d}(x, y) = d(x, y)$.

which is a standard basis element and, thus, an open set of the $\overline{d}$-metric topology. That is, $\overline{\mathcal{B}}$ is a subcollection of the $\overline{d}$-metric topology.

Fix an open set U of the $\overline{d}$-metric topology and fix $x \in U$. By 2.0.2, there exists $\delta > 0$ such that $x \in B^{\overline{d}}_\delta(x) \subset U$. By considering $\epsilon := \frac{\min\{\delta, 1\}}{2} \in (0, 1)$, we obtain that $B^d_\epsilon(x) \in \overline{\mathcal{B}}$ and

$$x \in B^d_\epsilon(x) = B^{\overline{d}}_\epsilon(x) \subset B^{\overline{d}}_\delta(x) \subset U.$$

Now, fix an open set U of the d-metric topology and fix $x \in U$. By 2.0.2, there exists $\delta > 0$ such that $x \in B^d_\delta(x) \subset U$. Consider $\delta' := \frac{\min\{\delta, 1\}}{2}$. Since $B^{\overline{d}}_{\delta'}(x)$ is a $\overline{d}$-neighborhood of x by definition, the preceding case ensures there exists some $B \in \overline{\mathcal{B}}$ such that

$$x \in B \subset B^{\overline{d}}_{\delta'}(x) \subset B^d_\delta(x) \subset U.$$

> Alternatively, we may just repeat the preceding argument almost verbatim.

Via 1.2.4, $\overline{\mathcal{B}}$ is a basis for both the d-metric topology and $\overline{d}$-metric topology of X. Indeed, the two topologies coincide. That is, $\overline{d}$ is compatible with the d-metric topology. □

Lemma 2.0.12 (Uniform Metric on Product)**.** *Suppose* $\{(X_\alpha, d_\alpha)\}_{\alpha\in J}$ *is a collection of metric spaces for some index set* J*. Then, the map* $\overline{d}\colon \prod_{\alpha\in J} X_\alpha \times \prod_{\alpha\in J} X_\alpha \to \mathbb{R}$ *defined by*

$$\overline{d}(\tilde{x},\tilde{y}) := \sup_{\alpha\in J} \overline{d_\alpha}(\tilde{x}_\alpha,\tilde{y}_\alpha),\ \forall \tilde{x},\tilde{y} \in \prod_{\alpha\in J} X_\alpha$$

is a metric of $\prod_{\alpha\in J} X_\alpha$*.*

Proof. For brevity, denote $X := \prod_{\alpha\in J} X_\alpha$.

For any $\alpha \in J$, the range of the metric $\overline{d}_\alpha$ is contained in $[0,1]$ by definition. Particularly, for any $\tilde{x},\tilde{y} \in X$, $\left\{\overline{d}_\alpha(\tilde{x}_\alpha,\tilde{y}_\alpha) : \alpha \in J\right\} \subset [0,1]$, which ensures $\overline{d}(\tilde{x},\tilde{y}) \in \mathbb{R}$ and is non-negative.

Fix $\tilde{x},\tilde{y},\tilde{z} \in X$.

By defintion, $\overline{d}$ is symmetric since $\overline{d}(\tilde{x},\tilde{y}) := \sup_{\alpha\in J} \overline{d_\alpha}(\tilde{x}_\alpha,\tilde{y}_\alpha) = \sup_{\alpha\in J} \overline{d_\alpha}(\tilde{y}_\alpha,\tilde{x}_\alpha) =: \overline{d}(\tilde{y},\tilde{x})$ as $\forall\alpha \in J$ $\overline{d_\alpha}$ is symmetric.

If $\tilde{x} = \tilde{y}$, then $\overline{d}(\tilde{x},\tilde{y}) = \sup_{\alpha\in J} \overline{d_\alpha}(\tilde{x}_\alpha,\tilde{x}_\alpha) = \sup_{\alpha\in J} 0 = 0$. If $\overline{d}(\tilde{x},\tilde{y}) = 0$, then $\forall\alpha \in J$, $\overline{d}(\tilde{x},\tilde{y}) \geq \overline{d_\alpha}(\tilde{x}_\alpha,\tilde{y}_\alpha) \geq 0$ ensures that $\overline{d_\alpha}(\tilde{x}_\alpha,\tilde{y}_\alpha) = 0$ and, thus, $\tilde{x}_\alpha = \tilde{y}_\alpha$, which implies $\tilde{x} = \tilde{y}$. Indeed, $\overline{d}$ vanishes on and only on reflexive pairs.

For each $\alpha \in J$, $\overline{d_\alpha}$ satisfies the triangle inequality ensures

$$\overline{d_\alpha}(\tilde{x}_\alpha,\tilde{z}_\alpha) \leq \overline{d_\alpha}(\tilde{x}_\alpha,\tilde{y}_\alpha) + \overline{d_\alpha}(\tilde{y}_\alpha,\tilde{z}_\alpha) \leq \overline{d}(\tilde{x},\tilde{y}) + \overline{d}(\tilde{y},\tilde{z}).$$

Particularly, $\overline{d}$ satisfies the triangle inequality $\overline{d}(\tilde{x},\tilde{z}) \leq \overline{d}(\tilde{x},\tilde{y}) + \overline{d}(\tilde{y},\tilde{z})$ by its construction. By definition, $\overline{d}$ is a metric on X. □

Definition 2.0.8 (Uniform metric on Cartesian product)**.** *Suppose* $\{(X_\alpha, d_\alpha)\}_{\alpha\in J}$ *is a collection of metric spaces for some index set* J*.*

(a) The uniform metric of $\prod_{\alpha\in J} X_\alpha$ *is the metric* $\overline{d}\colon \prod_{\alpha\in J} X_\alpha \times \prod_{\alpha\in J} X_\alpha \to \mathbb{R}$ *defined by*

$$\overline{d}(\tilde{x},\tilde{y}) := \sup_{\alpha\in J} \overline{d_\alpha}(\tilde{x}_\alpha,\tilde{y}_\alpha),\ \forall \tilde{x},\tilde{y} \in \prod_{\alpha\in J} X_\alpha.$$

(b) The uniform topology of $\prod_{\alpha\in J} X_\alpha$ *is the metric topology induced by its uniform metric.*

Theorem 2.0.13 (Ordering of Topologies on Products of Metric Spaces)**.** *Suppose* $\{(X_\alpha, d_\alpha)\}_{\alpha\in J}$ *is a collection of metric spaces for some index set* J*. For* $\prod_{\alpha\in J} X_\alpha$*,*

(a) its box topology is finer than its uniform topology, which is finer than its product topology;

(a.1) if J *is finite, then the above mentioned topologies coincide.*

Proof. For brevity, denote the product $\prod_{\alpha\in J} X_\alpha$, box topology, uniform topology, and product topology of $\prod_{\alpha\in J} X_\alpha$, by X, τ_{box}, τ_{unif}, and τ_{prod}, respectively.

We make use of 1.2.5. Fix $x \in X$.

[$\tau_{box} \supset \tau_{unif}$] Fix a standard basis element B of τ_{unif} containing x. Since B is open in the metric topology τ_{unif}, there exists $\delta > 0$ with $x \in B_\delta(x) \subset B$.

Consider $\delta' := \frac{\delta}{2}$ and the standard basis element $U := \prod_{\alpha\in J} B^{\overline{d_\alpha}}_{\delta'}(x_\alpha)$ of τ_{box}.

For each $\alpha \in J$, the uniform metric $\overline{d_\alpha}$ is compatible with the metric topology of (X_α, d_α), which ensures that $B_{\delta'}^{\overline{d_\alpha}}(x_\alpha)$ is open in (X_α, d_α). Indeed, U is a standard basis element of the box topology.

By construction, $x \in U$, where every $y \in U$ satisfies

$$\forall \alpha \in J,\ \overline{d_\alpha}(y_\alpha, x_\alpha) < \delta'$$

and, thus, $\overline{d}(y, x) \leq \delta' < \delta$. That is, $x \in U \subset B_\delta(x) \subset B$. By 1.2.5, τ_{box} is finer than τ_{unif}.

$[\tau_{unif} \supset \tau_{prod}]$ Fix a standard basis element U of τ_{prod} containing x, say

$$U = \left[\prod_{\alpha \in J/\{\beta_1,\ldots,\beta_n\}} X_\alpha\right] \times \prod_{j=1}^{n} U_{\beta_j}$$

For each $\beta \in \{\beta_1, \ldots, \beta_n\}$, $x_\beta \in U_\beta$ and U_β is open in the metric space (X_β, d_β) and, thus, $(X_\beta, \overline{d_\beta})$ ensure $x_\beta \in B_{\epsilon_\beta}^{\overline{d_\beta}}(x_\beta) \subset U_\beta$ for some $\epsilon_\beta > 0$.

Consider $\delta := \min\{\epsilon_{\beta_1}, \ldots, \epsilon_{\beta_n}\}$ and the standard basis element $B := B_\delta(x)$ of τ_{unif}. By construction $x \in B$, where, for each $y \in B$, $\overline{d}(x, y) < \delta$ ensures that

$$\forall \beta \in \{\beta_1, \ldots, \beta_n\},\ \overline{d_\beta}(x_\beta, y_\beta) \leq \overline{d}(x, y) < \delta \leq \epsilon_\beta$$

and, thus, $y_\beta \in B_{\epsilon_\beta}^{\overline{d_\beta}}(x_\beta) \subset U_\beta$. Particularly, $x \in B \subset U$. By 1.2.5, τ_{unif} is finer than τ_{prod}.

(a.1) If J is finite, then the box topology and product topology of X coincide, so part (a) ensures all three topologies coincide. □

Definition 2.0.9 (Uniform metric on $\mathbb{R}^X$)**.** *Suppose X is a set.*

(a) The uniform metric ρ on $\mathbb{R}^X$ is the uniform metric of $\mathbb{R}^X$ induced by the standard metric of $\mathbb{R}$; that is, the metric $\rho\colon \mathbb{R}^X \times \mathbb{R}^X \to \mathbb{R}$ defined by

$$\rho(f, g) := \sup_{x \in X} \overline{d_{\mathbb{R}}}(f(x), g(x)) = \sup_{x \in X}[\min\{|f(x) - g(x)|, 1\}],\ \forall f, g \in \mathbb{R}^X.$$

(b) The uniform topology of $\mathbb{R}^X$ is the metric topology on $\mathbb{R}^X$ induced by its uniform metric.

Corollary 2.0.14 (Ordering of Topologies on $\mathbb{R}^\omega$)**.** *For $\mathbb{R}^\omega$, its box topology is strictly finer than its uniform topology, which is strictly finer than its product topology.*

Proof. Via 2.0.13, it remains to show that the fineness is strict.

$[\tau_{box} \neq \tau_{unif}]$ The set $U := \prod_{n \in \mathbb{N}}(-\frac{1}{n}, \frac{1}{n})$ is open in the box topology but not the uniform topology. Indeed, τ_{box} is strictly finer than τ_{unif}.

Suppose, to the contrary, that U is open in the uniform topology.

Consider the constant sequence $\tilde{0}$ of value 0, which lives in U. Since U is a neighborhood of $\tilde{0}$ in τ_{unif} by assumtpion, $B_\epsilon(\tilde{0}) \subset U$ for some $\epsilon > 0$.

It follows that the constant sequence $\tilde{\frac{\epsilon}{2}}$ of value $\frac{\epsilon}{2}$ lives in the open ball $B_\epsilon(\tilde{0})$ and, thus, U. As a result, $\forall n \in \mathbb{N}$, $\left|\frac{\epsilon}{2}\right| < \frac{1}{n}$, which is a contradiction since $\frac{\epsilon}{2} > 0$.

$[\tau_{unif} \neq \tau_{prod}]$ Denote by $\tilde{0}$ the constant sequence of value 0. The set $B_1(\overline{0})$ is open in the uniform topology but not the product topology, which ensures that τ_{unif} is strictly finer than τ_{prod}.

Suppose, to the contrary, that $B_1(\tilde{0})$ is open in the product topology. Since $\tilde{0} \in B_1(\tilde{0})$, there is a standard basis element U of τ_{prod} satisfying $\tilde{0} \in U \subset B_1(\tilde{0})$, say

$$U = \left[\prod_{n\in\mathbb{N}/\{n_1,\ldots,n_N\}} \mathbb{R}\right] \times \prod_{j=1}^{N} U_{n_j}.$$

For each $n \in \{n_1, \ldots, n_N\}$, U_n is a neighborhood of 0 ensures $B_{\epsilon_n}(0) \subset U_n$ for some $\epsilon_n > 0$. Particularly, $U' := \left[\prod_{n\in\mathbb{N}/\{n_1,\ldots,n_N\}} \mathbb{R}\right] \times \prod_{j=1}^{N} B_{\epsilon_{n_j}}(0)$ satisfies $U' \subset U \subset B_1(\tilde{0})$. It follows that $\tilde{x} \in \mathbb{R}^\omega$ defined by

$$\tilde{x}_n := \begin{cases} \frac{\epsilon_n}{2} & \text{if } n \in \{n_1, \ldots, n_N\} \\ 2 & \text{if } n \notin \{n_1, \ldots, n_N\} \end{cases}, \ \forall n \in \mathbb{N}$$

lives in U' and, thus, $B_1(\tilde{0})$.

Consider $m \in \mathbb{N}/\{n_1, \ldots, n_N\}$. By construction, $\tilde{x}_m := 2$, where

$$\overline{d_{\mathbb{R}}}(\tilde{x}_m, \tilde{0}_m) := \min\left\{\left|\tilde{x}_m - \tilde{0}_m\right|, 1\right\} = 1.$$

However, $\tilde{x} \in B_1(\tilde{0})$ ensures that

$$1 > \rho(\tilde{x}, \tilde{0}) \geq \overline{d_{\mathbb{R}}}(\tilde{x}_m, \tilde{0}_m) = 1,$$

which is a contradiction.

□

Remark 2.0.1. *When considering topologies of $\mathbb{R}^\omega$, the set $\prod_{n\in\mathbb{N}}(-\frac{1}{n}, \frac{1}{n})$ and the constant sequences are useful in obtaining contradictions.*